

COURSE NOTES

GROUP THEORY

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Chapter 1

Cyclic Groups and Their Automorphism Groups

1.1 Basic Facts about Cyclic Groups

Definition 1.1.1 (Cyclic group). *Let G be a group. We say that G is **cyclic** if there exists an element $g \in G$ such that*

$$G = \langle g \rangle = \{g^k : k \in \mathbb{Z}\}.$$

If G has order n , then it is usually denoted by C_n .

A finite cyclic group is often written additively:

$$C_n \cong \mathbb{Z}/n\mathbb{Z} = \{\bar{0}, \bar{1}, \dots, \overline{n-1}\}.$$

Since cyclic groups are abelian, both multiplicative and additive notation are commonly used.

Proposition 1.1.2 (Basic properties of cyclic groups). *Let G be a cyclic group.*

- (1) A finite cyclic group is uniquely determined up to isomorphism by its order.*
- (2) Every subgroup of a cyclic group is cyclic.*
- (3) Every quotient group of a cyclic group is cyclic.*

Proof. Suppose $G = \langle g \rangle$ and $|G| = n$. Then the map

$$\langle g \rangle \longrightarrow \mathbb{Z}/n\mathbb{Z}, \quad g^i \longmapsto \bar{i}$$

is an isomorphism. Hence every cyclic group of order n is isomorphic to $\mathbb{Z}/n\mathbb{Z}$.

Now let $H \leq \langle g \rangle$. If $H = \{1\}$, then H is cyclic. Otherwise, choose the smallest positive

integer d such that $g^d \in H$. We claim that

$$H = \langle g^d \rangle.$$

Indeed, if $g^m \in H$, write $m = qd + r$ with $0 \leq r < d$. Then

$$g^r = g^m (g^d)^{-q} \in H.$$

By the minimality of d , we must have $r = 0$. Hence $g^m \in \langle g^d \rangle$, and so $H = \langle g^d \rangle$.

Finally, if $N \triangleleft G = \langle g \rangle$, then

$$G/N = \langle gN \rangle,$$

so G/N is cyclic. □

1.2 Automorphisms of Finite Cyclic Groups

Definition 1.2.1 (The unit group modulo n). *The group of units modulo n is*

$$U(n) = (\mathbb{Z}/n\mathbb{Z})^\times = \{\bar{a} \in \mathbb{Z}/n\mathbb{Z} : \gcd(a, n) = 1\}.$$

Its group operation is multiplication modulo n , and

$$|U(n)| = \varphi(n),$$

where φ is Euler's phi function.

Theorem 1.2.2 (Automorphisms of a finite cyclic group). *Let $C_n = \langle g \rangle$. Then*

$$\text{Aut}(C_n) \cong (\mathbb{Z}/n\mathbb{Z})^\times = U(n).$$

In particular,

$$|\text{Aut}(C_n)| = \varphi(n).$$

Proof. Every automorphism of C_n is completely determined by the image of the generator g .

Let $\sigma \in \text{Aut}(C_n)$. Since $C_n = \langle g \rangle$, there exists an integer i such that

$$\sigma(g) = g^i.$$

Because automorphisms preserve orders, $\sigma(g) = g^i$ must also have order n . This happens exactly when

$$\gcd(i, n) = 1.$$

Thus each automorphism determines a unit $\bar{i} \in (\mathbb{Z}/n\mathbb{Z})^\times$.

Conversely, if $\gcd(a, n) = 1$, then g^a is also a generator of C_n . Hence the assignment

$$g \mapsto g^a$$

extends uniquely to an automorphism of C_n . Denote this automorphism by σ_a .

Now define

$$\Psi : \text{Aut}(C_n) \longrightarrow (\mathbb{Z}/n\mathbb{Z})^\times, \quad \sigma_a \mapsto \bar{a}.$$

This map is bijective by the previous discussion. Moreover,

$$(\sigma_a \circ \sigma_b)(g) = \sigma_a(g^b) = (g^a)^b = g^{ab},$$

so

$$\sigma_a \circ \sigma_b = \sigma_{ab}.$$

Therefore Ψ is a group isomorphism. □

Remark 1.2.3. *Theorem 1.2.2 reduces the computation of $\text{Aut}(C_n)$ to the computation of the unit group $U(n)$.*

1.3 Odd Prime Powers

1.3.1 The prime case

Remark 1.3.1 (Finite fields). *Let $\mathbb{F}_{p^e}$ be the finite field with p^e elements. Then*

$$\mathbb{F}_{p^e}^+ \cong \underbrace{C_p \oplus \cdots \oplus C_p}_{e \text{ copies}}$$

and

$$\mathbb{F}_{p^e}^\times \cong C_{p^e-1}.$$

In particular,

$$\mathbb{F}_p^\times \cong C_{p-1}.$$

Remark 1.3.2. *Let p be a prime and let $e \geq 1$. Then*

$$\mathbb{F}_{p^e}^\times \cong C_{p^e-1}.$$

Proof. Put $q = p^e$ and $F = \mathbb{F}_q$. Let

$$G = F^\times.$$

Then G is a finite abelian group under multiplication, and

$$|G| = q - 1 = p^e - 1.$$

Let m be the exponent of G , namely

$$m = \text{lcm}\{\text{ord}(a) : a \in G\}.$$

By Lagrange's theorem, $\text{ord}(a) \mid |G| = q - 1$ for every $a \in G$. Hence

$$m \mid q - 1.$$

In particular,

$$m \leq q - 1.$$

We first show that there exists an element of G whose order is exactly m . Write

$$m = \prod_{\ell \mid m} \ell^{a_\ell}$$

as a product of prime powers. For each prime $\ell \mid m$, by the definition of m , there exists $g_\ell \in G$ such that the ℓ -part of $\text{ord}(g_\ell)$ is ℓ^{a_ℓ} . Define

$$h_\ell = g_\ell^{\text{ord}(g_\ell)/\ell^{a_\ell}}.$$

Then

$$\text{ord}(h_\ell) = \ell^{a_\ell}.$$

Since G is abelian, the elements h_ℓ commute. Therefore the product

$$g = \prod_{\ell \mid m} h_\ell$$

has order

$$\text{ord}(g) = \prod_{\ell \mid m} \ell^{a_\ell} = m,$$

because the orders of the h_ℓ 's are pairwise coprime.

Now, by the definition of m , every element $a \in G$ satisfies

$$a^m = 1.$$

Thus every element of $G = F^\times$ is a root of the polynomial

$$X^m - 1 \in F[X].$$

But a nonzero polynomial of degree m over a field has at most m roots. Hence

$$q - 1 = |F^\times| \leq m.$$

Together with $m \leq q - 1$, we get

$$m = q - 1.$$

Therefore G contains an element g of order $q - 1 = |G|$. Hence

$$G = \langle g \rangle$$

is cyclic. Consequently,

$$\mathbb{F}_{p^e}^\times = F^\times \cong C_{q-1} = C_{p^e-1}.$$

□

Corollary 1.3.3. *If p is prime, then*

$$\text{Aut}(C_p) \cong \text{U}(p) \cong \mathbb{F}_p^\times \cong C_{p-1}.$$

Proof. By Theorem 1.2.2,

$$\text{Aut}(C_p) \cong (\mathbb{Z}/p\mathbb{Z})^\times.$$

But $(\mathbb{Z}/p\mathbb{Z})^\times = \mathbb{F}_p^\times$, and the multiplicative group of a finite field is cyclic. Hence

$$\text{Aut}(C_p) \cong C_{p-1}.$$

□

1.3.2 Odd prime powers

Theorem 1.3.4 (Units modulo odd prime powers). *Let p be an odd prime and let $e \geq 2$. Then*

$$\text{U}(p^e) \cong C_{p^{e-1}(p-1)}.$$

Equivalently,

$$\text{Aut}(C_{p^e}) \cong C_{p^{e-1}(p-1)}.$$

Proof. We show that $\text{U}(p^e)$ contains an element of order

$$\varphi(p^e) = p^{e-1}(p-1).$$

First choose an integer λ whose image modulo p is a primitive root in $\mathbb{F}_p^\times$. Thus λ has order $p-1$ modulo p .

Consider λ modulo p^2 . Since its image modulo p has order $p-1$, its order modulo p^2 is either $p-1$ or $p(p-1)$.

If

$$\lambda^{p-1} \not\equiv 1 \pmod{p^2},$$

then λ has order $p(p-1)$ modulo p^2 .

If instead

$$\lambda^{p-1} \equiv 1 \pmod{p^2},$$

then consider $\lambda + p$. By the binomial theorem,

$$(\lambda + p)^{p-1} \equiv \lambda^{p-1} + p(p-1)\lambda^{p-2} \pmod{p^2}.$$

Using $\lambda^{p-1} \equiv 1 \pmod{p^2}$, we get

$$(\lambda + p)^{p-1} \equiv 1 + p(p-1)\lambda^{p-2} \not\equiv 1 \pmod{p^2},$$

because $p \nmid \lambda$. Hence $\lambda + p$ has order $p(p-1)$ modulo p^2 .

Therefore there exists an integer r that is a primitive root modulo p^2 .

Now we show that the same r is a primitive root modulo p^e for every $e \geq 2$.

Since r has order $p(p-1)$ modulo p^2 , we have

$$r^{p-1} \equiv 1 + cp \pmod{p^2}$$

for some integer c with $p \nmid c$.

We claim that for every $k \geq 2$,

$$r^{p^{k-2}(p-1)} \equiv 1 + c_k p^{k-1} \pmod{p^k}, \quad p \nmid c_k.$$

The case $k = 2$ is exactly the congruence above.

Assume the congruence holds for some $k \geq 2$. Then

$$r^{p^{k-1}(p-1)} = \left(r^{p^{k-2}(p-1)} \right)^p \equiv (1 + c_k p^{k-1})^p \pmod{p^{k+1}}.$$

By the binomial theorem,

$$(1 + c_k p^{k-1})^p \equiv 1 + c_k p^k \pmod{p^{k+1}},$$

because the higher-order terms are divisible by p^{k+1} . Since $p \nmid c_k$, the induction continues.

Thus, modulo p^e ,

$$r^{p^{e-2}(p-1)} \not\equiv 1 \pmod{p^e},$$

while Euler's theorem gives

$$r^{p^{e-1}(p-1)} \equiv 1 \pmod{p^e}.$$

Therefore

$$\text{ord}_{p^e}(r) = p^{e-1}(p-1).$$

Hence $U(p^e)$ is cyclic of order $p^{e-1}(p-1)$. □

Proposition 1.3.5 (Alternative proof using Sylow subgroups). *Let p be an odd prime and let $d \geq 1$. Then $U(p^d)$ is cyclic.*

Proof. We use the structure theorem for finite abelian groups.

First, the Sylow p -subgroup of $U(p^d)$ is cyclic. Indeed, one checks that $1 + p$ has order p^{d-1} modulo p^d . More precisely,

$$(1 + p)^{p^{d-1}} \equiv 1 \pmod{p^d},$$

but

$$(1 + p)^{p^{d-2}} \not\equiv 1 \pmod{p^d}$$

when $d \geq 2$. Hence the Sylow p -subgroup is cyclic.

Now let $q \neq p$ be a prime. Consider the natural reduction map

$$\mathbb{Z}/p^d\mathbb{Z} \longrightarrow \mathbb{Z}/p\mathbb{Z}, \quad a + (p^d) \longmapsto a + (p).$$

It induces a surjective group homomorphism

$$U(p^d) \longrightarrow U(p).$$

The kernel has order p^{d-1} . Therefore, for $q \neq p$, every Sylow q -subgroup of $U(p^d)$ injects into $U(p)$.

But

$$U(p) \cong C_{p-1}$$

is cyclic. Hence every Sylow q -subgroup of $U(p^d)$ is cyclic.

Thus all Sylow subgroups of the finite abelian group $U(p^d)$ are cyclic. Since a finite abelian group is the direct product of its Sylow subgroups, and cyclic groups of pairwise coprime orders have cyclic direct product, $U(p^d)$ is cyclic. $\square$

1.4 Powers of Two

Theorem 1.4.1 (Units modulo powers of two). *For powers of 2, the unit groups are:*

$$U(2) = \{1\}, \quad U(4) \cong C_2,$$

and for $e \geq 3$,

$$U(2^e) \cong C_{2^{e-2}} \times C_2.$$

Consequently,

$$\text{Aut}(C_{2^e}) \cong C_{2^{e-2}} \times C_2 \quad (e \geq 3).$$

Proof. The cases $e = 1$ and $e = 2$ are immediate:

$$U(2) = \{1\}, \quad U(4) = \{\bar{1}, \bar{3}\} \cong C_2.$$

Now assume $e \geq 3$.

We first prove that 5 has order 2^{e-2} in $U(2^e)$. Since $5 = 1 + 4$, we claim that for every $m \geq 0$,

$$5^{2^m} \equiv 1 + 2^{m+2} \pmod{2^{m+3}}.$$

For $m = 0$, this says

$$5 \equiv 1 + 4 \pmod{8},$$

which is true. If the congruence holds for m , then

$$5^{2^{m+1}} = (5^{2^m})^2 \equiv (1 + 2^{m+2})^2 \pmod{2^{m+4}}.$$

Expanding gives

$$(1 + 2^{m+2})^2 = 1 + 2^{m+3} + 2^{2m+4} \equiv 1 + 2^{m+3} \pmod{2^{m+4}}.$$

So the claim follows by induction.

Taking $m = e - 2$ gives

$$5^{2^{e-2}} \equiv 1 \pmod{2^e},$$

while taking $m = e - 3$ gives

$$5^{2^{e-3}} \equiv 1 + 2^{e-1} \not\equiv 1 \pmod{2^e}.$$

Therefore

$$\text{ord}_{2^e}(5) = 2^{e-2}.$$

Now define

$$K = \{\bar{a} \in \text{U}(2^e) : a \equiv 1 \pmod{4}\}.$$

Exactly half of the odd residues modulo 2^e are congruent to 1 modulo 4, so

$$|K| = 2^{e-2}.$$

Since $5 \equiv 1 \pmod{4}$, all powers of 5 lie in K . But $\langle 5 \rangle$ has order 2^{e-2} , so

$$K = \langle 5 \rangle.$$

Every odd residue is congruent either to 1 or to 3 modulo 4. Thus every unit modulo 2^e lies either in K or in $-K$. Hence

$$\text{U}(2^e) = K \langle -1 \rangle.$$

Also,

$$K \cap \langle -1 \rangle = \{1\},$$

because $-1 \equiv 3 \pmod{4}$ and hence $-1 \notin K$. Therefore

$$\text{U}(2^e) = \langle 5 \rangle \times \langle -1 \rangle \cong C_{2^{e-2}} \times C_2.$$

□

Remark 1.4.2 (Three involutions in $\text{Aut}(C_{2^e})$). *For $e \geq 3$, the group*

$$\text{Aut}(C_{2^e}) \cong \text{U}(2^e)$$

has exactly three elements of order 2. If $C_{2^e} = \langle g \rangle$, these correspond to the following three units modulo 2^e :

$$-1, \quad 1 + 2^{e-1}, \quad 2^{e-1} - 1.$$

Thus the three automorphisms of order 2 are

$$g \mapsto g^{-1}, \quad g \mapsto g^{1+2^{e-1}}, \quad g \mapsto g^{2^{e-1}-1}.$$

1.5 Chinese Remainder Theorem

1.5.1 Ring version

Theorem 1.5.1 (Chinese Remainder Theorem). *Let R be a commutative ring with identity, and let*

$$A_1, A_2, \dots, A_k$$

be ideals of R . Define

$$\Phi : R \longrightarrow R/A_1 \times R/A_2 \times \cdots \times R/A_k$$

by

$$\Phi(r) = (r + A_1, r + A_2, \dots, r + A_k).$$

Then Φ is a ring homomorphism and

$$\text{Ker } \Phi = A_1 \cap A_2 \cap \cdots \cap A_k.$$

If the ideals are pairwise comaximal, meaning

$$A_i + A_j = R \quad (i \neq j),$$

then Φ is surjective and

$$A_1 \cap A_2 \cap \cdots \cap A_k = A_1 A_2 \cdots A_k.$$

Consequently,

$$R/(A_1 A_2 \cdots A_k) \cong \prod_{i=1}^k R/A_i.$$

Proof. It is clear that Φ is a ring homomorphism. Also,

$$r \in \text{Ker } \Phi \iff r \in A_i \text{ for every } i \iff r \in A_1 \cap \cdots \cap A_k.$$

Now assume that the ideals are pairwise comaximal.

We first prove surjectivity. Fix i . Since A_i is comaximal with every A_j for $j \neq i$,

repeated use of the identity

$$(I + J = R, I + K = R) \implies I + JK = R$$

shows that

$$A_i + \prod_{j \neq i} A_j = R.$$

Thus we may choose

$$u_i \in A_i, \quad e_i \in \prod_{j \neq i} A_j$$

such that

$$u_i + e_i = 1.$$

Then

$$e_i \equiv 1 \pmod{A_i}, \quad e_i \equiv 0 \pmod{A_j} \quad (j \neq i).$$

Given an arbitrary element

$$(r_1 + A_1, \dots, r_k + A_k) \in \prod_{i=1}^k R/A_i,$$

set

$$r = r_1 e_1 + r_2 e_2 + \dots + r_k e_k.$$

Then

$$\Phi(r) = (r_1 + A_1, \dots, r_k + A_k),$$

so Φ is surjective.

It remains to prove

$$A_1 \cap \dots \cap A_k = A_1 \cdots A_k.$$

The inclusion

$$A_1 A_2 \cdots A_k \subseteq A_1 \cap \dots \cap A_k$$

is immediate.

For the reverse inclusion, it suffices by induction to prove the case of two comaximal ideals. Suppose $A + B = R$. Choose $a \in A$ and $b \in B$ such that

$$a + b = 1.$$

If $x \in A \cap B$, then

$$x = x(a + b) = xa + xb.$$

Since $x \in B$ and $a \in A$, we have $xa \in AB$. Since $x \in A$ and $b \in B$, we have $xb \in AB$. Therefore $x \in AB$, so

$$A \cap B = AB.$$

The general case follows by induction. □

1.5.2 Integer version

Corollary 1.5.2 (Chinese Remainder Theorem for integers). *If $m_1, m_2, \dots, m_k$ are pairwise coprime positive integers, then*

$$\mathbb{Z}/(m_1 m_2 \cdots m_k) \mathbb{Z} \cong \mathbb{Z}/m_1 \mathbb{Z} \times \mathbb{Z}/m_2 \mathbb{Z} \times \cdots \times \mathbb{Z}/m_k \mathbb{Z}.$$

Proof. Apply Theorem 1.5.1 to $R = \mathbb{Z}$ and

$$A_i = m_i \mathbb{Z}.$$

Pairwise coprimality of the m_i is exactly the condition

$$m_i \mathbb{Z} + m_j \mathbb{Z} = \mathbb{Z} \quad (i \neq j).$$

□

Corollary 1.5.3 (Unit group version). *If $\gcd(m, n) = 1$, then*

$$(\mathbb{Z}/mn\mathbb{Z})^\times \cong (\mathbb{Z}/m\mathbb{Z})^\times \times (\mathbb{Z}/n\mathbb{Z})^\times.$$

More generally, if $m_1, \dots, m_k$ are pairwise coprime, then

$$U(m_1 m_2 \cdots m_k) \cong \prod_{i=1}^k U(m_i).$$

Proof. The Chinese Remainder Theorem gives a ring isomorphism

$$\mathbb{Z}/(m_1 m_2 \cdots m_k) \mathbb{Z} \cong \prod_{i=1}^k \mathbb{Z}/m_i \mathbb{Z}.$$

A ring isomorphism sends units to units and induces an isomorphism between the corresponding unit groups. Therefore

$$U(m_1 m_2 \cdots m_k) \cong \prod_{i=1}^k U(m_i).$$

□

1.6 The General Case

Theorem 1.6.1 (General structure of $\text{Aut}(C_n)$). *Let*

$$n = p_1^{e_1} p_2^{e_2} \cdots p_t^{e_t}$$

be the prime factorization of n . Then

$$\text{Aut}(C_n) \cong \text{U}(n) \cong \text{U}(p_1^{e_1}) \times \text{U}(p_2^{e_2}) \times \cdots \times \text{U}(p_t^{e_t}).$$

Moreover,

$$\text{U}(p^e) \cong \begin{cases} C_{p^{e-1}(p-1)}, & p \text{ odd}, \\ \{1\}, & p = 2, e = 1, \\ C_2, & p = 2, e = 2, \\ C_{2^{e-2}} \times C_2, & p = 2, e \geq 3. \end{cases}$$

Proof. By Theorem 1.2.2,

$$\text{Aut}(C_n) \cong \text{U}(n).$$

By the Chinese Remainder Theorem for unit groups,

$$\text{U}(n) \cong \prod_{i=1}^t \text{U}(p_i^{e_i}).$$

The structure of each odd prime-power factor is given by Theorem 1.3.4, and the structure of each power-of-two factor is given by Theorem 1.4.1. $\square$

Remark 1.6.2 (Summary table).

<i>Cyclic group</i>	<i>Automorphism group</i>	<i>Order</i>
C_n	$\text{U}(n)$	$\varphi(n)$
C_p	C_{p-1}	$p - 1$
$C_{p^e}, p \text{ odd}$	$C_{p^{e-1}(p-1)}$	$p^{e-1}(p - 1)$
$C_{2^e}, e \geq 3$	$C_{2^{e-2}} \times C_2$	2^{e-1}

Chapter 2

Dihedral Groups

2.1 Groups Generated by Two Involutions

Definition 2.1.1 (Involution). *An element of order 2 in a group is called an **involution**.*

Definition 2.1.2 (Dihedral group). *Let*

$$G = \langle a, b \rangle, \quad |a| = |b| = 2.$$

Put

$$x = ab.$$

*If $|x| = n < \infty$, then G is called a **dihedral group** of order $2n$, and we write*

$$G = D_{2n}.$$

*If $|x| = \infty$, then G is called the **infinite dihedral group**.*

Example 2.1.3. *Let*

$$a = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad b = \begin{pmatrix} 0 & 2 \\ \frac{1}{2} & 0 \end{pmatrix} \in \mathrm{GL}_2(\mathbb{R}).$$

Then

$$a^2 = b^2 = I_2,$$

so both a and b are involutions. Moreover,

$$ab = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 2 \end{pmatrix}.$$

Hence

$$(ab)^m = \begin{pmatrix} 2^{-m} & 0 \\ 0 & 2^m \end{pmatrix} \neq I_2 \quad (m \neq 0).$$

Thus $|ab| = \infty$, and therefore

$$G = \langle a, b \rangle$$

is an infinite dihedral group.

Proposition 2.1.4 (Normal form). *Let*

$$G = \langle a, b \rangle, \quad |a| = |b| = 2,$$

and put $x = ab$. Then every element of G has the form

$$x^i \quad \text{or} \quad x^i b$$

for some $i \in \mathbb{Z}$.

Proof. Since

$$a^2 = b^2 = 1,$$

any word in a and b can be reduced by deleting adjacent subwords aa and bb .

Hence every reduced word is alternating. Thus the possible words are of the following types:

$$a, ab, aba, abab, \dots$$

and

$$b, ba, bab, baba, \dots$$

Equivalently,

$$G = \{(ab)^i, (ba)^j, (ab)^i a, (ba)^j b : i, j \geq 0\}.$$

Now

$$(ba)^j = b(ab)^j b = ((ab)^j)^{-1} = x^{-j}.$$

Also,

$$(ab)^i a = (ab)^i abb = (ab)^{i+1} b = x^{i+1} b,$$

and

$$(ba)^j b = b(ab)^j = x^{-j} b.$$

Therefore every element of G is of the form

$$x^i \quad \text{or} \quad x^i b$$

for some $i \in \mathbb{Z}$. □

Theorem 2.1.5. *Let*

$$G = \langle a, b \rangle, \quad |a| = |b| = 2.$$

Then G is finite if and only if ab has finite order.

Moreover, if

$$|ab| = n,$$

then

$$|G| = 2n.$$

Proof. Put

$$x = ab.$$

If G is finite, then every element of G has finite order, so $x = ab$ has finite order.

Conversely, suppose that $|x| = n < \infty$. By the normal form above,

$$G = \{x^i, x^i b : i \in \mathbb{Z}\}.$$

Since $x^n = 1$, we may reduce the exponent i modulo n . Hence

$$G = \{1, x, x^2, \dots, x^{n-1}\} \cup \{b, xb, x^2b, \dots, x^{n-1}b\}.$$

Thus G is finite and $|G| \leq 2n$.

Let

$$H = \langle x \rangle.$$

Then $|H| = n$. We claim that

$$H \cap Hb = \emptyset.$$

Indeed, if $x^i = x^j b$, then

$$b = x^{j-i} \in H.$$

It follows that both $a = xb$ and b lie in the cyclic group H . Since a and b are both involutions, this would force $a = b$, and hence

$$x = ab = 1,$$

contradicting $|x| = n$ unless $n = 1$. In the case $n = 1$, we have $a = b$ and $H = \{1\}$, so again $b \notin H$. Therefore H and Hb are disjoint.

Hence

$$|G| = |H| + |Hb| = n + n = 2n.$$

□

2.2 Semidirect Product Description

Definition 2.2.1 (Internal semidirect product). *Let G be a group, and let $A, B \leq G$. We say that G is an **internal semidirect product** of A by B if*

$$(1) \ A \triangleleft G;$$

$$(2) \ A \cap B = 1;$$

$$(3) \ G = AB.$$

In this case, we write

$$G = A : B \quad \text{or} \quad G = A \rtimes B.$$

Proposition 2.2.2. *Let*

$$G = D_{2n} = \langle a, b \rangle, \quad |a| = |b| = 2, \quad |ab| = n.$$

Put

$$x = ab.$$

Then

$$G = \langle x \rangle : \langle b \rangle.$$

In particular,

$$D_{2n} \cong C_n : C_2.$$

Proof. We have

$$x = ab, \quad a = xb.$$

Therefore

$$G = \langle a, b \rangle = \langle x, b \rangle.$$

Moreover,

$$bxb = b(ab)b = ba = (ab)^{-1} = x^{-1}.$$

Hence b normalizes $\langle x \rangle$, and so

$$\langle x \rangle \triangleleft G.$$

Also,

$$\langle x \rangle \cap \langle b \rangle = 1,$$

and

$$G = \langle x \rangle \langle b \rangle.$$

Thus

$$G = \langle x \rangle : \langle b \rangle.$$

Since

$$\langle x \rangle \cong C_n, \quad \langle b \rangle \cong C_2,$$

we get

$$D_{2n} \cong C_n : C_2.$$

Here the element b acts on $C_n = \langle x \rangle$ by inversion:

$$x \mapsto x^{-1}.$$

□

2.3 Automorphisms of D_{2n} for Odd n

Throughout this section, write

$$D_{2n} = \langle g, b \rangle = \langle g \rangle : \langle b \rangle,$$

where

$$|g| = n, \quad |b| = 2, \quad bgb = g^{-1}.$$

Thus

$$D_{2n} = \{g^i, g^i b : 0 \leq i \leq n-1\}.$$

Theorem 2.3.1 (Automorphism group of D_{2n} for n odd). *Assume that n is odd. Then every automorphism of D_{2n} has the form*

$$\theta_{i,j} : \begin{cases} g \mapsto g^i, \\ b \mapsto g^j b, \end{cases}$$

where

$$\gcd(i, n) = 1, \quad j \in \mathbb{Z}/n\mathbb{Z}.$$

Consequently,

$$|\text{Aut}(D_{2n})| = n\varphi(n),$$

and

$$\text{Aut}(D_{2n}) \cong C_n : \text{Aut}(C_n).$$

Proof. Let

$$\theta \in \text{Aut}(D_{2n}).$$

Since $|g| = n$ and n is odd, the element $\theta(g)$ also has odd order n .

But every element outside $\langle g \rangle$ has the form $g^j b$, and

$$(g^j b)^2 = g^j b g^j b = g^j g^{-j} b^2 = 1.$$

Hence every element outside $\langle g \rangle$ has order 2. Therefore

$$\theta(g) \in \langle g \rangle.$$

So

$$\theta(g) = g^i$$

for some i with

$$\gcd(i, n) = 1.$$

Next, $\theta(b)$ has order 2. Since n is odd, the only elements of order 2 in D_{2n} are the elements

$$g^j b, \quad 0 \leq j \leq n-1.$$

Therefore

$$\theta(b) = g^j b$$

for some $j \in \mathbb{Z}/n\mathbb{Z}$.

Conversely, for any i, j satisfying

$$\gcd(i, n) = 1, \quad j \in \mathbb{Z}/n\mathbb{Z},$$

define

$$\theta_{i,j}(g) = g^i, \quad \theta_{i,j}(b) = g^j b.$$

We check that the defining relations are preserved:

$$(g^i)^n = 1,$$

and

$$(g^j b)^2 = g^j b g^j b = g^j g^{-j} b^2 = 1.$$

Also,

$$(g^j b) g^i (g^j b)^{-1} = g^j b g^i b g^{-j} = g^j g^{-i} g^{-j} = g^{-i}.$$

Thus $\theta_{i,j}$ gives a homomorphism $D_{2n} \rightarrow D_{2n}$.

Since $\gcd(i, n) = 1$, the element g^i generates $\langle g \rangle$. Hence the image of $\theta_{i,j}$ contains $\langle g \rangle$. It also contains $g^j b$, and therefore contains

$$b = g^{-j} (g^j b).$$

Thus $\theta_{i,j}$ is surjective, hence an automorphism.

Therefore

$$\text{Aut}(D_{2n}) = \{\theta_{i,j} : \gcd(i, n) = 1, j \in \mathbb{Z}/n\mathbb{Z}\},$$

so

$$|\text{Aut}(D_{2n})| = n\varphi(n).$$

Finally, define

$$N = \{\theta_{1,j} : j \in \mathbb{Z}/n\mathbb{Z}\}.$$

Then

$$N \cong C_n.$$

Also define

$$U = \{\theta_{i,0} : \gcd(i, n) = 1\}.$$

Then

$$U \cong \text{Aut}(C_n).$$

Moreover,

$$\theta_{i,j} \circ \theta_{k,\ell} = \theta_{ik, j+i\ell}.$$

This shows that

$$\text{Aut}(D_{2n}) = N : U \cong C_n : \text{Aut}(C_n).$$

□

Corollary 2.3.2. *Let p be an odd prime. Then*

$$\text{Aut}(D_{2p}) \cong C_p : C_{p-1}.$$

More generally, for $e \geq 1$,

$$\text{Aut}(D_{2p^e}) \cong C_{p^e} : C_{p^{e-1}(p-1)}.$$

Proof. By the theorem,

$$\text{Aut}(D_{2n}) \cong C_n : \text{Aut}(C_n)$$

for odd n .

If $n = p$, then

$$\text{Aut}(C_p) \cong C_{p-1}.$$

Hence

$$\text{Aut}(D_{2p}) \cong C_p : C_{p-1}.$$

If $n = p^e$, then

$$\text{Aut}(C_{p^e}) \cong C_{p^{e-1}(p-1)}.$$

Therefore

$$\text{Aut}(D_{2p^e}) \cong C_{p^e} : C_{p^{e-1}(p-1)}.$$

□

Theorem 2.3.3 (Automorphism group of D_{2n} for n even). *Assume that n is even*

and $n \geq 4$. Then every automorphism of D_{2n} has the form

$$\theta_{i,j} : \begin{cases} g \mapsto g^i, \\ b \mapsto g^j b, \end{cases}$$

where

$$\gcd(i, n) = 1, \quad j \in \mathbb{Z}/n\mathbb{Z}.$$

Consequently,

$$|\text{Aut}(D_{2n})| = n\varphi(n),$$

and

$$\text{Aut}(D_{2n}) \cong C_n : \text{Aut}(C_n).$$

Proof. Let

$$D_{2n} = \langle g, b \rangle, \quad |g| = n, \quad |b| = 2, \quad bgb = g^{-1}.$$

Then

$$D_{2n} = \{g^i, g^i b : 0 \leq i \leq n-1\}.$$

Let

$$\theta \in \text{Aut}(D_{2n}).$$

Since g has order n , the element $\theta(g)$ also has order n .

But every element outside $\langle g \rangle$ has the form $g^j b$, and

$$(g^j b)^2 = g^j b g^j b = g^j g^{-j} b^2 = 1.$$

Hence every element outside $\langle g \rangle$ has order 2.

Since $n \geq 4$, an element of order n cannot lie outside $\langle g \rangle$. Therefore

$$\theta(g) \in \langle g \rangle.$$

So

$$\theta(g) = g^i$$

for some i satisfying

$$\gcd(i, n) = 1.$$

Next, $\theta(b)$ has order 2. Since θ is surjective, $\theta(b)$ cannot lie in $\langle g \rangle$; otherwise the image of θ would be contained in $\langle g \rangle$. Hence

$$\theta(b) = g^j b$$

for some

$$j \in \mathbb{Z}/n\mathbb{Z}.$$

Conversely, take i, j such that

$$\gcd(i, n) = 1, \quad j \in \mathbb{Z}/n\mathbb{Z}.$$

Define

$$\theta_{i,j}(g) = g^i, \quad \theta_{i,j}(b) = g^j b.$$

We check that the defining relations are preserved.

First,

$$(g^i)^n = 1.$$

Second,

$$(g^j b)^2 = g^j b g^j b = g^j g^{-j} b^2 = 1.$$

Finally,

$$(g^j b) g^i (g^j b)^{-1} = g^j b g^i b g^{-j} = g^j g^{-i} g^{-j} = g^{-i}.$$

Thus $\theta_{i,j}$ defines a homomorphism

$$D_{2n} \longrightarrow D_{2n}.$$

Since $\gcd(i, n) = 1$, the element g^i generates $\langle g \rangle$. Hence the image of $\theta_{i,j}$ contains $\langle g \rangle$. Also, it contains $g^j b$, so it contains

$$b = g^{-j} (g^j b).$$

Therefore the image contains both g and b , and hence $\theta_{i,j}$ is surjective. Since D_{2n} is finite, $\theta_{i,j}$ is an automorphism.

Thus

$$\text{Aut}(D_{2n}) = \{\theta_{i,j} : \gcd(i, n) = 1, j \in \mathbb{Z}/n\mathbb{Z}\}.$$

There are $\varphi(n)$ choices for i and n choices for j , so

$$|\text{Aut}(D_{2n})| = n\varphi(n).$$

Moreover, put

$$N = \{\theta_{1,j} : j \in \mathbb{Z}/n\mathbb{Z}\}, \quad U = \{\theta_{i,0} : \gcd(i, n) = 1\}.$$

Then

$$N \cong C_n, \quad U \cong \text{Aut}(C_n).$$

Also,

$$\theta_{i,j} \circ \theta_{k,\ell} = \theta_{ik, j+i\ell}.$$

Hence

$$\text{Aut}(D_{2n}) = N : U \cong C_n : \text{Aut}(C_n).$$

□

Corollary 2.3.4. *Let $m \geq 2$. Then*

$$\text{Aut}(D_{4m}) \cong C_{2m} : \text{Aut}(C_{2m}),$$

and hence

$$|\text{Aut}(D_{4m})| = 2m \varphi(2m).$$

Proof. This follows from the theorem by taking

$$n = 2m.$$

□

Corollary 2.3.5. *Let $e \geq 2$. Then*

$$\text{Aut}(D_{2^{e+1}}) \cong C_{2^e} : \text{Aut}(C_{2^e}).$$

In particular,

$$|\text{Aut}(D_{2^{e+1}})| = 2^e \varphi(2^e) = 2^{2e-1}.$$

More explicitly,

$$\text{Aut}(C_{2^e}) \cong \begin{cases} C_2, & e = 2, \\ C_2 \times C_{2^{e-2}}, & e \geq 3. \end{cases}$$

Therefore

$$\text{Aut}(D_8) \cong C_4 : C_2,$$

and for $e \geq 3$,

$$\text{Aut}(D_{2^{e+1}}) \cong C_{2^e} : (C_2 \times C_{2^{e-2}}).$$

Remark 2.3.6. *The case $e = 1$ is exceptional. In that case,*

$$D_4 \cong C_2 \times C_2,$$

so

$$\text{Aut}(D_4) \cong S_3.$$

Thus the formula above starts from $e \geq 2$.

2.4 Conjugacy Classes in D_{2n}

Let

$$D_{2n} = \langle g \rangle : \langle b \rangle = \{g^i, g^i b : 0 \leq i \leq n-1\},$$

where

$$|g| = n, \quad |b| = 2, \quad bgb = g^{-1}.$$

Lemma 2.4.1. *For every integer i ,*

$$bg^ib = g^{-i}.$$

Consequently,

$$b^{-1}g^ib = g^{-i}.$$

Also, for all integers i, j ,

$$g^j(g^ib)g^{-j} = g^{i+2j}b.$$

Proof. Since

$$bgb = g^{-1},$$

we get

$$bg^ib = (bgb)^i = (g^{-1})^i = g^{-i}.$$

Because $b^{-1} = b$, this also gives

$$b^{-1}g^ib = g^{-i}.$$

For the second identity,

$$g^j(g^ib)g^{-j} = g^{i+j}bg^{-j}.$$

Using

$$bg^{-j} = g^jb,$$

we get

$$g^j(g^ib)g^{-j} = g^{i+j}g^jb = g^{i+2j}b.$$

□

2.4.1 The case where n is odd

Theorem 2.4.2 (Conjugacy classes for odd n). *Assume that n is odd. Then the conjugacy classes of D_{2n} are*

$$\{1\},$$

$$\{g^i, g^{-i}\}, \quad 1 \leq i \leq \frac{n-1}{2},$$

and

$$\{b, gb, g^2b, \dots, g^{n-1}b\}.$$

Equivalently, the elements of odd order have conjugacy classes

$$\{g^i, g^{-i}\},$$

and all elements of even order are conjugate.

Proof. First consider the powers of g . Since $\langle g \rangle$ is abelian, conjugation by powers of g fixes

every g^i . Conjugation by b gives

$$b^{-1}g^ib = g^{-i}.$$

Therefore the conjugacy class of g^i is

$$\{g^i, g^{-i}\}.$$

Since n is odd,

$$g^i = g^{-i} \iff g^{2i} = 1 \iff n \mid 2i \iff n \mid i.$$

Thus the identity is its own conjugacy class, and the remaining rotations occur in pairs

$$\{g^i, g^{-i}\}.$$

Now consider the elements outside $\langle g \rangle$. They are

$$g^ib, \quad 0 \leq i \leq n-1.$$

Each has order 2, because

$$(g^ib)^2 = g^ibg^ib = g^ig^{-i}b^2 = 1.$$

So D_{2n} has exactly n elements of order 2, namely

$$g^ib, \quad 0 \leq i \leq n-1.$$

We claim that all these elements are conjugate. Indeed,

$$g^jb g^{-j} = g^{2j}b.$$

Since n is odd, the map

$$j \mapsto 2j \pmod{n}$$

is a bijection of $\mathbb{Z}/n\mathbb{Z}$. Hence the elements

$$g^jb g^{-j}, \quad 0 \leq j \leq n-1,$$

are exactly

$$b, gb, g^2b, \dots, g^{n-1}b.$$

Thus all elements of the form g^ib are conjugate.

Therefore, when n is odd, the elements of odd order are precisely the rotations, whose conjugacy classes are

$$\{g^i, g^{-i}\},$$

and the elements of even order are precisely the reflections g^ib , which form a single conjugacy class. $\square$

Remark 2.4.3. *Equivalently, since 2 is invertible modulo n , choose t such that*

$$2t \equiv 1 \pmod{n}.$$

Then

$$g^t(g^i b)g^{-t} = g^{i+2t}b = g^{i+1}b.$$

Hence consecutive reflections

$$g^i b \quad \text{and} \quad g^{i+1} b$$

are conjugate, and therefore all reflections are conjugate.

2.4.2 The case where n is even

Theorem 2.4.4 (Conjugacy classes for even n). *Assume that n is even. Then the conjugacy classes of D_{2n} are*

$$\begin{aligned} &\{1\}, \\ &\{g^{n/2}\}, \\ &\{g^i, g^{-i}\}, \quad 1 \leq i \leq \frac{n}{2} - 1, \end{aligned}$$

and the following two conjugacy classes of reflections:

$$\begin{aligned} &\{b, g^2 b, g^4 b, \dots, g^{n-2} b\}, \\ &\{gb, g^3 b, g^5 b, \dots, g^{n-1} b\}. \end{aligned}$$

Equivalently, the reflections split into two conjugacy classes according to whether the exponent of g is even or odd.

Proof. Recall that

$$D_{2n} = \langle g \rangle : \langle b \rangle = \{g^i, g^i b : 0 \leq i \leq n-1\},$$

where

$$|g| = n, \quad |b| = 2, \quad bgb = g^{-1}.$$

First consider the powers of g . Since $\langle g \rangle$ is abelian, conjugation by powers of g fixes every g^i . On the other hand,

$$b^{-1}g^i b = bg^i b = g^{-i}.$$

Therefore the conjugacy class of g^i is contained in

$$\{g^i, g^{-i}\}.$$

Moreover, b conjugates g^i to g^{-i} , so the conjugacy class of g^i is exactly

$$\{g^i, g^{-i}\},$$

unless

$$g^i = g^{-i}.$$

Now

$$g^i = g^{-i} \iff g^{2i} = 1 \iff n \mid 2i.$$

Since n is even, this happens precisely when

$$i \equiv 0 \pmod{n} \quad \text{or} \quad i \equiv \frac{n}{2} \pmod{n}.$$

Hence

$$\{1\}$$

and

$$\{g^{n/2}\}$$

are single conjugacy classes. The remaining rotations occur in pairs

$$\{g^i, g^{-i}\}, \quad 1 \leq i \leq \frac{n}{2} - 1.$$

Now consider the elements outside $\langle g \rangle$, namely the reflections

$$g^i b, \quad 0 \leq i \leq n-1.$$

For any integer j , we have

$$g^j(g^i b)g^{-j} = g^{i+2j}b.$$

Thus conjugation by a power of g changes the exponent i by an even integer.

Also,

$$b^{-1}(g^i b)b = b(g^i b)b = bg^i = g^{-i}b.$$

This again preserves the parity of i .

Hence conjugation preserves the parity of the exponent i in $g^i b$. Therefore a reflection with even exponent cannot be conjugate to a reflection with odd exponent.

Conversely, suppose i and k have the same parity. Then

$$k - i$$

is even. Since n is even, the congruence

$$2j \equiv k - i \pmod{n}$$

has a solution j . For such a j ,

$$g^j(g^i b)g^{-j} = g^{i+2j}b = g^k b.$$

Therefore any two reflections whose exponents have the same parity are conjugate.

Thus the reflections split into exactly two conjugacy classes:

$$\{g^{2j}b : 0 \leq j \leq \frac{n}{2} - 1\} = \{b, g^2b, g^4b, \dots, g^{n-2}b\},$$

and

$$\{g^{2j+1}b : 0 \leq j \leq \frac{n}{2} - 1\} = \{gb, g^3b, g^5b, \dots, g^{n-1}b\}.$$

This proves the theorem. $\square$

Corollary 2.4.5. *Assume that n is even. Then the involutions in D_{2n} are*

$$g^{n/2}$$

and

$$g^ib, \quad 0 \leq i \leq n-1.$$

They split into three conjugacy classes:

$$\{g^{n/2}\},$$

$$\{b, g^2b, g^4b, \dots, g^{n-2}b\},$$

and

$$\{gb, g^3b, g^5b, \dots, g^{n-1}b\}.$$

Proof. The element $g^{n/2}$ has order 2, and every reflection g^ib has order 2 because

$$(g^ib)^2 = g^ibg^ib = g^ig^{-i}b^2 = 1.$$

The conjugacy classes are exactly those described in the theorem. $\square$

Chapter 3

Constructing Groups and Group Actions

3.1 Constructing Groups from Automorphisms

Proposition 3.1.1 (The construction $G : H$). *Let G be a group and let*

$$H \leq \text{Aut}(G).$$

For $h \in H$ and $g \in G$, write g^h for the image of g under the automorphism h . Define

$$X = G : H = \{gh : g \in G, h \in H\}$$

with multiplication

$$(g_1 h_1)(g_2 h_2) = (g_1 g_2^{h_1^{-1}})(h_1 h_2).$$

Then X is a group. Moreover,

$$G \triangleleft X, \quad X/G \cong H.$$

Proof. Equivalently, as a set $X = G \times H$, and the multiplication is

$$(g_1, h_1)(g_2, h_2) = (g_1 g_2^{h_1^{-1}}, h_1 h_2).$$

The identity element is $(1, 1)$, and

$$(g, h)^{-1} = ((g^{-1})^h, h^{-1}).$$

Associativity follows from the fact that H acts on G by automorphisms.

Identifying G with $G \times 1$ and H with $1 \times H$, we get

$$X = GH, \quad G \cap H = 1.$$

Also,

$$h^{-1}gh = g^h \in G \quad (g \in G, h \in H),$$

so $G \triangleleft X$. Finally, the quotient map $X \rightarrow X/G$ kills G and identifies the quotient with H . $\square$

Remark 3.1.2. The group $G : \text{Aut}(G)$ is called the **holomorph** of G :

$$\text{Hol}(G) = G : \text{Aut}(G).$$

3.2 Examples from Cyclic Groups

Example 3.2.1 (The group $\mathbb{Z}_7 : \mathbb{Z}_2$). Let

$$G = \mathbb{Z}_7 = \langle g \rangle.$$

Then

$$\text{Aut}(G) \cong \mathbb{Z}_6.$$

Let $b \in \text{Aut}(G)$ have order 2. Then b acts by inversion:

$$g^b = g^{-1}.$$

Hence

$$X = G : \langle b \rangle = \langle g \rangle : \langle b \rangle$$

has presentation

$$X = \langle g, b \mid |g| = 7, |b| = 2, g^b = g^{-1} \rangle.$$

Since

$$(gb)^2 = gg^b = gg^{-1} = 1,$$

we get

$$X \cong \mathbb{Z}_7 : \mathbb{Z}_2 \cong D_{14}.$$

Example 3.2.2 (The group $\mathbb{Z}_7 : \mathbb{Z}_3$). Let

$$G = \mathbb{Z}_7 = \langle g \rangle.$$

Choose $z \in \text{Aut}(G)$ of order 3. For example, we may take

$$g^z = g^2,$$

because 2 has order 3 in $(\mathbb{Z}/7\mathbb{Z})^\times$.

Then

$$X = G : \langle z \rangle = \langle g, z \mid |g| = 7, |z| = 3, g^z = g^2 \rangle.$$

Thus

$$X \cong \mathbb{Z}_7 : \mathbb{Z}_3.$$

This group is nonabelian.

Example 3.2.3 (General cyclic group of prime order). *Let*

$$G = \langle g \rangle \cong \mathbb{Z}_p,$$

where p is prime. Then

$$\text{Aut}(G) \cong \mathbb{Z}_{p-1}.$$

Let $b \in \text{Aut}(G)$ satisfy

$$|b| = \ell.$$

Then

$$\ell \mid (p-1), \quad \langle b \rangle \cong \mathbb{Z}_\ell.$$

The corresponding semidirect product is

$$X = G : \langle b \rangle \cong \mathbb{Z}_p : \mathbb{Z}_\ell.$$

More explicitly, if

$$g^b = g^\lambda,$$

where $\lambda \in (\mathbb{Z}/p\mathbb{Z})^\times$ has order ℓ , then

$$X = \langle g, b \mid |g| = p, |b| = \ell, g^b = g^\lambda \rangle.$$

If $\ell = 2$, then $\lambda = -1$, and

$$X \cong D_{2p}.$$

If $\ell = p-1$, then

$$X \cong \mathbb{Z}_p : \mathbb{Z}_{p-1} = \text{AGL}(1, p),$$

the affine general linear group of degree 1 over $\mathbb{F}_p$.

Remark 3.2.4 (The case $p = 7$). *Since*

$$\text{Aut}(\mathbb{Z}_7) \cong \mathbb{Z}_6,$$

the possible images of a cyclic complement $\mathbb{Z}_6$ in $\text{Aut}(\mathbb{Z}_7)$ have order

$$1, 2, 3, 6.$$

Thus one obtains the following groups:

$$\mathbb{Z}_7 \times \mathbb{Z}_6, \quad (\mathbb{Z}_7 : \mathbb{Z}_3) \times \mathbb{Z}_2, \quad (\mathbb{Z}_7 : \mathbb{Z}_2) \times \mathbb{Z}_3, \quad \mathbb{Z}_7 : \mathbb{Z}_6.$$

Remark 3.2.5 (General cyclic case). *In general, for*

$$G = \mathbb{Z}_n,$$

we have

$$\text{Aut}(G) \cong \text{U}(n).$$

Thus every subgroup

$$H \leq \text{U}(n)$$

gives a group

$$X = \mathbb{Z}_n : H.$$

3.3 The Case $\mathbb{Z}_{35}$

Example 3.3.1. *Let*

$$G = \mathbb{Z}_{35} \cong \mathbb{Z}_5 \times \mathbb{Z}_7.$$

Then

$$\text{Aut}(G) \cong \text{Aut}(\mathbb{Z}_5) \times \text{Aut}(\mathbb{Z}_7) \cong \mathbb{Z}_4 \times \mathbb{Z}_6.$$

Write

$$\text{Aut}(G) = \langle b_1 \rangle \times \langle b_2 \rangle,$$

where

$$|b_1| = 4, \quad |b_2| = 6.$$

For any subgroup $H \leq \text{Aut}(G)$, we can form

$$X = \mathbb{Z}_{35} : H.$$

Moreover,

$$X < (\mathbb{Z}_5 : \mathbb{Z}_4) \times (\mathbb{Z}_7 : \mathbb{Z}_6).$$

Take

$$h = b_1^2 b_2^3.$$

Then h has order 2 and acts by inversion on both factors $\mathbb{Z}_5$ and $\mathbb{Z}_7$. Hence h acts by inversion on $\mathbb{Z}_{35}$, and therefore

$$X_0 = \mathbb{Z}_{35} : \langle h \rangle \cong D_{70}.$$

Remark 3.3.2. *If*

$$G = D_{2n} \cong \mathbb{Z}_n : \mathbb{Z}_2,$$

then

$$\text{Aut}(G) \cong \mathbb{Z}_n : \text{U}(n) = \mathbb{Z}_n : \text{Aut}(\mathbb{Z}_n).$$

3.4 Finite Field Example

Example 3.4.1 (Affine groups over finite fields). *Let*

$$F = \text{GF}(p^f) = \mathbb{F}_{p^f}, \quad q = p^f.$$

Let

$$G = F^+.$$

Then

$$G \cong \mathbb{Z}_p^f.$$

Also,

$$F^\times \cong \mathbb{Z}_{q-1}.$$

For each $h \in F^\times$, multiplication by h induces an automorphism of the additive group F^+ :

$$a \mapsto a^h = ah.$$

Indeed,

$$(a + b)^h = (a + b)h = ah + bh = a^h + b^h.$$

Thus

$$F^\times \leq \text{Aut}(F^+).$$

Define

$$X = F^+ : F^\times.$$

Then

$$X \cong \mathbb{Z}_p^f : \mathbb{Z}_{p^f-1}.$$

*The subgroup $F^\times \leq \text{GL}(f, p)$ is a **Singer cycle**.*

Proposition 3.4.2 (Automorphisms of the affine group). *Let*

$$X = F^+ : F^\times, \quad F = \mathbb{F}_{p^f}.$$

Then

$$\text{Aut}(X) \cong X : \text{Gal}(F/\mathbb{F}_p).$$

Since

$$\text{Gal}(F/\mathbb{F}_p) \cong \mathbb{Z}_f,$$

we have

$$\text{Aut}(X) \cong (\mathbb{Z}_p^f : \mathbb{Z}_{p^f-1}) : \mathbb{Z}_f.$$

Equivalently,

$$\text{Aut}(X) \cong (\mathbb{Z}_p^f : \langle g \rangle) : \langle z \rangle,$$

where

$$|g| = p^f - 1, \quad |z| = f, \quad g^z = g^p.$$

Here z is the Frobenius automorphism

$$z : x \mapsto x^p.$$

Proof. Put

$$q = p^f, \quad V = F^+, \quad H = F^\times.$$

Then

$$X = V : H.$$

We write elements of X as pairs

$$(u, a), \quad u \in F, \quad a \in F^\times,$$

with multiplication

$$(u, a)(v, b) = (u + av, ab).$$

Thus V is identified with $\{(u, 1) : u \in F\}$, and H is identified with

$$\{(0, a) : a \in F^\times\}.$$

We assume $q > 2$. The case $q = 2$ is exceptional, since then

$$X \cong \mathbb{Z}_2, \quad \text{Aut}(X) = 1.$$

First observe that V is the unique Sylow p -subgroup of X . Indeed,

$$|V| = q = p^f, \quad |H| = q - 1,$$

and $p \nmid (q - 1)$. Since $V \triangleleft X$, it is the unique Sylow p -subgroup. Hence V is characteristic in X .

We first compute the normalizer of $H = F^\times$ in

$$\text{GL}_{\mathbb{F}_p}(V).$$

Here H acts on $V = F^+$ by scalar multiplication:

$$a : v \mapsto av.$$

We claim that

$$N_{\text{GL}_{\mathbb{F}_p}(V)}(H) = H : \text{Gal}(F/\mathbb{F}_p).$$

It is clear that H normalizes itself. Also, every

$$\sigma \in \text{Gal}(F/\mathbb{F}_p)$$

normalizes H , because

$$\sigma(av) = \sigma(a)\sigma(v).$$

Thus

$$H : \text{Gal}(F/\mathbb{F}_p) \leq N_{\text{GL}_{\mathbb{F}_p}(V)}(H).$$

Conversely, let

$$A \in N_{\text{GL}_{\mathbb{F}_p}(V)}(H).$$

Let

$$c = A(1) \in F^\times.$$

Put

$$B = c^{-1}A,$$

where c^{-1} denotes scalar multiplication by c^{-1} . Then B still normalizes H , and

$$B(1) = 1.$$

Since B normalizes H , for each $a \in F^\times$, there exists $a^* \in F^\times$ such that

$$B(ax) = a^*B(x) \quad \text{for all } x \in F.$$

Putting $x = 1$, we obtain

$$B(a) = a^*.$$

Therefore

$$B(ax) = B(a)B(x) \quad \text{for all } a \in F^\times, x \in F.$$

This also holds for $a = 0$. Since B is $\mathbb{F}_p$ -linear, it preserves addition:

$$B(x + y) = B(x) + B(y).$$

Thus B is a field automorphism of F fixing $\mathbb{F}_p$. Hence

$$B \in \text{Gal}(F/\mathbb{F}_p).$$

So

$$A = cB \in H : \text{Gal}(F/\mathbb{F}_p).$$

This proves

$$N_{\text{GL}_{\mathbb{F}_p}(V)}(H) = H : \text{Gal}(F/\mathbb{F}_p).$$

Now let

$$\alpha \in \text{Aut}(X).$$

Since V is characteristic in X , we have

$$\alpha(V) = V.$$

The subgroup $\alpha(H)$ is a complement to V in X . We show that every complement to V is conjugate to H by an element of V .

Let K be a complement to V . Then the projection

$$X \rightarrow X/V \cong H$$

restricts to an isomorphism

$$K \cong H.$$

Hence each element of K has the form

$$(c(a), a), \quad a \in H,$$

for some map

$$c : H \rightarrow V.$$

Since K is a subgroup, we have

$$(c(a), a)(c(b), b) = (c(ab), ab),$$

and therefore

$$c(ab) = c(a) + ac(b).$$

Let r be a generator of $H = F^\times$. Since $q > 2$, we have $r \neq 1$. Define

$$t = (1 - r)^{-1}c(r) \in F.$$

Then

$$c(r) = t - rt.$$

Using

$$c(r^{n+1}) = c(r^n) + r^n c(r),$$

we get by induction

$$c(r^n) = t - r^n t \quad \text{for all } n.$$

Thus

$$K = (t, 1)H(t, 1)^{-1}.$$

Therefore every complement to V is conjugate to H by an element of V .

Applying this to $K = \alpha(H)$, there exists $t \in V$ such that after composing α with conjugation by $(t, 1)^{-1}$, we may assume

$$\alpha(H) = H.$$

So it remains to describe the automorphisms of X that stabilize both V and H .

Assume now that

$$\alpha(V) = V, \quad \alpha(H) = H.$$

Then

$$\alpha|_V = A \in \mathrm{GL}_{\mathbb{F}_p}(V),$$

and

$$\alpha|_H = \theta \in \mathrm{Aut}(H).$$

For $a \in H$ and $u \in V$, we have in X

$$(0, a)(u, 1)(0, a)^{-1} = (au, 1).$$

Applying α , we get

$$(0, \theta(a))(A(u), 1)(0, \theta(a))^{-1} = (A(au), 1).$$

Hence

$$A(au) = \theta(a)A(u).$$

Equivalently,

$$AHA^{-1} = H.$$

Thus

$$A \in N_{\mathrm{GL}_{\mathbb{F}_p}(V)}(H).$$

Conversely, if

$$A \in N_{\mathrm{GL}_{\mathbb{F}_p}(V)}(H),$$

then A induces an automorphism θ_A of H by

$$A \circ a \circ A^{-1} = \theta_A(a).$$

Define

$$\alpha_A : X \rightarrow X$$

by

$$\alpha_A(u, a) = (A(u), \theta_A(a)).$$

Then

$$\begin{aligned} \alpha_A((u, a)(v, b)) &= \alpha_A(u + av, ab) \\ &= (A(u + av), \theta_A(ab)) \\ &= (A(u) + A(av), \theta_A(a)\theta_A(b)) \\ &= (A(u) + \theta_A(a)A(v), \theta_A(a)\theta_A(b)) \\ &= (A(u), \theta_A(a))(A(v), \theta_A(b)) \\ &= \alpha_A(u, a)\alpha_A(v, b). \end{aligned}$$

So $\alpha_A \in \mathrm{Aut}(X)$.

Therefore the subgroup of $\mathrm{Aut}(X)$ stabilizing H is

$$N_{\mathrm{GL}_{\mathbb{F}_p}(V)}(H) = H : \mathrm{Gal}(F/\mathbb{F}_p).$$

The automorphisms induced by conjugation with elements of V form a subgroup iso-

morphic to V . Indeed, if $(t, 1) \in V$ centralizes H , then

$$t - at = 0 \quad \text{for all } a \in F^\times.$$

Since $q > 2$, we can choose $a \neq 1$, and hence $t = 0$. Thus this copy of V intersects the stabilizer of H trivially.

It follows that

$$\text{Aut}(X) \cong V : (H : \text{Gal}(F/\mathbb{F}_p)).$$

Since

$$X = V : H = F^+ : F^\times,$$

we obtain

$$\text{Aut}(X) \cong X : \text{Gal}(F/\mathbb{F}_p).$$

Finally, for a finite field

$$F = \mathbb{F}_{p^f},$$

the Galois group is cyclic:

$$\text{Gal}(F/\mathbb{F}_p) = \langle z \rangle \cong \mathbb{Z}_f,$$

where z is the Frobenius automorphism

$$z : x \mapsto x^p.$$

Also,

$$F^+ \cong \mathbb{Z}_p^f, \quad F^\times \cong \mathbb{Z}_{p^f-1}.$$

Hence

$$\text{Aut}(X) \cong (\mathbb{Z}_p^f : \mathbb{Z}_{p^f-1}) : \mathbb{Z}_f.$$

If

$$F^\times = \langle g \rangle,$$

then

$$|g| = p^f - 1, \quad |z| = f.$$

The Frobenius automorphism acts on $F^\times$ by

$$a \mapsto a^p.$$

Therefore

$$g^z = g^p.$$

Thus equivalently

$$\text{Aut}(X) \cong (\mathbb{Z}_p^f : \langle g \rangle) : \langle z \rangle,$$

where

$$|g| = p^f - 1, \quad |z| = f, \quad g^z = g^p.$$

□

Corollary 3.4.3. *For p odd,*

$$\text{Aut}(\mathbb{Z}_p : \mathbb{Z}_{p-1}) \cong \mathbb{Z}_p : \mathbb{Z}_{p-1}.$$

Proof. Take

$$F = \mathbb{F}_p.$$

Then

$$f = 1, \quad F^+ \cong \mathbb{Z}_p, \quad F^\times \cong \mathbb{Z}_{p-1}.$$

Therefore

$$X = F^+ : F^\times \cong \mathbb{Z}_p : \mathbb{Z}_{p-1}.$$

Since p is odd, we have $p > 2$, so the proposition applies.

Moreover,

$$\text{Gal}(\mathbb{F}_p/\mathbb{F}_p) = 1.$$

Hence

$$\text{Aut}(X) \cong X : \text{Gal}(\mathbb{F}_p/\mathbb{F}_p) \cong X.$$

Therefore

$$\text{Aut}(\mathbb{Z}_p : \mathbb{Z}_{p-1}) \cong \mathbb{Z}_p : \mathbb{Z}_{p-1}.$$

□

3.5 The Case $p = 3$ and $f = 2$

Example 3.5.1. *Assume*

$$p = 3, \quad f = 2.$$

Then

$$q = 9, \quad F = \mathbb{F}_9.$$

Let

$$X = \mathbb{F}_9^+ : \mathbb{F}_9^\times.$$

Since

$$\mathbb{F}_9^+ \cong \mathbb{Z}_3^2, \quad \mathbb{F}_9^\times \cong \mathbb{Z}_8,$$

we have

$$X \cong \mathbb{Z}_3^2 : \mathbb{Z}_8.$$

By the previous proposition,

$$\text{Aut}(X) \cong (\mathbb{Z}_3^2 : \mathbb{Z}_8) : \mathbb{Z}_2.$$

Equivalently,

$$\text{Aut}(X) \cong \mathbb{Z}_3^2 : (\mathbb{Z}_8 : \mathbb{Z}_2).$$

Let

$$\mathbb{Z}_8 = \langle g \rangle,$$

and let z be the Frobenius automorphism. Since $p = 3$, we have

$$g^z = g^3.$$

Now define

$$a = g^2, \quad b = gz.$$

Then

$$|a| = 4.$$

Also,

$$b^2 = (gz)(gz) = gzg z = gg^z z^2 = gg^3 = g^4.$$

Hence

$$|b| = 4, \quad b^2 = g^4.$$

But

$$a^2 = (g^2)^2 = g^4,$$

so

$$a^2 = b^2.$$

Moreover,

$$a^b = (g^2)^{gz} = (g^2)^z = g^6 = (g^2)^{-1} = a^{-1}.$$

Therefore

$$\langle a, b \rangle \cong Q_8,$$

where

$$Q_8 = \{1, -1, i, -i, j, -j, k, -k\}.$$

Thus

$$\text{Aut}(X) \cong (\mathbb{Z}_3^2 : \mathbb{Z}_8) : \mathbb{Z}_2 = \mathbb{Z}_3^2 : (\mathbb{Z}_8 : \mathbb{Z}_2) > \mathbb{Z}_3^2 : Q_8.$$

Exercise 3.5.2. Show that all elements of order 3 in

$$\mathbb{Z}_3^2 : Q_8$$

are conjugate.

Proof. Let

$$G = \mathbb{Z}_3^2 : Q_8, \quad V = \mathbb{Z}_3^2.$$

Then

$$V \triangleleft G, \quad G/V \cong Q_8.$$

First we show that every element of order 3 in G lies in V . Let $x \in G$ have order 3. Then the image $xV \in G/V \cong Q_8$ has order dividing 3. But

$$|Q_8| = 8,$$

so Q_8 has no nontrivial element of order 3. Hence

$$xV = V,$$

and therefore

$$x \in V.$$

Since $V \cong \mathbb{Z}_3^2$, every nonidentity element of V has order 3. Thus the elements of order 3 in G are exactly

$$V^\# = V \setminus \{0\}.$$

It remains to show that all elements of $V^\#$ are conjugate in G . Since $Q_8 \leq G$, it suffices to show that Q_8 acts transitively on $V^\#$.

Recall that in this construction

$$V = \mathbb{F}_9^+$$

and

$$Q_8 = \langle a, b \rangle \leq \mathbb{F}_9^\times : \text{Gal}(\mathbb{F}_9/\mathbb{F}_3),$$

where

$$a^2 = b^2$$

is the central element of order 2 in Q_8 . On $V = \mathbb{F}_9^+$, this central element acts as multiplication by -1 . Hence it has no nonzero fixed point, because if

$$-v = v,$$

then

$$2v = 0,$$

and over $\mathbb{F}_3$ this implies

$$v = 0.$$

Now let $1 \neq q \in Q_8$. If q has order 2, then q is the central element and we have just shown that it fixes no nonzero vector. If q has order 4, then

$$q^2$$

is the central element of order 2. If q fixed some $0 \neq v \in V$, then q^2 would also fix v , contradicting the previous paragraph. Therefore no nonidentity element of Q_8 fixes a nonzero element of V .

Thus the stabilizer in Q_8 of any $0 \neq v \in V$ is trivial. By the orbit-stabilizer theorem,

$$|v^{Q_8}| = |Q_8 : C_{Q_8}(v)| = |Q_8| = 8.$$

But

$$|V^\#| = |\mathbb{Z}_3^2| - 1 = 9 - 1 = 8.$$

Therefore

$$v^{Q_8} = V^\#.$$

So Q_8 is transitive on $V^\#$.

Consequently all nonidentity elements of V , equivalently all elements of order 3 in G , are conjugate in G . $\square$

3.6 Group Actions

Definition 3.6.1 (Group action). *Let G be a group and let Ω be a set. We say that G **acts** on Ω if the following conditions hold.*

(1) *Each element $g \in G$ gives a one-to-one map*

$$\Omega \longrightarrow \Omega.$$

The image of $\omega \in \Omega$ under g is denoted by

$$\omega^g.$$

(2) *The identity element acts trivially:*

$$\omega^1 = \omega \quad \text{for all } \omega \in \Omega.$$

(3) *The action is compatible with multiplication:*

$$(\omega^{g_1})^{g_2} = \omega^{g_1 g_2} \quad \text{for all } \omega \in \Omega, \ g_1, g_2 \in G.$$

Example 3.6.2 (Linear action). *The group $\text{GL}(n, F)$ acts on*

$$\Omega = F^n.$$

Viewing vectors as row vectors, the action is

$$v^A = vA \quad (v \in F^n, \ A \in \text{GL}(n, F)).$$

Example 3.6.3 (Conjugation action). A group G acts on itself by conjugation:

$$\Omega = G, \quad \omega^g = g^{-1}\omega g.$$

The conjugacy class of an element $x \in G$ is

$$x^G = \{x^g : g \in G\} = \{g^{-1}xg : g \in G\}.$$

Example 3.6.4 (Right multiplication action). A group G acts on itself by right multiplication:

$$\Omega = G, \quad \omega^g = \omega g.$$

3.7 The N/C Lemma

Lemma 3.7.1 (N/C -lemma). Let $H \leq G$. Then

$$C_G(H) \triangleleft N_G(H),$$

and

$$N_G(H)/C_G(H) \leq \text{Aut}(H).$$

In particular, if $H \triangleleft G$, then

$$G/C_G(H) \leq \text{Aut}(H).$$

Proof. The normalizer $N_G(H)$ acts on H by conjugation:

$$h^x = x^{-1}hx \quad (h \in H, x \in N_G(H)).$$

Since $x \in N_G(H)$, we have $x^{-1}Hx = H$, so this conjugation map is an automorphism of H .

Thus we get a homomorphism

$$\theta : N_G(H) \longrightarrow \text{Aut}(H), \quad x \longmapsto (h \mapsto h^x).$$

Its kernel is

$$\text{Ker } \theta = \{x \in N_G(H) : h^x = h \text{ for all } h \in H\} = C_G(H).$$

Therefore

$$C_G(H) \triangleleft N_G(H),$$

and by the First Isomorphism Theorem,

$$N_G(H)/C_G(H) \cong \text{Im } \theta \leq \text{Aut}(H).$$

If $H \triangleleft G$, then

$$N_G(H) = G,$$

so

$$G/C_G(H) \leq \text{Aut}(H).$$

□

Chapter 4

Sylow Theorems

4.1 Cauchy's Theorem and the Class Equation

Theorem 4.1.1 (Cauchy's theorem). *Let G be a finite group, and let p be a prime. If*

$$p \mid |G|,$$

then there exists an element $g \in G$ such that

$$|g| = p.$$

Proof. Let

$$\Omega = \{(x_1, x_2, \dots, x_p) \in G^p : x_1 x_2 \cdots x_p = 1\}.$$

Once $x_1, \dots, x_{p-1}$ are chosen, the last entry is uniquely determined by

$$x_p = (x_1 x_2 \cdots x_{p-1})^{-1}.$$

Hence

$$|\Omega| = |G|^{p-1}.$$

Since $p \mid |G|$, we have

$$p \mid |\Omega|.$$

Let the cyclic group $C_p = \langle \sigma \rangle$ act on Ω by cyclic permutation:

$$\sigma \cdot (x_1, x_2, \dots, x_p) = (x_2, x_3, \dots, x_p, x_1).$$

This is well-defined. Indeed, if $x_1 x_2 \cdots x_p = 1$, then

$$x_2 x_3 \cdots x_p x_1 = x_1^{-1} (x_1 x_2 \cdots x_p) x_1 = 1.$$

The orbits of this action have size either 1 or p , because C_p has prime order. Let N be the number of orbits of size 1. Since all remaining orbits have size p , and $p \mid |\Omega|$, it follows

that

$$p \mid N.$$

Now an element $(x_1, \dots, x_p) \in \Omega$ is fixed by σ if and only if

$$x_1 = x_2 = \dots = x_p.$$

Thus the fixed points are precisely the tuples

$$(x, x, \dots, x)$$

such that

$$x^p = 1.$$

Hence N is the number of elements $x \in G$ satisfying $x^p = 1$.

The identity element satisfies $1^p = 1$, so $N \geq 1$. Since $p \mid N$, we must have

$$N \geq p.$$

Therefore there exists some $g \neq 1$ such that

$$g^p = 1.$$

Thus $|g|$ divides p . Since p is prime and $g \neq 1$, we conclude that

$$|g| = p.$$

□

Proposition 4.1.2 (Orbit decomposition). *Let G act on a set Ω . Then G partitions Ω into orbits:*

$$\Omega = \Omega_1 \sqcup \Omega_2 \sqcup \dots \sqcup \Omega_m.$$

Proof. Define a relation $\sim$ on Ω by

$$\omega \sim \omega' \iff \omega' = \omega^g \text{ for some } g \in G.$$

We show that $\sim$ is an equivalence relation.

First, for every $\omega \in \Omega$, since $e \in G$ acts trivially, we have

$$\omega^e = \omega.$$

Hence $\omega \sim \omega$, so $\sim$ is reflexive.

Next, suppose $\omega \sim \omega'$. Then there exists $g \in G$ such that

$$\omega' = \omega^g.$$

Applying g^{-1} , we get

$$\omega = (\omega')^{g^{-1}}.$$

Thus $\omega' \sim \omega$, so $\sim$ is symmetric.

Finally, suppose $\omega \sim \omega'$ and $\omega' \sim \omega''$. Then there exist $g, h \in G$ such that

$$\omega' = \omega^g, \quad \omega'' = (\omega')^h.$$

Therefore

$$\omega'' = (\omega^g)^h = \omega^{gh}.$$

Since $gh \in G$, we have $\omega \sim \omega''$. Hence $\sim$ is transitive.

Thus $\sim$ is an equivalence relation on Ω . Its equivalence classes are exactly the orbits of the action, because the equivalence class of ω is

$$\{\omega' \in \Omega : \omega' \sim \omega\} = \{\omega^g : g \in G\}.$$

Equivalence classes of an equivalence relation form a partition of the underlying set. Therefore the orbits of G partition Ω . If the distinct orbits are denoted by

$$\Omega_1, \Omega_2, \dots, \Omega_m,$$

then

$$\Omega = \Omega_1 \sqcup \Omega_2 \sqcup \dots \sqcup \Omega_m.$$

□

Theorem 4.1.3 (Orbit-stabilizer theorem). *Let G be a finite group acting on a set Ω . If Ω_i is an orbit and $\omega_i \in \Omega_i$, then*

$$|\Omega_i| = |G : G_{\omega_i}|,$$

where

$$G_{\omega_i} = \{g \in G : \omega_i^g = \omega_i\}$$

is the stabilizer of ω_i . Equivalently,

$$|G| = |\Omega_i| |G_{\omega_i}|.$$

Proof. Let

$$H = G_{\omega_i} = \{g \in G : \omega_i^g = \omega_i\}.$$

First we note that $H \leq G$. Indeed, $e \in H$, and if $g, h \in H$, then

$$\omega_i^{gh^{-1}} = (\omega_i^g)^{h^{-1}} = \omega_i^{h^{-1}} = \omega_i,$$

so $gh^{-1} \in H$. Hence H is a subgroup of G .

Now consider the set of left cosets $H \backslash G$. Define a map

$$\varphi : H \backslash G \longrightarrow \Omega_i$$

by

$$\varphi(Hg) = \omega_i^g.$$

We first show that φ is well-defined. Suppose

$$Hg = Hh.$$

Then $h = ag$ for some $a \in H$. Hence

$$\omega_i^h = \omega_i^{ag} = (\omega_i^a)^g = \omega_i^g,$$

because $a \in H = G_{\omega_i}$. Thus $\varphi(Hg) = \varphi(Hh)$.

The map φ is surjective by the definition of the orbit:

$$\Omega_i = \{\omega_i^g : g \in G\}.$$

It remains to show that φ is injective. Suppose

$$\varphi(Hg) = \varphi(Hh).$$

Then

$$\omega_i^g = \omega_i^h.$$

Applying h^{-1} to both sides gives

$$\omega_i^{gh^{-1}} = \omega_i.$$

Thus

$$gh^{-1} \in H.$$

Hence

$$Hg = Hh.$$

Therefore φ is injective.

Thus φ is a bijection, so

$$|\Omega_i| = |H \backslash G| = |G : H| = |G : G_{\omega_i}|.$$

Since G is finite, this is equivalent to

$$|G| = |\Omega_i| |G_{\omega_i}|.$$

□

Example 4.1.4 (Conjugation action and class equation). *Let G act on itself by conjugation:*

$$x^g = g^{-1}xg.$$

For $x \in G$, write

$$C(x) = x^G = \{g^{-1}xg : g \in G\}$$

for the conjugacy class of x . Then

$$|C(x)| = |G : C_G(x)|,$$

where

$$C_G(x) = \{g \in G : gx = xg\}.$$

Thus

$$|G| = |C(x)| |C_G(x)|.$$

In particular,

$$|C(x)| \mid |G|.$$

If $x_1, \dots, x_t$ are representatives of the noncentral conjugacy classes of G , then

$$G = Z(G) \sqcup C(x_1) \sqcup \dots \sqcup C(x_t).$$

Hence the class equation is

$$|G| = |Z(G)| + |C(x_1)| + \dots + |C(x_t)|.$$

Theorem 4.1.5 (Center of a finite p -group). *Let G be a finite group of order*

$$|G| = p^e,$$

where p is prime. Then

$$Z(G) \neq 1.$$

Proof. Use the class equation:

$$G = Z(G) \sqcup C(x_1) \sqcup \dots \sqcup C(x_t),$$

where $x_1, \dots, x_t$ represent the noncentral conjugacy classes. Then

$$|G| = |Z(G)| + |C(x_1)| + \dots + |C(x_t)|.$$

For each noncentral element x_i , we have

$$|C(x_i)| = |G : C_G(x_i)|.$$

Since $x_i \notin Z(G)$, the class $C(x_i)$ is not a singleton. Therefore

$$|C(x_i)| > 1.$$

Also $|C(x_i)|$ divides $|G| = p^e$, so

$$|C(x_i)| = p^{e_i}$$

for some $e_i > 0$. Hence each $|C(x_i)|$ is divisible by p .

Thus

$$|G| = |Z(G)| + \sum_{i=1}^t |C(x_i)|$$

implies

$$0 \equiv |Z(G)| \pmod{p}.$$

So $p \mid |Z(G)|$. In particular, $|Z(G)| > 1$, and hence

$$Z(G) \neq 1.$$

□

Remark 4.1.6. *If x and y are conjugate in G , then*

$$|x| = |y|.$$

The converse is not true in general. For example, in the cyclic group

$$C_m = \langle g \rangle,$$

the elements g and g^{-1} have the same order, but they are not conjugate unless $g = g^{-1}$, because C_m is abelian.

Exercise 4.1.7. *Let G be a finite group with the property that any two elements of the same order are conjugate. Show that*

$$G \cong S_2 \quad \text{or} \quad G \cong S_3.$$

4.2 Sylow Theorems

Throughout this section, let G be a finite group and write

$$|G| = p^e m,$$

where p is prime and

$$\gcd(p, m) = 1.$$

Definition 4.2.1 (Sylow p -subgroup). A subgroup $P \leq G$ is called a **Sylow p -subgroup** of G if

$$|P| = p^e.$$

The set of all Sylow p -subgroups of G is denoted by

$$\text{Syl}_p(G).$$

Remark 4.2.2. If a Sylow p -subgroup exists, then Cauchy's theorem follows. Indeed, if

$$P \in \text{Syl}_p(G),$$

then $|P| = p^e$. Choose $1 \neq x \in P$. Since P is a p -group, $|x| = p^r$ for some $r \geq 1$. Then

$$x^{p^{r-1}}$$

has order p .

Theorem 4.2.3 (First Sylow theorem). Let G be a finite group. If

$$|G| = p^e m, \quad \gcd(p, m) = 1,$$

then G has a Sylow p -subgroup. That is,

$$\text{Syl}_p(G) \neq \emptyset.$$

Proof. We argue by induction on $|G|$.

Use the class equation:

$$|G| = |Z(G)| + |C(x_1)| + \cdots + |C(x_t)|,$$

where $x_1, \dots, x_t$ are representatives of the noncentral conjugacy classes.

Case 1: $p \nmid |Z(G)|$.

Since

$$p \mid |G|$$

but

$$p \nmid |Z(G)|,$$

not all of the noncentral conjugacy class sizes can be divisible by p . Hence there exists some x_i such that

$$p \nmid |C(x_i)|.$$

By the orbit-stabilizer theorem,

$$|G| = |C(x_i)| |C_G(x_i)|.$$

Since $p^e \mid |G|$ and $p \nmid |C(x_i)|$, we get

$$p^e \mid |C_G(x_i)|.$$

Also $x_i \notin Z(G)$, so

$$C_G(x_i) < G.$$

By induction, $C_G(x_i)$ has a subgroup P of order p^e . Then P is a Sylow p -subgroup of G .

Case 2: $p \mid |Z(G)|$.

First we prove Cauchy's theorem for finite abelian groups.

Let A be a finite abelian group and suppose $p \mid |A|$. We prove that A contains an element of order p by induction on $|A|$.

Choose $1 \neq x \in A$. If

$$p \mid |x|,$$

then

$$x^{|x|/p}$$

has order p .

Now suppose

$$p \nmid |x|.$$

Then p divides

$$|A/\langle x \rangle|.$$

By induction, $A/\langle x \rangle$ contains an element

$$a\langle x \rangle$$

of order p . The order of $a\langle x \rangle$ divides the order of a , so

$$p \mid |a|.$$

Hence

$$a^{|a|/p}$$

has order p .

Thus every finite abelian group whose order is divisible by p contains an element of order p .

Now return to G . Since $Z(G)$ is abelian and

$$p \mid |Z(G)|,$$

there exists

$$x \in Z(G)$$

such that

$$|x| = p.$$

Then

$$\langle x \rangle \triangleleft G$$

and

$$|G/\langle x \rangle| = p^{e-1}m.$$

If $e = 1$, then $\langle x \rangle$ itself is a Sylow p -subgroup of G .

If $e > 1$, then by induction $G/\langle x \rangle$ has a subgroup

$$H/\langle x \rangle$$

of order p^{e-1} . Therefore

$$|H| = p^e,$$

so H is a Sylow p -subgroup of G . □

Theorem 4.2.4 (Second Sylow theorem). *Sylow p -subgroups are conjugate. More generally, if $H \leq G$ is a p -subgroup and*

$$P \in \text{Syl}_p(G),$$

then there exists $x \in G$ such that

$$H \leq P^x.$$

In particular, if H is also a Sylow p -subgroup, then

$$H = P^x.$$

Proof. Let

$$\Delta = \{P^x : x \in G\}$$

be the conjugacy class of P under conjugation. Then G acts transitively on Δ by

$$(P^x)^g = P^{xg}.$$

By the orbit-stabilizer theorem,

$$|\Delta| = |G : N_G(P)|.$$

Since

$$P \leq N_G(P),$$

we have

$$|\Delta| = |G : N_G(P)| \mid |G : P| = m.$$

Hence

$$p \nmid |\Delta|.$$

Now let $H \leq G$ be a p -subgroup, say

$$|H| = p^d.$$

Restrict the action of G on Δ to an action of H . Then Δ is a disjoint union of H -orbits:

$$\Delta = \Delta_1 \sqcup \cdots \sqcup \Delta_r.$$

Each orbit size divides $|H|$, so each $|\Delta_i|$ is a power of p . Since

$$p \nmid |\Delta|,$$

at least one orbit has size 1.

Thus there exists $x \in G$ such that

$$(P^x)^h = P^x \quad \text{for every } h \in H.$$

Equivalently,

$$H \leq N_G(P^x).$$

But P^x is a normal Sylow p -subgroup of $N_G(P^x)$. Therefore every p -subgroup of $N_G(P^x)$ is contained in P^x . Hence

$$H \leq P^x.$$

If H is itself a Sylow p -subgroup, then

$$|H| = |P^x| = p^e,$$

so

$$H = P^x.$$

Therefore all Sylow p -subgroups are conjugate. □

Theorem 4.2.5 (Third Sylow theorem). *Let*

$$|G| = p^e m, \quad \gcd(p, m) = 1,$$

and let

$$n_p = |\text{Syl}_p(G)|$$

be the number of Sylow p -subgroups of G . Then

$$n_p \mid m$$

and

$$n_p \equiv 1 \pmod{p}.$$

Proof. Let

$$P \in \text{Syl}_p(G).$$

By the second Sylow theorem,

$$\text{Syl}_p(G) = \{P^x : x \in G\}.$$

Therefore

$$n_p = |G : N_G(P)|.$$

Since

$$P \leq N_G(P),$$

we get

$$n_p = |G : N_G(P)| \mid |G : P| = m.$$

Hence

$$n_p \mid m.$$

Now let P act on $\text{Syl}_p(G)$ by conjugation. The orbit sizes are powers of p . We determine the fixed points.

Suppose

$$Q \in \text{Syl}_p(G)$$

is fixed by P . Then

$$Q^h = Q \quad \text{for every } h \in P,$$

so

$$P \leq N_G(Q).$$

But Q is a normal Sylow p -subgroup of $N_G(Q)$. Hence every p -subgroup of $N_G(Q)$ is contained in Q . Thus

$$P \leq Q.$$

Since P and Q have the same order, we get

$$P = Q.$$

Therefore the only fixed point of this action is P itself.

All other P -orbits have sizes divisible by p . Hence

$$n_p = 1 + (\text{a sum of multiples of } p),$$

so

$$n_p \equiv 1 \pmod{p}.$$

□

Chapter 5

Minimal Normal Subgroups and Transfer

5.1 Minimal Normal Subgroups

Let G be a finite group and let

$$1 \neq N \triangleleft G.$$

Then

$$\overline{G} = G/N$$

is a group with

$$|\overline{G}| < |G|.$$

Thus we have a short exact sequence

$$1 \longrightarrow N \longrightarrow G \longrightarrow \overline{G} \longrightarrow 1.$$

In this sense, G is an extension of N by $\overline{G}$.

Lemma 5.1.1 (Structure of minimal normal subgroups). *Let G be a finite group and let $N \triangleleft_{\min} G$ be a minimal normal subgroup of G . Then*

$$N = S_1 \times S_2 \times \cdots \times S_r,$$

where

$$S_1 \cong S_2 \cong \cdots \cong S_r$$

are simple groups. Equivalently,

$$N \cong S^r$$

for some simple group S .

In particular, either

$$N \cong C_p^r$$

for some prime p , or

$$N \cong T^r$$

for some nonabelian simple group T .

Proof. We argue by induction on $|N|$.

If N is simple, then there is nothing to prove. So assume that N is not simple. Choose a minimal normal subgroup

$$M \triangleleft_{\min} N.$$

By induction applied to the group N and its minimal normal subgroup M , we may write

$$M = T_1 \times T_2 \times \cdots \times T_m,$$

where

$$T_1 \cong T_2 \cong \cdots \cong T_m$$

are simple groups.

For every $g \in G$, since $N \triangleleft G$, the conjugate

$$M^g = g^{-1} M g$$

is again a minimal normal subgroup of N . Hence, by minimality,

$$M^g = M \quad \text{or} \quad M^g \cap M = 1.$$

More generally, any two distinct conjugates of M under G have trivial intersection.

Let

$$M^G = \langle M^g : g \in G \rangle$$

be the normal closure of M in G . Choose distinct conjugates

$$M^{g_1}, M^{g_2}, \dots, M^{g_s}.$$

Since these subgroups are normal in N , if $i \neq j$, then

$$[M^{g_i}, M^{g_j}] \leq M^{g_i} \cap M^{g_j} = 1.$$

Therefore the distinct conjugates commute with one another, and so

$$M^G = M^{g_1} \times M^{g_2} \times \cdots \times M^{g_s}.$$

Also,

$$1 \neq M^G \triangleleft G \quad \text{and} \quad M^G \leq N.$$

Since $N \triangleleft_{\min} G$, we must have

$$M^G = N.$$

Thus N is a direct product of simple groups, all isomorphic to the simple factors of M .

Finally, if the common simple factor is abelian, then it is cyclic of prime order, so

$$N \cong C_p^r.$$

Otherwise the common simple factor is nonabelian simple. □

Corollary 5.1.2. *If G is solvable and $N \triangleleft_{\min} G$, then*

$$N \cong C_p^r$$

for some prime p and some $r \geq 1$.

Proof. Since G is solvable, every subgroup of G is solvable. Hence N is solvable.

By the lemma,

$$N \cong S^r$$

for some simple group S . Since S is a quotient of N , the group S is also solvable. A solvable simple group must be cyclic of prime order. Hence

$$S \cong C_p$$

for some prime p , and therefore

$$N \cong C_p^r.$$

□

5.2 Automorphisms of Nonabelian Simple Groups Products

Topic 5.2.1 (Question). *Let*

$$M = T_1 \times T_2 \times \cdots \times T_r,$$

where

$$T_1 \cong T_2 \cong \cdots \cong T_r \cong T$$

and T is nonabelian simple. We want to determine

$$\text{Aut}(M).$$

Lemma 5.2.2. *Let*

$$M = T_1 \times T_2 \times \cdots \times T_r,$$

where each T_i is nonabelian simple. Then the minimal normal subgroups of M are

precisely

$$T_1, T_2, \dots, T_r.$$

Proof. Each T_i is normal in M , and since T_i is simple, it is a minimal normal subgroup of M .

Conversely, let $L \triangleleft_{\min} M$. Since $L \neq 1$, there exists some i such that

$$[L, T_i] \neq 1.$$

Indeed, if $[L, T_i] = 1$ for every i , then L centralizes all of M , so

$$L \leq Z(M) = 1,$$

because each T_i is nonabelian simple and hence has trivial center. This is impossible.

Now

$$[L, T_i] \leq L \cap T_i.$$

Thus

$$L \cap T_i \neq 1.$$

But $L \cap T_i \triangleleft M$ and L is minimal normal in M , so

$$L \cap T_i = L.$$

Hence $L \leq T_i$. Since T_i is simple and $L \neq 1$, we get

$$L = T_i.$$

Therefore the minimal normal subgroups of M are exactly the direct factors T_i . □

Theorem 5.2.3 (Automorphisms of T^r). *Let T be a nonabelian simple group and let*

$$M = T_1 \times T_2 \times \cdots \times T_r, \quad T_i \cong T.$$

Then

$$\text{Aut}(M) \cong \text{Aut}(T) \wr S_r = \text{Aut}(T)^r \rtimes S_r,$$

where S_r acts on $\text{Aut}(T)^r$ by permuting the r coordinates.

More explicitly, $\text{Aut}(M)$ acts on the set

$$\{T_1, T_2, \dots, T_r\},$$

and the kernel of this action is

$$K = \text{Aut}(T_1) \times \text{Aut}(T_2) \times \cdots \times \text{Aut}(T_r) \cong \text{Aut}(T)^r.$$

Moreover,

$$\text{Aut}(M)/K \cong S_r.$$

Proof. Let $\alpha \in \text{Aut}(M)$. Since automorphisms preserve minimal normal subgroups, and since the minimal normal subgroups of M are precisely

$$T_1, T_2, \dots, T_r,$$

the automorphism α permutes the direct factors. Thus we get a homomorphism

$$\text{Aut}(M) \longrightarrow S_r.$$

The kernel consists of those automorphisms that stabilize every factor T_i . If α lies in the kernel, then

$$\alpha|_{T_i} \in \text{Aut}(T_i)$$

for each i , and for

$$m = t_1 t_2 \cdots t_r \quad (t_i \in T_i),$$

we have

$$\alpha(m) = \alpha(t_1)\alpha(t_2) \cdots \alpha(t_r).$$

Therefore

$$\text{Ker}(\text{Aut}(M) \rightarrow S_r) = \text{Aut}(T_1) \times \cdots \times \text{Aut}(T_r) = K.$$

Conversely, every permutation of the factors gives an automorphism of M , since all the factors are isomorphic. Hence the map

$$\text{Aut}(M) \longrightarrow S_r$$

is surjective, and so

$$\text{Aut}(M)/K \cong S_r.$$

Thus

$$\text{Aut}(M) \cong K \rtimes S_r \cong \text{Aut}(T)^r \rtimes S_r = \text{Aut}(T) \wr S_r.$$

□

Example 5.2.4 (Minimal normal subgroups in groups of order 72). *Let*

$$|G| = 72 = 2^3 \cdot 3^2.$$

Suppose that G is solvable, and let

$$N \triangleleft_{\min} G.$$

Then

$$N \cong C_p^r$$

for some prime p .

Since $|N| \mid |G| = 72$, the possibilities are

$$N \cong C_2, C_2^2, C_2^3, \quad \text{or} \quad N \cong C_3, C_3^2.$$

Equivalently, after viewing N as a vector space over $\mathbb{F}_p$, we have

$$N \leq C_2^3 \quad \text{or} \quad N \leq C_3^2.$$

5.3 Burnside's Theorem and Transfer

Theorem 5.3.1 (Burnside). *Let G be a finite group. If all Sylow subgroups of G are cyclic, then G is solvable.*

Remark 5.3.2. *A standard proof of this theorem uses the **transfer map**. We now define transfer and record its basic properties.*

Definition 5.3.3 (Transfer). *Let G be a finite group and let $H \leq G$. Put*

$$\Omega = H \backslash G = \{Hx : x \in G\}.$$

Choose a right transversal

$$\Omega = \{Hx_1, Hx_2, \dots, Hx_n\}, \quad n = [G : H].$$

For $g \in G$, right multiplication by g gives a permutation of Ω :

$$Hx_i \longmapsto Hx_i g.$$

Thus there exists a permutation $\tau_g \in S_n$ such that

$$Hx_i g = Hx_{i\tau_g}.$$

Hence

$$x_i g x_{i\tau_g}^{-1} \in H.$$

Define

$$h_i(g) = x_i g x_{i\tau_g}^{-1} \in H.$$

*The **transfer map** from G to H/H' is*

$$V_{G \rightarrow H} : G \longrightarrow H/H',$$

defined by

$$V_{G \rightarrow H}(g) = \prod_{i=1}^n h_i(g) H' = \prod_{i=1}^n (x_i g x_{i^{\tau_g}}^{-1}) H'.$$

Here $H' = [H, H]$ is the commutator subgroup of H .

Lemma 5.3.4 (Basic properties of transfer). *Let G be finite and let $H \leq G$.*

(1) *The map*

$$V_{G \rightarrow H} : G \longrightarrow H/H'$$

is a group homomorphism.

(2) *The map $V_{G \rightarrow H}$ does not depend on the choice of right coset representatives*

$$x_1, x_2, \dots, x_n.$$

(3) *Let $g \in G$. Suppose the action of $\langle g \rangle$ on*

$$\Omega = H \backslash G$$

has orbits

$$(Hx_1, Hx_1g, \dots, Hx_1g^{e_1-1}),$$

$$(Hx_2, Hx_2g, \dots, Hx_2g^{e_2-1}),$$

$$\dots,$$

$$(Hx_t, Hx_tg, \dots, Hx_tg^{e_t-1}).$$

Then

$$x_i g^{e_i} x_i^{-1} \in H$$

for each i , and

$$V_{G \rightarrow H}(g) = \prod_{i=1}^t (x_i g^{e_i} x_i^{-1}) H'.$$

Proof. For each $g \in G$, write

$$Hx_i g = Hx_{i^{\tau_g}},$$

and put

$$h_i(g) = x_i g x_{i^{\tau_g}}^{-1} \in H.$$

(1) Let $g_1, g_2 \in G$. Since right multiplication gives an action on Ω , we have

$$i^{\tau_{g_1 g_2}} = (i^{\tau_{g_1}})^{\tau_{g_2}}.$$

Hence

$$\begin{aligned}
h_i(g_1 g_2) &= x_i g_1 g_2 x_{i^{\tau_{g_1} g_2}}^{-1} \\
&= x_i g_1 x_{i^{\tau_{g_1}}}^{-1} \cdot x_{i^{\tau_{g_1}}} g_2 x_{(i^{\tau_{g_1}})^{\tau_{g_2}}}^{-1} \\
&= h_i(g_1) h_{i^{\tau_{g_1}}}(g_2).
\end{aligned}$$

Therefore, in the abelian group H/H' ,

$$\begin{aligned}
V_{G \rightarrow H}(g_1 g_2) &= \prod_{i=1}^n h_i(g_1 g_2) H' \\
&= \prod_{i=1}^n h_i(g_1) h_{i^{\tau_{g_1}}}(g_2) H' \\
&= \left(\prod_{i=1}^n h_i(g_1) H' \right) \left(\prod_{i=1}^n h_{i^{\tau_{g_1}}}(g_2) H' \right).
\end{aligned}$$

Since τ_{g_1} is a permutation of $\{1, \dots, n\}$,

$$\prod_{i=1}^n h_{i^{\tau_{g_1}}}(g_2) H' = \prod_{i=1}^n h_i(g_2) H'.$$

Thus

$$V_{G \rightarrow H}(g_1 g_2) = V_{G \rightarrow H}(g_1) V_{G \rightarrow H}(g_2).$$

So $V_{G \rightarrow H}$ is a homomorphism.

(2) Let

$$y_1, y_2, \dots, y_n$$

be another choice of right coset representatives, with

$$Hy_i = Hx_i.$$

Then there exist $t_i \in H$ such that

$$y_i = t_i x_i.$$

For the new representatives, the corresponding element of H is

$$\begin{aligned}
\tilde{h}_i(g) &= y_i g y_{i^{\tau_g}}^{-1} \\
&= t_i x_i g x_{i^{\tau_g}}^{-1} t_{i^{\tau_g}}^{-1} \\
&= t_i h_i(g) t_{i^{\tau_g}}^{-1}.
\end{aligned}$$

Hence in H/H' ,

$$\begin{aligned} \prod_{i=1}^n \tilde{h}_i(g)H' &= \prod_{i=1}^n t_i h_i(g) t_{i\tau_g}^{-1} H' \\ &= \left(\prod_{i=1}^n h_i(g) H' \right) \left(\prod_{i=1}^n t_i H' \right) \left(\prod_{i=1}^n t_{i\tau_g}^{-1} H' \right). \end{aligned}$$

Since τ_g is a permutation,

$$\prod_{i=1}^n t_i H' = \prod_{i=1}^n t_{i\tau_g} H'.$$

Therefore

$$\prod_{i=1}^n \tilde{h}_i(g)H' = \prod_{i=1}^n h_i(g)H',$$

so the transfer does not depend on the chosen transversal.

(3) By part (2), we may choose a convenient transversal adapted to the $\langle g \rangle$ -orbits. For an orbit

$$(Hx_i, Hx_i g, \dots, Hx_i g^{e_i-1}),$$

choose representatives

$$x_i, x_i g, \dots, x_i g^{e_i-1}.$$

For the first $e_i - 1$ representatives in this orbit, the corresponding transfer terms are

$$(x_i g^k)g(x_i g^{k+1})^{-1} = 1 \quad (0 \leq k \leq e_i - 2).$$

For the last representative, the term is

$$(x_i g^{e_i-1})g x_i^{-1} = x_i g^{e_i} x_i^{-1}.$$

Since

$$Hx_i g^{e_i} = Hx_i,$$

we indeed have

$$x_i g^{e_i} x_i^{-1} \in H.$$

Multiplying over all orbits gives

$$V_{G \rightarrow H}(g) = \prod_{i=1}^t (x_i g^{e_i} x_i^{-1}) H'.$$

□

5.4 p -Nilpotence and Burnside's Transfer Theorem

Definition 5.4.1 (p -nilpotent group). Let G be a finite group and let p be a prime divisor of $|G|$. We say that G is **p -nilpotent** if

$$G = NP,$$

where

$$N \triangleleft G, \quad P \in \text{Syl}_p(G), \quad N \cap P = 1.$$

Equivalently, G is p -nilpotent if and only if G has a normal subgroup N such that

$$|G/N| = |G|_p = |P|,$$

where P is a Sylow p -subgroup of G .

Theorem 5.4.2 (Burnside normal p -complement theorem). Let G be a finite group and let $P \in \text{Syl}_p(G)$. If

$$N_G(P) = C_G(P),$$

then G is p -nilpotent.

Proof. We use transfer. Since

$$N_G(P) = C_G(P),$$

and $P \leq N_G(P)$, we get

$$P \leq C_G(P).$$

Hence P is abelian, so

$$P' = 1.$$

Thus the transfer map gives a homomorphism

$$V_{G \rightarrow P} : G \longrightarrow P.$$

Let $g \in P \setminus \{1\}$. Consider the action of $\langle g \rangle$ on

$$P \backslash G.$$

Let the $\langle g \rangle$ -orbits be represented by

$$Px_1, \dots, Px_t,$$

with orbit lengths

$$\ell_1, \dots, \ell_t.$$

By the orbit formula for transfer,

$$V_{G \rightarrow P}(g) = \prod_{i=1}^t x_i g^{\ell_i} x_i^{-1}.$$

Here each factor lies in P :

$$x_i g^{\ell_i} x_i^{-1} \in P.$$

We claim that

$$x_i g^{\ell_i} x_i^{-1} = g^{\ell_i}$$

for every i .

Indeed, since $g^{\ell_i} \in P$ and P is abelian, we have

$$P \leq C_G(g^{\ell_i}).$$

Also, since

$$x_i g^{\ell_i} x_i^{-1} \in P$$

and P is abelian, we have

$$P \leq C_G(x_i g^{\ell_i} x_i^{-1}).$$

But

$$C_G(x_i g^{\ell_i} x_i^{-1}) = x_i C_G(g^{\ell_i}) x_i^{-1}.$$

Hence

$$x_i^{-1} P x_i \leq C_G(g^{\ell_i}).$$

Therefore both

$$P \quad \text{and} \quad x_i^{-1} P x_i$$

are Sylow p -subgroups of $C_G(g^{\ell_i})$. By Sylow's theorem, there exists

$$u_i \in C_G(g^{\ell_i})$$

such that

$$u_i^{-1} x_i^{-1} P x_i u_i = P.$$

Equivalently,

$$(x_i u_i)^{-1} P (x_i u_i) = P,$$

so

$$x_i u_i \in N_G(P) = C_G(P).$$

In particular, $x_i u_i$ centralizes g^{ℓ_i} . Since $u_i \in C_G(g^{\ell_i})$, it follows that x_i also centralizes g^{ℓ_i} .

Thus

$$x_i g^{\ell_i} x_i^{-1} = g^{\ell_i}.$$

Therefore

$$\begin{aligned} V_{G \rightarrow P}(g) &= \prod_{i=1}^t g^{\ell_i} \\ &= g^{\ell_1 + \dots + \ell_t}. \end{aligned}$$

Since the $\langle g \rangle$ -orbits partition $P \setminus G$,

$$\ell_1 + \dots + \ell_t = [G : P].$$

Hence

$$V_{G \rightarrow P}(g) = g^{[G:P]}.$$

Because g has p -power order and $p \nmid [G : P]$, we have

$$g^{[G:P]} \neq 1.$$

Thus

$$V_{G \rightarrow P}|_P : P \longrightarrow P$$

is injective. Since P is finite, it is also surjective. Hence

$$V_{G \rightarrow P}(P) = P.$$

Therefore

$$V_{G \rightarrow P}(G) = P.$$

Let

$$K = \text{Ker } V_{G \rightarrow P}.$$

Then

$$K \triangleleft G,$$

and by the first isomorphism theorem,

$$G/K \cong P.$$

Hence

$$|K| = \frac{|G|}{|P|}.$$

Moreover, since $V_{G \rightarrow P}|_P$ is injective,

$$K \cap P = 1.$$

Thus

$$|KP| = \frac{|K||P|}{|K \cap P|} = |G|,$$

so

$$G = KP.$$

Therefore G is p -nilpotent. □

Theorem 5.4.3 (Burnside). *Let G be a finite group whose Sylow subgroups are all cyclic. Let p be the smallest prime divisor of $|G|$, and let*

$$P \in \text{Syl}_p(G).$$

Then:

(1) G is p -nilpotent.

(2) G is solvable.

Proof. Since P is cyclic, write

$$|P| = p^e.$$

Then

$$|\text{Aut}(P)| = p^{e-1}(p-1).$$

Hence every prime divisor of $|\text{Aut}(P)|$ is at most p .

Conjugation induces an embedding

$$N_G(P)/C_G(P) \hookrightarrow \text{Aut}(P).$$

Thus every prime divisor of

$$|N_G(P)/C_G(P)|$$

is at most p .

On the other hand, since P is cyclic, it is abelian. Hence

$$P \leq C_G(P).$$

Therefore

$$|N_G(P)/C_G(P)| \mid |N_G(P)/P|.$$

Also

$$|N_G(P)/P| \mid [G : P].$$

Since p is the smallest prime divisor of $|G|$ and $p \nmid [G : P]$, every prime divisor of $[G : P]$ is strictly larger than p .

Thus every prime divisor of $|N_G(P)/C_G(P)|$ is both at most p and strictly larger than p , which is impossible unless

$$|N_G(P)/C_G(P)| = 1.$$

Hence

$$N_G(P) = C_G(P).$$

By Burnside's normal p -complement theorem, G is p -nilpotent.

So there exists a normal subgroup

$$K \triangleleft G$$

such that

$$|K| = \frac{|G|}{|P|}.$$

Every Sylow subgroup of K is also a Sylow subgroup of G for the corresponding prime, and hence is cyclic.

Now let p_1 be the smallest prime divisor of $|K|$. By the same argument, K is p_1 -nilpotent. Hence K has a normal subgroup

$$K_1 \triangleleft K$$

such that

$$|K_1| = \frac{|K|}{|P_1|},$$

where

$$P_1 \in \text{Syl}_{p_1}(K).$$

Repeating this process, we obtain a normal series

$$G = K_0 \triangleright K_1 \triangleright K_2 \triangleright \cdots \triangleright K_m = Q \triangleright 1,$$

where Q is a Sylow q -subgroup of G and q is the largest prime divisor of $|G|$.

Each factor

$$K_i/K_{i+1}$$

is cyclic, and Q is cyclic. Hence this is a normal series with abelian factors. Therefore G is solvable. $\square$

Remark 5.4.4. *In the above process, the final Sylow subgroup*

$$Q \in \text{Syl}_q(G)$$

is actually normal in G .

Proof. Suppose that

$$Q \triangleleft K_n \triangleleft K_{n-1}.$$

For every $g \in K_{n-1}$, since $K_n \triangleleft K_{n-1}$, we have

$$g^{-1}Qg \leq g^{-1}K_n g = K_n.$$

Thus $g^{-1}Qg$ is a Sylow q -subgroup of K_n . Since

$$Q \triangleleft K_n,$$

the group K_n has a unique Sylow q -subgroup. Hence

$$g^{-1}Qg = Q.$$

Therefore

$$Q \triangleleft K_{n-1}.$$

Repeating this argument upward along the series gives

$$Q \triangleleft G.$$

□

Corollary 5.4.5. *If a Sylow 2-subgroup of G is cyclic, then G is 2-nilpotent. In particular, G is solvable.*

Proof. Let

$$P \in \text{Syl}_2(G)$$

be cyclic.

If $|G|$ is odd, then G is solvable by the Feit–Thompson theorem. So assume that $2 \mid |G|$.

Since P is cyclic, it is abelian, so

$$P \leq C_G(P).$$

Again conjugation induces

$$N_G(P)/C_G(P) \hookrightarrow \text{Aut}(P).$$

But $\text{Aut}(P)$ is a 2-group. Hence

$$N_G(P)/C_G(P)$$

is a 2-group.

On the other hand,

$$|N_G(P)/C_G(P)| \mid |N_G(P)/P| \mid [G : P],$$

and $[G : P]$ is odd. Thus

$$N_G(P)/C_G(P) = 1.$$

Therefore

$$N_G(P) = C_G(P).$$

By Burnside’s normal p -complement theorem, G is 2-nilpotent. Hence there exists

$$N \triangleleft G$$

such that

$$G = NP, \quad N \cap P = 1.$$

Then $|N|$ is odd, so N is solvable by the Feit–Thompson theorem. Also

$$G/N \cong P$$

is cyclic, hence solvable. Since both N and G/N are solvable, G is solvable. $\square$

5.5 Consequences of Burnside's Normal Complement Theorem

Proposition 5.5.1. *Let G be a finite nonabelian simple group, and let p be a prime divisor of $|G|$.*

(1) *If p is the smallest prime divisor of $|G|$, then*

$$p^3 \mid |G| \quad \text{or} \quad 12 \mid |G|.$$

(2) *If*

$$|G| = pm, \quad (p, m) = 1,$$

and $P \in \text{Syl}_p(G)$, then

$$C_G(P) \neq N_G(P) \neq G,$$

and

$$|N_G(P)/C_G(P)| \mid (p-1).$$

Proof. We first prove (2). Since $|G| = pm$ with $(p, m) = 1$, a Sylow p -subgroup has order p . Hence

$$P \cong C_p.$$

Conjugation gives a homomorphism

$$N_G(P) \longrightarrow \text{Aut}(P),$$

whose kernel is $C_G(P)$. Therefore

$$N_G(P)/C_G(P) \hookrightarrow \text{Aut}(P).$$

Since

$$\text{Aut}(C_p) \cong C_{p-1},$$

we get

$$|N_G(P)/C_G(P)| \mid (p-1).$$

Also $N_G(P) \neq G$, because otherwise $P \triangleleft G$, contradicting the simplicity of G .

If $N_G(P) = C_G(P)$, then by Burnside's normal p -complement theorem, G is p -nilpotent.

Thus G has a normal p -complement, say $K \triangleleft G$, with

$$|K| = \frac{|G|}{|P|} = m.$$

Since G is nonabelian simple, this is impossible. Hence

$$C_G(P) \neq N_G(P).$$

This proves (2).

Now prove (1). Assume that p is the smallest prime divisor of $|G|$, and suppose that

$$p^3 \nmid |G|.$$

We shall prove that $12 \mid |G|$.

First, p^2 must divide $|G|$. Indeed, if $p^2 \nmid |G|$, then $|G| = pm$ with $(p, m) = 1$. By part (2),

$$C_G(P) \neq N_G(P),$$

and

$$|N_G(P)/C_G(P)| \mid (p-1).$$

But $P \leq C_G(P)$, so

$$|N_G(P)/C_G(P)| \mid |N_G(P)/P| \mid \frac{|G|}{p}.$$

Since p is the smallest prime divisor of $|G|$ and $p \nmid |G|/p$, every prime divisor of $|G|/p$ is strictly larger than p . This contradicts the fact that

$$|N_G(P)/C_G(P)|$$

is a nontrivial divisor of $p-1$. Therefore

$$p^2 \mid |G|.$$

Since we assumed $p^3 \nmid |G|$, we have

$$p^2 \parallel |G|.$$

Let

$$P \in \text{Syl}_p(G).$$

Then

$$|P| = p^2.$$

Hence either

$$P \cong C_{p^2} \quad \text{or} \quad P \cong C_p \times C_p.$$

Suppose first that

$$P \cong C_{p^2}.$$

Then

$$|\operatorname{Aut}(P)| = \varphi(p^2) = p(p-1).$$

Again,

$$N_G(P)/C_G(P) \hookrightarrow \operatorname{Aut}(P).$$

Since P is cyclic, it is abelian, so

$$P \leq C_G(P).$$

Thus

$$|N_G(P)/C_G(P)| \mid \frac{|G|}{p^2}.$$

Every prime divisor of $|G|/p^2$ is strictly larger than p , whereas every prime divisor of $p(p-1)$ is at most p . Hence

$$N_G(P) = C_G(P).$$

By Burnside's normal p -complement theorem, G is p -nilpotent, contradicting the simplicity of G . Therefore

$$P \not\cong C_{p^2}.$$

So

$$P \cong C_p \times C_p.$$

Now P is abelian, hence

$$P \leq C_G(P).$$

If

$$N_G(P) = C_G(P),$$

then again Burnside's normal p -complement theorem implies that G is p -nilpotent, impossible. Thus

$$N_G(P) \neq C_G(P).$$

Hence the quotient

$$N_G(P)/C_G(P)$$

is nontrivial.

Since

$$P \cong C_p \times C_p,$$

we have

$$\operatorname{Aut}(P) \cong \operatorname{GL}(2, p).$$

Thus

$$N_G(P)/C_G(P) \hookrightarrow \operatorname{GL}(2, p).$$

Moreover,

$$|\operatorname{GL}(2, p)| = (p^2 - 1)(p^2 - p) = p(p-1)^2(p+1).$$

Let r be a prime divisor of

$$|N_G(P)/C_G(P)|.$$

Since $P \leq C_G(P)$, we have

$$|N_G(P)/C_G(P)| \mid \frac{|G|}{p^2}.$$

Thus r is a prime divisor of $|G|/p^2$. Since p is the smallest prime divisor of $|G|$ and $p^3 \nmid |G|$, we get

$$r > p.$$

On the other hand, r divides

$$|\mathrm{GL}(2, p)| = p(p-1)^2(p+1).$$

Since $r > p$, it cannot divide p or $p-1$. Therefore

$$r \mid p+1.$$

Thus

$$p < r \leq p+1.$$

Hence

$$r = p+1.$$

Since r is prime, this forces

$$p = 2, \quad r = 3.$$

Therefore

$$|P| = p^2 = 4$$

and $3 \mid |G|/4$. Hence

$$12 \mid |G|.$$

This proves (1). □

Example 5.5.2. *If*

$$|G| = 5 \cdot 7 \cdot 13,$$

then G is cyclic.

Let G_p denote a Sylow p -subgroup of G .

First consider $p = 13$. Let

$$G_{13} \in \mathrm{Syl}_{13}(G).$$

The number n_{13} of Sylow 13-subgroups satisfies

$$n_{13} \equiv 1 \pmod{13}, \quad n_{13} \mid 35.$$

The divisors of 35 are 1, 5, 7, 35, and only 1 is congruent to 1 modulo 13. Hence

$$n_{13} = 1.$$

Thus

$$G_{13} \triangleleft G.$$

Now conjugation gives

$$G/C_G(G_{13}) = N_G(G_{13})/C_G(G_{13}) \hookrightarrow \text{Aut}(G_{13}).$$

Since

$$G_{13} \cong C_{13},$$

we have

$$\text{Aut}(G_{13}) \cong C_{12}.$$

Also $G_{13} \leq C_G(G_{13})$, so

$$|G/C_G(G_{13})| \mid \frac{|G|}{13} = 35.$$

Therefore

$$|G/C_G(G_{13})| \mid \gcd(35, 12) = 1.$$

Hence

$$C_G(G_{13}) = G.$$

So

$$G_{13} \leq Z(G).$$

By Burnside's normal p -complement theorem applied to $p = 13$, the group G is 13-nilpotent. Hence there exists a normal subgroup

$$K \triangleleft G$$

such that

$$|K| = 5 \cdot 7 = 35$$

and

$$G = KG_{13}, \quad K \cap G_{13} = 1.$$

Since $G_{13} \leq Z(G)$, this product is direct:

$$G = K \times G_{13}.$$

It remains to show that K is cyclic. Since

$$|K| = 35 = 5 \cdot 7,$$

let

$$Q \in \text{Syl}_7(K).$$

The number $n_7(K)$ of Sylow 7-subgroups of K satisfies

$$n_7(K) \equiv 1 \pmod{7}, \quad n_7(K) \mid 5.$$

Hence

$$n_7(K) = 1.$$

Thus

$$Q \triangleleft K.$$

Conjugation gives

$$K/C_K(Q) = N_K(Q)/C_K(Q) \hookrightarrow \text{Aut}(Q) \cong C_6.$$

Since $Q \leq C_K(Q)$, the order of $K/C_K(Q)$ divides

$$|K/Q| = 5.$$

Thus

$$|K/C_K(Q)| \mid \gcd(5, 6) = 1,$$

so

$$C_K(Q) = K.$$

Therefore

$$Q \leq Z(K).$$

Let

$$R \in \text{Syl}_5(K).$$

Then R centralizes Q , and

$$K = QR, \quad Q \cap R = 1.$$

Thus

$$K \cong Q \times R \cong C_7 \times C_5 \cong C_{35}.$$

Consequently

$$G \cong K \times G_{13} \cong C_{35} \times C_{13} \cong C_{455}.$$

Hence G is cyclic.

Chapter 6

Solvable Groups and p -Groups

6.1 Derived Series and Solvable Groups

Definition 6.1.1 (Commutator subgroup). *Let G be a group. The **commutator subgroup** of G is*

$$G' = [G, G] = \langle [x, y] : x, y \in G \rangle,$$

where

$$[x, y] = x^{-1}y^{-1}xy.$$

Equivalently, G' is the smallest normal subgroup $N \triangleleft G$ such that

$$G/N$$

is abelian.

Definition 6.1.2 (Derived series). *Define the derived series of G by*

$$G^{(0)} = G, \quad G^{(1)} = G',$$

and inductively

$$G^{(n)} = (G^{(n-1)})' \quad (n \geq 1).$$

Thus we have a descending normal series

$$G \supseteq G' \supseteq G'' \supseteq \cdots \supseteq G^{(n)} \supseteq \cdots.$$

Example 6.1.3 (Some commutator subgroups). (1) *If G is simple and nonabelian, then*

$$G' = G.$$

Indeed, $G' \triangleleft G$. Since G is nonabelian, $G' \neq 1$, and hence by simplicity $G' = G$.

(2) For the symmetric group S_n , one has

$$S'_n = A_n \quad (n \geq 3).$$

In particular,

$$S'_4 = A_4, \quad S'_n = A_n \quad (n \geq 5).$$

This is because A_n is generated by 3-cycles, and 3-cycles occur as commutators.

Remark 6.1.4. If G is finite, then the descending chain

$$G \supseteq G' \supseteq G^{(2)} \supseteq \dots$$

must eventually stabilize. Hence either

$$G^{(n)} = 1$$

for some n , or else

$$G^{(n)} = G^{(n+1)} = G^{(n+2)} = \dots$$

for all sufficiently large n . The stable term is sometimes denoted by

$$G^{(\infty)}.$$

Definition 6.1.5 (Solvable and perfect groups). Let G be a group.

(1) We say that G is **solvable** if

$$G^{(n)} = 1$$

for some $n \geq 0$.

(2) We say that G is **perfect** if

$$G = G'.$$

Example 6.1.6 (Perfect groups). The following are important examples of perfect groups.

(1) Every nonabelian simple group is perfect.

(2) The alternating group A_n is perfect for $n \geq 5$. For example,

$$(123)(345)(123)^{-1}(345)^{-1} = (143).$$

Since A_n is generated by 3-cycles, this implies

$$A'_n = A_n.$$

(3) The group $\mathrm{SL}_d(q)$ is perfect whenever it is nonsolvable; more precisely,

$$\mathrm{SL}_d(q)' = \mathrm{SL}_d(q)$$

for $d \geq 2$, except in the small cases

$$(d, q) = (2, 2), (2, 3).$$

Its center is

$$Z(\mathrm{SL}_d(q)) = \{\lambda I_d : \lambda^d = 1, \lambda \in \mathbb{F}_q^\times\},$$

and hence

$$|Z(\mathrm{SL}_d(q))| = \gcd(d, q - 1).$$

6.2 Basic Properties of Solvable Groups

Proposition 6.2.1 (Properties of solvable groups). *Let G be a group.*

- (1) *If G is solvable, then every subgroup of G and every quotient group of G is solvable.*
- (2) *If $N \triangleleft G$, and both N and G/N are solvable, then G is solvable.*
- (3) *Every finite p -group is solvable.*

Proof. (1) Let $H \leq G$. Since

$$H' \leq G',$$

induction gives

$$H^{(n)} \leq G^{(n)}$$

for all n . Thus if $G^{(n)} = 1$, then $H^{(n)} = 1$, so H is solvable.

Now let $N \triangleleft G$. The natural map $G \rightarrow G/N$ sends $G^{(i)}$ onto a subgroup of $(G/N)^{(i)}$, and in fact

$$(G/N)^{(i)} = G^{(i)}N/N.$$

Hence if $G^{(n)} = 1$, then

$$(G/N)^{(n)} = 1,$$

so G/N is solvable.

(2) Since G/N is solvable, there exists r such that

$$(G/N)^{(r)} = 1.$$

Hence

$$G^{(r)} \leq N.$$

Since N is solvable, there exists s such that

$$N^{(s)} = 1.$$

Therefore

$$G^{(r+s)} \leq N^{(s)} = 1.$$

Thus G is solvable.

(3) We argue by induction on $|G|$. If $G = 1$, there is nothing to prove.

If $G \neq 1$ is a finite p -group, then

$$Z(G) \neq 1.$$

Since $Z(G) \triangleleft G$ is abelian, it is solvable. Also $G/Z(G)$ is a smaller p -group, hence solvable by induction. By part (2), G is solvable. □

6.3 Double Cosets and a Normalizer Lemma

Definition 6.3.1 (Double coset). *Let $P \leq G$. For $x \in G$, the subset*

$$PxP = \{h_1 x h_2 : h_1, h_2 \in P\}$$

*is called a **double coset** of P in G .*

Lemma 6.3.2 (Right cosets inside a double coset). *Let $P \leq G$, and write*

$$P^x = x^{-1}Px.$$

Then the number of right cosets of P contained in the double coset PxP is

$$\frac{|P|}{|P^x \cap P|} = [P : P^x \cap P].$$

Proof. We may write

$$PxP = \bigcup_{h \in P} P x h.$$

Two right cosets $P x h_1$ and $P x h_2$ are equal if and only if

$$P x h_1 = P x h_2.$$

This is equivalent to

$$x h_1 h_2^{-1} x^{-1} \in P,$$

or equivalently

$$h_1 h_2^{-1} \in x^{-1} P x = P^x.$$

Since $h_1 h_2^{-1} \in P$, this condition becomes

$$h_1 h_2^{-1} \in P^x \cap P.$$

Therefore the distinct right cosets Pxh are parametrized by the right cosets of $P^x \cap P$ in P . Hence the number is

$$[P : P^x \cap P] = \frac{|P|}{|P^x \cap P|}.$$

□

Lemma 6.3.3 (Normalizer lemma). *Let G be finite, and let P be a p -subgroup of G . If P is not a Sylow p -subgroup of G , then*

$$P < N_G(P).$$

Proof. Decompose G into double cosets of P :

$$G = \bigsqcup_{i=0}^t P x_i P,$$

where we take

$$x_0 = 1, \quad P x_0 P = P.$$

Thus

$$P$$

is the unique right coset of P inside P .

Dividing the double coset decomposition by right cosets of P , we obtain

$$[G : P] = 1 + \sum_{i=1}^t [P : P^{x_i} \cap P].$$

Since P is not Sylow, the index $[G : P]$ is divisible by p .

Each summand

$$[P : P^{x_i} \cap P]$$

is a power of p . Hence it is either 1, or divisible by p . If all of the summands with $i \geq 1$ were divisible by p , then the right-hand side would be congruent to 1 (mod p), contradicting $p \mid [G : P]$. Therefore for some $i \geq 1$,

$$[P : P^{x_i} \cap P] = 1.$$

Thus

$$P^{x_i} \cap P = P,$$

so

$$P^{x_i} = P.$$

Therefore

$$x_i \in N_G(P).$$

Since $i \geq 1$, we may choose $x_i \notin P$. Hence

$$P < N_G(P).$$

□

6.4 Subgroups of p -Groups

Theorem 6.4.1 (Subgroups and normal subgroups of p -groups). *Let G be a finite p -group.*

(1) *If $H < G$, then*

$$H < N_G(H).$$

(2) *If H is a maximal subgroup of G , then*

$$H \triangleleft G \quad \text{and} \quad |G : H| = p.$$

(3) *The group G is solvable.*

(4) *The center of G is nontrivial:*

$$Z(G) > 1.$$

(5) *If $N \triangleleft G$ and $|N| = p$, then*

$$N \leq Z(G).$$

Proof. (1) Since G itself is a p -group, every proper subgroup $H < G$ is not a Sylow p -subgroup of G . By the normalizer lemma,

$$H < N_G(H).$$

(2) If H is maximal, then by part (1),

$$H < N_G(H) \leq G.$$

Hence maximality forces

$$N_G(H) = G.$$

Thus $H \triangleleft G$. The quotient G/H has no nontrivial proper subgroups, because H is maximal. Since G/H is a nontrivial p -group, it must have order p . Hence

$$|G : H| = p.$$

(3) This follows by induction on $|G|$. Choose a maximal subgroup $H < G$. By part (2),

$$H \triangleleft G \quad \text{and} \quad G/H \cong C_p.$$

By induction, H is solvable, and G/H is abelian, hence solvable. Therefore G is solvable.

(4) Let G act on itself by conjugation. The class equation is

$$|G| = |Z(G)| + \sum_i [G : C_G(x_i)],$$

where the x_i 's represent the noncentral conjugacy classes.

If $x_i \notin Z(G)$, then

$$C_G(x_i) < G,$$

so

$$[G : C_G(x_i)]$$

is divisible by p . Since $|G|$ is also divisible by p , it follows that

$$p \mid |Z(G)|.$$

Thus $Z(G) \neq 1$.

(5) Since $N \triangleleft G$, the normalizer-centralizer lemma gives

$$G/C_G(N) \hookrightarrow \text{Aut}(N).$$

But $|N| = p$, so

$$N \cong C_p \quad \text{and} \quad \text{Aut}(N) \cong C_{p-1}.$$

Hence

$$|G : C_G(N)| \mid p - 1.$$

On the other hand, $G/C_G(N)$ is a quotient of the p -group G , so its order is a power of p . Therefore

$$|G : C_G(N)| = 1.$$

Thus

$$G = C_G(N),$$

which means every element of G centralizes N . Hence

$$N \leq Z(G).$$

□

Lemma 6.4.2 (Cyclic quotient by the center). *Let G be a group. If*

$$G/Z(G)$$

is cyclic, then G is abelian.

Proof. Assume

$$G/Z(G) = \langle gZ(G) \rangle$$

for some $g \in G$. Then every element of G can be written in the form

$$g^m z$$

for some $m \in \mathbb{Z}$ and some $z \in Z(G)$.

Let $x, y \in G$. Then there exist $m, n \in \mathbb{Z}$ and $z_1, z_2 \in Z(G)$ such that

$$x = g^m z_1, \quad y = g^n z_2.$$

Since $z_1, z_2 \in Z(G)$, they commute with all elements of G . Hence

$$xy = (g^m z_1)(g^n z_2) = g^{m+n} z_1 z_2 = g^{n+m} z_2 z_1 = (g^n z_2)(g^m z_1) = yx.$$

Thus any two elements of G commute, so G is abelian. □

Corollary 6.4.3. *Every group of order p^2 , where p is prime, is abelian.*

Proof. Let $|G| = p^2$. Since G is a p -group,

$$Z(G) \neq 1.$$

Thus

$$|Z(G)| = p \quad \text{or} \quad |Z(G)| = p^2.$$

If $|Z(G)| = p^2$, then $Z(G) = G$, so G is abelian.

If $|Z(G)| = p$, then

$$|G/Z(G)| = p,$$

so $G/Z(G)$ is cyclic. But if $G/Z(G)$ is cyclic, then G is abelian. Hence $Z(G) = G$, contradiction. Therefore G must be abelian. □

6.5 Groups of Order p^3

Theorem 6.5.1 (Structure of groups of order p^3). *Let G be a group of order p^3 , where p is prime. Then either G is abelian, in which case*

$$G \cong C_{p^3}, \quad G \cong C_{p^2} \times C_p, \quad \text{or} \quad G \cong C_p \times C_p \times C_p,$$

or G is nonabelian. In the nonabelian case, the following possibilities occur.

(1) *If $p = 2$, then*

$$G \cong D_8 \quad \text{or} \quad G \cong Q_8.$$

(2) *If p is odd, then either*

$$G \cong \langle a, b \mid a^{p^2} = b^p = 1, a^b = a^{1+p} \rangle,$$

or

$$G \cong \langle a, b, c \mid a^p = b^p = c^p = 1, c = [a, b], [a, c] = [b, c] = 1 \rangle.$$

The first group has exponent p^2 , and the second group has exponent p .

Proof. Assume G is nonabelian.

Since G is a p -group, $Z(G) \neq 1$. If $|Z(G)| = p^2$, then

$$|G/Z(G)| = p,$$

so $G/Z(G)$ is cyclic, forcing G to be abelian. This is impossible. Hence

$$|Z(G)| = p.$$

Also, $G/Z(G)$ has order p^2 , so it is abelian. Therefore

$$G' \leq Z(G).$$

Since G is nonabelian, $G' \neq 1$, and hence

$$G' = Z(G) \cong C_p.$$

We split into two cases.

Case 1: G contains an element of order p^2 .

Let $a \in G$ with

$$|a| = p^2.$$

Then

$$\langle a \rangle$$

has order p^2 , so it is a maximal subgroup of G . Hence

$$\langle a \rangle \triangleleft G.$$

Choose

$$b \in G \setminus \langle a \rangle.$$

Then $G/\langle a \rangle$ has order p , so

$$b^p \in \langle a \rangle.$$

Conjugation by b gives an automorphism of $\langle a \rangle$. Thus

$$a^b = a^\lambda$$

for some integer λ . Since $b^p \in \langle a \rangle$, the induced automorphism has order dividing p , so

$$\lambda^p \equiv 1 \pmod{p^2}.$$

If $p = 2$, then the only nontrivial automorphism of C_4 is inversion. Hence

$$a^b = a^{-1}.$$

If there exists $b \notin \langle a \rangle$ with $b^2 = 1$, then

$$G \cong D_8.$$

Otherwise every element outside $\langle a \rangle$ has order 4, and one obtains

$$G \cong Q_8.$$

Now assume p is odd. Since G is nonabelian, the action is nontrivial, so

$$\lambda \not\equiv 1 \pmod{p^2}.$$

From

$$\lambda^p \equiv 1 \pmod{p^2},$$

we get

$$\lambda \equiv 1 \pmod{p}.$$

Thus

$$\lambda = 1 + kp$$

for some k with $1 \leq k \leq p-1$.

Write

$$b^p = a^{\ell p}$$

for some integer ℓ . A direct computation gives

$$(ba^{-\ell})^p = 1.$$

Replacing b by $ba^{-\ell}$, we may assume

$$b^p = 1.$$

Finally, replacing b by a suitable power of b , we may arrange that

$$a^b = a^{1+p}.$$

Therefore

$$G \cong \langle a, b \mid a^{p^2} = b^p = 1, a^b = a^{1+p} \rangle.$$

Case 2: G has no element of order p^2 .

Then every nonidentity element of G has order p . If $p = 2$, then every element has order 2, and hence G is abelian, a contradiction. Therefore p is odd.

Choose $z \in Z(G)$ with $z \neq 1$. Then

$$|\langle z \rangle| = p,$$

and

$$|G/\langle z \rangle| = p^2.$$

Hence $G/\langle z \rangle$ is abelian, so

$$G' \leq \langle z \rangle.$$

Since G is nonabelian, $G' \neq 1$. Thus

$$G' = \langle z \rangle = Z(G).$$

Choose $a, b \in G$ such that

$$[a, b] \neq 1.$$

Let

$$c = [a, b].$$

Then

$$c \in G' = Z(G),$$

so c is central. Since every nonidentity element has order p ,

$$a^p = b^p = c^p = 1.$$

Moreover, $\langle a, b \rangle$ is not abelian, so it cannot have order p or p^2 . Hence

$$\langle a, b \rangle = G.$$

Thus

$$G \cong \langle a, b, c \mid a^p = b^p = c^p = 1, c = [a, b], [a, c] = [b, c] = 1 \rangle.$$

This is the nonabelian group of order p^3 and exponent p . □

6.6 Automorphisms of Nonabelian Groups of Order p^3

Remark 6.6.1. *For the two nonabelian groups of order 8, one has*

$$\text{Aut}(D_8) \cong D_8,$$

and

$$\text{Aut}(Q_8) \cong S_4.$$

Definition 6.6.2 (The groups P_-^{1+2} and P_+^{1+2}). *For odd p , denote the two nonabelian groups of order p^3 by*

$$P_-^{1+2} = \langle a, b \mid a^{p^2} = b^p = 1, a^b = a^{1+p} \rangle,$$

and

$$P_+^{1+2} = \langle a, b, c \mid a^p = b^p = c^p = 1, c = [a, b], [a, c] = [b, c] = 1 \rangle.$$

Then in both cases

$$G' = Z(G) \cong C_p.$$

Remark 6.6.3 (Induced automorphisms on G/G'). *Let G be one of the two nonabelian groups of order p^3 , with p odd. Since*

$$G' = Z(G) \cong C_p,$$

every automorphism $\alpha \in \text{Aut}(G)$ satisfies

$$(G')^\alpha = G', \quad Z(G)^\alpha = Z(G).$$

Hence α induces a linear automorphism

$$\bar{\alpha} \in \text{Aut}(G/G') \cong \text{GL}(2, p).$$

For

$$G = P_-^{1+2},$$

we have

$$G = \langle a \rangle : \langle b \rangle, \quad G' = \langle a^p \rangle.$$

Indeed, from

$$a^b = a^{1+p},$$

we get

$$(a^p)^b = (a^b)^p = (a^{1+p})^p = a^p.$$

The automorphism group has the form

$$\text{Aut}(P_-^{1+2}) \cong P : C_{p-1},$$

where P is a p -group.

For

$$G = P_+^{1+2},$$

the commutator map induces a nondegenerate alternating bilinear form

$$(G/G') \times (G/G') \longrightarrow G', \quad (\bar{x}, \bar{y}) \longmapsto [x, y].$$

Automorphisms fixing G' pointwise induce elements of

$$\text{Sp}(2, p) \cong \text{SL}(2, p).$$

Allowing scalar multiplication on G' extends this to

$$\text{GL}(2, p).$$

Equivalently,

$$\text{Aut}(P_+^{1+2})/O_p(\text{Aut}(P_+^{1+2})) \cong \text{GL}(2, p).$$

More explicitly,

$$\text{Aut}(P_+^{1+2}) \cong C_p^2 : \text{GL}(2, p).$$

Chapter 7

Free Groups and Minimal Normal Subgroups

7.1 Free Groups

Definition 7.1.1 (Free group). *Let*

$$X_0 = \{x_1, \dots, x_n\}$$

be a set, and let

$$X_0^{-1} = \{x_1^{-1}, \dots, x_n^{-1}\}$$

be a disjoint set of formal inverses. Set

$$S = X_0 \sqcup X_0^{-1}.$$

A **word** is a finite sequence of elements of S ,

$$w = a_1 a_2 \cdots a_r, \quad a_i \in S.$$

When $r = 0$, we obtain the empty word, denoted by 1.

A word is called a **reduced word** if it contains no adjacent pair

$$x_i x_i^{-1} \quad \text{or} \quad x_i^{-1} x_i.$$

Two words are called **equivalent** if one can be obtained from the other by finitely many insertions or deletions of

$$x_i x_i^{-1}, \quad x_i^{-1} x_i.$$

Write $[w]$ for the equivalence class of a word w . Define multiplication by

$$[u][v] = [uv],$$

where uv denotes the concatenation of the two words, followed by taking the equivalence class. All equivalence classes form a group, called the **free group** generated by

$$x_1, \dots, x_n,$$

denoted by

$$F_n = \langle x_1, \dots, x_n \rangle.$$

The number n is called the **rank** of the free group F_n .

Remark 7.1.2. The free group can also be defined equivalently as the set of all reduced words, with multiplication given by

concatenate, then reduce.

Hence every element of a free group can be written uniquely as a reduced word.

Example 7.1.3 (The free group of rank 2). Let

$$F_2 = \langle x_1, x_2 \rangle.$$

Then every nontrivial element $g \in F_2$ can be written uniquely in a form where the generators alternate, for example

$$g = x_1^{m_1} x_2^{m_2} x_1^{m_3} \dots$$

or

$$g = x_2^{m_1} x_1^{m_2} x_2^{m_3} \dots,$$

where

$$m_i \in \mathbb{Z} \setminus \{0\}.$$

The key point is that adjacent generators cannot be formal inverses of one another; otherwise further reduction is possible.

7.2 Generators and Relations

A free group carries no relations other than those forced by the group axioms. Imposing additional relations on a free group yields many important groups.

Example 7.2.1 (Dihedral groups). We have the presentation

$$\langle x_1, x_2 \mid x_1^2 = 1, x_2^2 = 1 \rangle.$$

This is the **infinite dihedral group**.

Moreover,

$$D_{2m} = \langle x_1, x_2 \mid x_1^2 = 1, x_2^2 = 1, (x_1 x_2)^m = 1 \rangle$$

is the dihedral group of order $2m$.

Example 7.2.2 (Another presentation of the dihedral group). *Let*

$$a = x_1 x_2, \quad y = x_2.$$

From

$$x_1^2 = x_2^2 = 1, \quad (x_1 x_2)^m = 1$$

we obtain

$$a^m = 1, \quad y^2 = 1, \quad y a y^{-1} = a^{-1}.$$

Therefore

$$D_{2m} = \langle a, y \mid a^m = 1, y^2 = 1, y a y^{-1} = a^{-1} \rangle.$$

Definition 7.2.3 (Triangle group). *Let m, n, ℓ be positive integers. The **triangle group** is defined by*

$$T_{m,n,\ell} = \langle x, y \mid x^m = 1, y^n = 1, (xy)^\ell = 1 \rangle.$$

Geometrically, it is often viewed as the group generated by two elements x, y with

$$|x| = m, \quad |y| = n, \quad |xy| = \ell.$$

Consider the angle sum

$$\left(\frac{1}{m} + \frac{1}{n} + \frac{1}{\ell} \right) \pi.$$

According to its size relative to π , triangle groups fall into three types:

$$\frac{1}{m} + \frac{1}{n} + \frac{1}{\ell} > 1 \quad \Longleftrightarrow \quad \text{spherical,}$$

$$\frac{1}{m} + \frac{1}{n} + \frac{1}{\ell} = 1 \quad \Longleftrightarrow \quad \text{Euclidean,}$$

$$\frac{1}{m} + \frac{1}{n} + \frac{1}{\ell} < 1 \quad \Longleftrightarrow \quad \text{hyperbolic.}$$

Without loss of generality, assume

$$m \leq n \leq \ell.$$

If $G \neq 1$, then $\ell > 1$.

Case 1: Spherical type

If

$$\frac{1}{m} + \frac{1}{n} + \frac{1}{\ell} > 1,$$

then the following cases may occur.

1. If $m = 1$, then $x = 1$ and the group reduces to the cyclic group generated by y ; when $|y| = n$,

$$G \cong C_n \cong \mathbb{Z}/n\mathbb{Z}.$$

2. If $m = 2$, $n = 2$, then

$$G \cong D_{2\ell}.$$

3. If $m = 2$, $n = 3$, then we must have

$$\ell = 3, 4, 5.$$

Further analysis yields:

(m, n, ℓ)	G	Geometric meaning
$(2, 3, 3)$	A_4	rotation group of the regular tetrahedron
$(2, 3, 4)$	S_4	rotation group of the cube or regular octahedron
$(2, 3, 5)$	A_5	rotation group of the regular dodecahedron or icosahedron

Case 2: Euclidean type

If

$$\frac{1}{m} + \frac{1}{n} + \frac{1}{\ell} = 1,$$

then the possible triples are

$$(m, n, \ell) = (3, 3, 3), (2, 3, 6), (2, 4, 4).$$

These correspond to tilings of the Euclidean plane:

(m, n, ℓ)	Corresponding tiling
$(3, 3, 3)$	equilateral triangle tiling
$(2, 3, 6)$	triangle/hexagon related tiling
$(2, 4, 4)$	square tiling

These groups are infinite and solvable.

Remark 7.2.4. *In the Euclidean plane, only three regular polygons can tile the plane by themselves:*

equilateral triangle, square, regular hexagon.

Case 3: Hyperbolic type

If

$$\frac{1}{m} + \frac{1}{n} + \frac{1}{\ell} < 1,$$

then $T_{m,n,\ell}$ is an infinite group and is generally nonsolvable. Such groups correspond to triangular tilings in hyperbolic geometry.

7.3 An Example Involving Symmetric Groups

Example 7.3.1. *Let*

$$S_n = \langle (12), (12 \cdots n) \rangle.$$

Set

$$x = (12), \quad y = (12 \cdots n).$$

Then

$$x^2 = 1, \quad y^n = 1.$$

Moreover,

$$xy = (23 \cdots n),$$

so

$$(xy)^{n-1} = 1.$$

Thus we certainly have the relations

$$x^2 = y^n = (xy)^{n-1} = 1.$$

However, it is generally not true that

$$S_n = \langle x, y \mid x^2 = 1, y^n = 1, (xy)^{n-1} = 1 \rangle;$$

more relations are needed. Otherwise, when n is sufficiently large,

$$\frac{1}{2} + \frac{1}{n} + \frac{1}{n-1} < 1,$$

and by the triangle-group discussion, the group given by this presentation would be infinite, contradicting the finiteness of S_n .

7.4 Finitely Generated Groups as Quotients of Free Groups

Theorem 7.4.1. *Every finitely generated group is a homomorphic image of some free group.*

Proof. Let

$$G = \langle g_1, \dots, g_n \rangle.$$

Let

$$F_n = \langle x_1, \dots, x_n \rangle$$

be the free group of rank n . Define a map

$$\varphi: F_n \longrightarrow G$$

by

$$\varphi(x_i) = g_i, \quad \varphi(x_i^{-1}) = g_i^{-1}.$$

Since elements of the free group are reduced words, φ extends naturally to a group homomorphism.

Because $g_1, \dots, g_n$ generate G , the map φ is surjective. Let

$$N = \ker \varphi.$$

By the first isomorphism theorem,

$$F_n/N \cong G.$$

Hence G is a homomorphic image of the free group F_n . □

Definition 7.4.2 (Defining relations). *Let*

$$G \cong F_n/N.$$

Then $x_1, \dots, x_n$ are called generators of G , and the elements of N are called relations. If N , as a normal subgroup, is generated by a set $R \subseteq F_n$, that is,

$$N = \langle R \rangle,$$

we write

$$G = \langle x_1, \dots, x_n \mid r = 1, r \in R \rangle.$$

7.5 Minimal Normal Subgroups

Let

$$\Omega = \{1, 2, \dots, n\}, \quad \text{Sym}(\Omega) = S_n.$$

Definition 7.5.1 (Minimal normal subgroup). *Let G be a group. A nontrivial normal subgroup $M \triangleleft G$ is called a **minimal normal subgroup** of G if there is no nontrivial normal subgroup $N \triangleleft G$ with*

$$1 < N < M.$$

Definition 7.5.2 (Transitive and regular actions). *Let $G \leq \text{Sym}(\Omega)$.*

If for all $\alpha, \beta \in \Omega$ there exists $g \in G$ such that

$$\alpha^g = \beta,$$

then G is said to act **transitively** on Ω .

If G is transitive and the point stabilizer of every point is trivial, that is,

$$G_\alpha = 1,$$

then G is said to act **regularly** on Ω .

Remark 7.5.3. If $H \leq S_n$ is not transitive on Ω , then it preserves a nontrivial orbit decomposition. In particular, if there are two orbits of sizes k and $n - k$, then

$$H \leq S_k \times S_{n-k}.$$

Definition 7.5.4 (Quasi-primitive permutation group). Let $G \leq \text{Sym}(\Omega)$. If every nontrivial normal subgroup of G acts transitively on Ω , then G is called a **quasi-primitive permutation group**, or a **quasi-primitive group**.

Example 7.5.5 (A counterexample). Consider the dihedral group in its natural action,

$$G = D_{2n} = \langle a, b \rangle \leq S_n,$$

where

$$a = (12 \cdots n)$$

is a rotation and

$$b = (1\ n)(2\ n-1) \cdots$$

is a reflection.

If n is even, then

$$N = \langle a^2 \rangle = \{1, a^2, \dots, a^{n-2}\}$$

is a normal subgroup of G , and

$$|N| = \frac{n}{2}.$$

It has two orbits on Ω , namely the odd and even points, so it is not transitive. Hence in this case D_{2n} is not quasi-primitive.

Proposition 7.5.6. Let $G \leq \text{Sym}(\Omega)$ be a quasi-primitive permutation group, and let M be a minimal normal subgroup of G . Then:

1. M acts transitively on Ω ;

2. M is a direct product of isomorphic simple groups, that is,

$$M \cong T_1 \times \cdots \times T_r,$$

where

$$r \geq 1, \quad T_1 \cong \cdots \cong T_r \cong T,$$

and T is a simple group.

Proof. Since G is quasi-primitive and $M \triangleleft G$ is nontrivial, M must be transitive by definition.

Since M is a minimal normal subgroup of G , it has no nontrivial proper characteristic subgroups. Otherwise, if K is a nontrivial proper characteristic subgroup of M , then because $M \triangleleft G$, we have

$$K \triangleleft G,$$

contradicting the minimality of M .

Therefore M is a finite characteristically simple group. Every finite characteristically simple group is a direct product of isomorphic simple groups, so

$$M \cong T_1 \times \cdots \times T_r,$$

where the T_i are pairwise isomorphic and simple. □

Remark 7.5.7. If G is an arbitrary permutation group, a minimal normal subgroup need not be transitive. For example,

$$G = \{1, (12)(34), (13)(24), (14)(23)\} \leq S_4.$$

This is the Klein four-group. Since G is abelian, every subgroup is normal. Let

$$A = \{1, (12)(34)\} \triangleleft G.$$

Then A is a minimal normal subgroup of G , but it is not transitive on $\{1, 2, 3, 4\}$.

Lemma 7.5.8. Let $M \triangleleft G$ and suppose M acts transitively on Ω . Then for every $\omega \in \Omega$,

$$G = MG_\omega,$$

where

$$G_\omega = \{g \in G \mid \omega^g = \omega\}$$

is the stabilizer of ω .

Proof. Take any $g \in G$ and set

$$\omega' = \omega^g.$$

Since M is transitive, there exists $m \in M$ such that

$$\omega^m = \omega'.$$

Then

$$\omega^{gm^{-1}} = (\omega^g)^{m^{-1}} = (\omega^m)^{m^{-1}} = \omega.$$

Therefore

$$gm^{-1} \in G_\omega.$$

Hence

$$g \in G_\omega M.$$

Since $M \triangleleft G$, we have

$$G_\omega M = MG_\omega.$$

Thus

$$G = MG_\omega.$$

□

7.6 Minimal Normal Subgroups of Quasi-Primitive Groups

Lemma 7.6.1. *Let $G \leq \text{Sym}(\Omega)$ be a quasi-primitive permutation group. Then G has at most two minimal normal subgroups.*

If M, N are two distinct minimal normal subgroups of G , then

$$M \cong N,$$

and both M and N are nonabelian regular groups.

Proof. Let M, N be two distinct minimal normal subgroups. Since

$$M \cap N \triangleleft G,$$

and since M, N are minimal normal and distinct, we have

$$M \cap N = 1.$$

Moreover,

$$[M, N] \leq M \cap N,$$

so

$$[M, N] = 1.$$

Hence M and N commute pointwise, and

$$MN = M \times N \triangleleft G.$$

Since G is quasi-primitive, both M and N are nontrivial normal subgroups, so both act transitively on Ω .

We now show that M is regular. Take any $\omega, \omega' \in \Omega$. Since N is transitive, there exists $x \in N$ such that

$$\omega^x = \omega'.$$

Then

$$M_{\omega'} = M_{\omega^x} = x^{-1}M_{\omega}x.$$

But M and N commute pointwise, so

$$x^{-1}M_{\omega}x = M_{\omega}.$$

Hence

$$M_{\omega'} = M_{\omega}$$

for every $\omega' \in \Omega$.

Therefore, if $m \in M_{\omega}$, then m fixes every point $\omega' \in \Omega$. Since the action of $G \leq \text{Sym}(\Omega)$ is faithful, we get

$$m = 1.$$

Thus

$$M_{\omega} = 1.$$

So M is regular. By the same argument, N is regular.

Next we show that M and N are nonabelian. Suppose for contradiction that M is abelian. Choose $1 \neq x \in N$, and let

$$H = \langle M, x \rangle.$$

Since M and N commute pointwise, H is abelian. Since M is transitive, H is also transitive.

An abelian transitive permutation group must be regular: if A is abelian and transitive, then

$$A_{\omega^a} = a^{-1}A_{\omega}a = A_{\omega}$$

for every $a \in A$, so a point stabilizer fixes all points; by faithfulness, the stabilizer is trivial.

Hence H is regular. Thus

$$|H| = |\Omega|.$$

But M is also regular, so

$$|M| = |\Omega|.$$

Since $x \notin M$, we have

$$M < H,$$

and therefore

$$|H| > |M|,$$

a contradiction. So M is nonabelian. By the same argument, N is nonabelian.

We now prove that $M \cong N$. Let

$$X = MN = M \times N.$$

Since M is transitive, X is transitive. Fix a point $\omega \in \Omega$ and consider the stabilizer X_ω .

Since M and N are regular,

$$|M| = |N| = |\Omega|.$$

Thus

$$|X_\omega| = \frac{|X|}{|\Omega|} = \frac{|M||N|}{|\Omega|} = |\Omega|.$$

Consider the natural projection

$$\pi_M: X_\omega \longrightarrow M.$$

Its kernel is

$$\ker \pi_M = X_\omega \cap N = N_\omega = 1,$$

so π_M is injective. Also,

$$|X_\omega| = |M|,$$

so π_M is bijective.

Therefore, for each $m \in M$, there is a unique $\psi(m) \in N$ such that

$$m\psi(m) \in X_\omega.$$

This defines a map

$$\psi: M \longrightarrow N.$$

Since X_ω is a subgroup, if

$$m_1\psi(m_1), \quad m_2\psi(m_2) \in X_\omega,$$

then

$$m_1m_2\psi(m_1)\psi(m_2) \in X_\omega.$$

By uniqueness,

$$\psi(m_1m_2) = \psi(m_1)\psi(m_2).$$

So ψ is a homomorphism. Similarly, the projection onto N is bijective, hence ψ is an isomorphism. Therefore

$$M \cong N.$$

Finally, we show that G has at most two minimal normal subgroups. Suppose L is a third minimal normal subgroup with

$$L \neq M.$$

By the preceding discussion, L is also regular and commutes pointwise with M , so

$$L \leq C_{\text{Sym}(\Omega)}(M).$$

On the other hand, if M is a regular group, then each element of $C_{\text{Sym}(\Omega)}(M)$ is determined uniquely by its value at one point, so

$$|C_{\text{Sym}(\Omega)}(M)| \leq |\Omega|.$$

Since N is regular and commutes pointwise with M ,

$$N \leq C_{\text{Sym}(\Omega)}(M), \quad |N| = |\Omega|.$$

Therefore

$$C_{\text{Sym}(\Omega)}(M) = N.$$

Similarly,

$$L \leq C_{\text{Sym}(\Omega)}(M) = N.$$

Since L and N are both nontrivial minimal normal subgroups,

$$L = N.$$

Hence G has at most two minimal normal subgroups. □

Corollary 7.6.2. *Let $G \leq \text{Sym}(\Omega)$ be transitive. Then*

$$C_{\text{Sym}(\Omega)}(G)$$

is semiregular on Ω .

Proof. Let

$$C = C_{\text{Sym}(\Omega)}(G).$$

We prove that

$$C_\omega = 1 \quad \text{for every } \omega \in \Omega.$$

Fix $\omega \in \Omega$, and let $c \in C_\omega$. Thus

$$\omega^c = \omega.$$

Let $\omega' \in \Omega$. Since G is transitive on Ω , there exists $g \in G$ such that

$$\omega^g = \omega'.$$

Since $c \in C_{\text{Sym}(\Omega)}(G)$, we have

$$cg = gc.$$

Therefore

$$(\omega')^c = (\omega^g)^c = \omega^{gc} = \omega^{cg} = (\omega^c)^g = \omega^g = \omega'.$$

Thus c fixes every point of Ω . Hence $c = 1$. Therefore

$$C_\omega = 1.$$

Since $\omega \in \Omega$ was arbitrary, C is semiregular on Ω . □

Definition 7.6.3 (Semiregular permutation group). *Let $L \leq \text{Sym}(\Omega)$. We say that L is **semiregular** on Ω if*

$$L_\omega = 1 \quad \text{for every } \omega \in \Omega,$$

where

$$L_\omega = \{\ell \in L \mid \omega^\ell = \omega\}$$

is the stabilizer of ω in L .

Equivalently, L is semiregular if no nonidentity element of L fixes any point of Ω ; that is,

$$1 \neq \ell \in L \implies \omega^\ell \neq \omega \quad \text{for all } \omega \in \Omega.$$

Chapter 8

Quasiprimitive Groups and the O’Nan–Scott–Praeger Types

Topic 8.0.1. *For a finite quasiprimitive group $G \leq \text{Sym}(\Omega)$, the structure of the minimal normal subgroups strongly restricts the whole permutation group.*

The O’Nan–Scott–Praeger theorem divides such groups into eight types. The first division is:

- *G has a unique minimal normal subgroup;*
- *G has exactly two minimal normal subgroups.*

The second case leads to the holomorph simple and holomorph compound types. The first case leads to affine, almost simple, twisted wreath, diagonal, and product action types.

8.1 Case I: A Unique Minimal Normal Subgroup

Assume first that $G \leq \text{Sym}(\Omega)$ is quasiprimitive and has a unique minimal normal subgroup N .

8.1.1 The Centralizer of a Nonabelian Minimal Normal Subgroup

Lemma 8.1.1. *Let G have a unique minimal normal subgroup N . If N is nonabelian, then*

$$C_G(N) = 1.$$

Consequently, conjugation gives an embedding

$$G \hookrightarrow \text{Aut}(N).$$

Proof. Since $N \trianglelefteq G$, its centralizer $C_G(N)$ is normal in G .

Suppose $C_G(N) \neq 1$. Since G is finite, $C_G(N)$ contains a minimal normal subgroup of G . But N is the unique minimal normal subgroup of G , so

$$N \leq C_G(N).$$

Thus every element of N centralizes N , meaning

$$N \leq Z(N).$$

Hence N is abelian, contradicting the assumption that N is nonabelian.

Therefore

$$C_G(N) = 1.$$

Now G acts on N by conjugation. The kernel of this action is exactly $C_G(N)$. Hence

$$G/C_G(N) \hookrightarrow \text{Aut}(N).$$

Since $C_G(N) = 1$, we get

$$G \hookrightarrow \text{Aut}(N).$$

□

8.1.2 The Abelian Case: HA Type

Proposition 8.1.2. *Let $G \leq \text{Sym}(\Omega)$ be quasiprimitive, and suppose that G has a unique minimal normal subgroup*

$$N \cong C_p^d$$

for some prime p and integer $d \geq 1$. Then N is regular on Ω .

Proof. Since $N \trianglelefteq G$ and G is quasiprimitive, N is transitive on Ω .

Fix $\omega \in \Omega$. Let $n \in N_\omega$. For any $\alpha \in \Omega$, since N is transitive, there exists $m \in N$ such that

$$\alpha = \omega^m.$$

Because N is abelian, we have

$$\alpha^n = (\omega^m)^n = (\omega^n)^m = \omega^m = \alpha.$$

Thus n fixes every point of Ω . Since the action is faithful, $n = 1$. Therefore

$$N_\omega = 1.$$

Hence N is regular. □

Definition 8.1.3 (HA type). *A quasiprimitive group $G \leq \text{Sym}(\Omega)$ is said to be of holomorph affine type, or HA type, if G has a unique minimal normal subgroup*

$$N \cong C_p^d$$

and N is regular on Ω .

Equivalently, after identifying Ω with the vector space $N \cong \mathbb{F}_p^d$, the group G embeds into the affine general linear group

$$\text{AGL}(d, p) = \mathbb{F}_p^d \rtimes \text{GL}(d, p).$$

Example 8.1.4 (An HA-type quasiprimitive group). *Let*

$$\Omega = \mathbb{F}_3$$

and let

$$G = \text{AGL}(1, 3) = \mathbb{F}_3^+ \rtimes \mathbb{F}_3^\times$$

act on Ω by affine transformations

$$x \mapsto ax + b, \quad a \in \mathbb{F}_3^\times, \quad b \in \mathbb{F}_3.$$

Thus

$$|G| = 3 \cdot 2 = 6,$$

and in fact

$$G \cong S_3.$$

Let

$$N = \{x \mapsto x + b \mid b \in \mathbb{F}_3\}.$$

Then

$$N \cong \mathbb{F}_3^+ \cong C_3.$$

The subgroup N is normal in G , since translations are normalized by affine transformations. Moreover, N acts regularly on Ω : for any $\alpha, \beta \in \mathbb{F}_3$, there is a unique translation

$$x \mapsto x + (\beta - \alpha)$$

sending α to β .

Since $G \cong S_3$, the subgroup corresponding to N is A_3 . Hence N is the unique minimal normal subgroup of G . Therefore every nontrivial normal subgroup of G contains N , and is transitive on Ω . Thus G is quasiprimitive.

Consequently, $G = \text{AGL}(1, 3)$ is of HA type, with regular abelian minimal normal subgroup

$$N \cong C_3.$$

8.1.3 The Nonabelian Simple Case: AS Type

Proposition 8.1.5. *Let $G \leq \text{Sym}(\Omega)$ be quasiprimitive, and suppose that G has a unique minimal normal subgroup N , where N is nonabelian simple. Then*

$$N \trianglelefteq G \leq \text{Aut}(N).$$

Proof. Since N is nonabelian and is the unique minimal normal subgroup of G , the previous lemma gives

$$C_G(N) = 1.$$

Therefore the conjugation action of G on N embeds G into $\text{Aut}(N)$. Under this embedding, N is identified with the inner automorphism group

$$\text{Inn}(N) \cong N.$$

Thus

$$N \trianglelefteq G \leq \text{Aut}(N).$$

□

Definition 8.1.6 (AS type). *A quasiprimitive group $G \leq \text{Sym}(\Omega)$ is said to be of almost simple type, or AS type, if G has a unique minimal normal subgroup N , where N is nonabelian simple, and*

$$N \trianglelefteq G \leq \text{Aut}(N).$$

8.1.4 The Nonabelian Characteristically Simple Case

Now suppose

$$N = T^k, \quad k \geq 2,$$

where T is nonabelian simple. Let $\omega \in \Omega$. Since $N \trianglelefteq G$ and G is quasiprimitive, N is transitive. Thus Ω may be identified with the set of right cosets of N_ω in N .

The possible structure of N_ω leads to four types.

Definition 8.1.7 (Full diagonal subgroup). *Let*

$$N = T_1 \times T_2 \times \cdots \times T_k,$$

where each $T_i \cong T$. A subgroup $D \leq N$ is called a full diagonal subgroup over a subset $I \subseteq \{1, \dots, k\}$ if, after choosing isomorphisms

$$\alpha_i : T \rightarrow T_i \quad (i \in I),$$

we have

$$D = \{(t^{\alpha_i})_{i \in I} \mid t \in T\} \leq \prod_{i \in I} T_i.$$

Thus $D \cong T$ and projects onto each direct factor T_i for $i \in I$.

Topic 8.1.8. Let $N = T^k$, $k \geq 2$, be the unique minimal normal subgroup of a quasiprimitive group $G \leq \text{Sym}(\Omega)$. For $\omega \in \Omega$, the point stabilizer N_ω determines the following four types:

(i) If

$$N_\omega = 1,$$

then N is regular. This is the TW type.

(ii) If

$$N_\omega \cong T$$

is a full diagonal subgroup of T^k , then this is the SD type.

(iii) If

$$N_\omega \cong T^m, \quad 2 \leq m \leq \frac{k}{2},$$

and N_ω is a product of full diagonal subgroups corresponding to a partition of the k simple direct factors, then this is the CD type.

(iv) If

$$1 \neq N_\omega$$

and N_ω is not of the diagonal forms above, then this is the PA type.

Definition 8.1.9 (TW type). A quasiprimitive group $G \leq \text{Sym}(\Omega)$ is said to be of twisted wreath type, or TW type, if its unique minimal normal subgroup is

$$N = T^k, \quad k \geq 2,$$

where T is nonabelian simple, and N is regular on Ω .

Definition 8.1.10 (SD type). A quasiprimitive group $G \leq \text{Sym}(\Omega)$ is said to be of simple diagonal type, or SD type, if its unique minimal normal subgroup is

$$N = T^k, \quad k \geq 2,$$

where T is nonabelian simple, and for $\omega \in \Omega$,

$$N_\omega \cong T$$

is a full diagonal subgroup of N .

Definition 8.1.11 (CD type). A quasiprimitive group $G \leq \text{Sym}(\Omega)$ is said to be of compound diagonal type, or CD type, if its unique minimal normal subgroup is

$$N = T^k, \quad k \geq 2,$$

where T is nonabelian simple, and for $\omega \in \Omega$,

$$N_\omega \cong T^m, \quad 2 \leq m \leq \frac{k}{2},$$

where N_ω is a product of full diagonal subgroups.

Definition 8.1.12 (PA type). A quasiprimitive group $G \leq \text{Sym}(\Omega)$ is said to be of product action type, or PA type, if its unique minimal normal subgroup is

$$N = T^k, \quad k \geq 2,$$

where T is nonabelian simple, and $N_\omega \neq 1$ is not a diagonal stabilizer of the SD or CD form.

8.2 Regular Representations and Central Products

This section collects the regular-action machinery used in the two-minimal-normal-subgroup case.

8.2.1 Right Regular, Left Regular, and Conjugation Actions

Let G be a finite group. Define three families of permutations of the set G .

For $g \in G$, define

$$x^{\widehat{g}} = xg,$$

$$x^{\check{g}} = g^{-1}x,$$

and

$$x^{\overline{g}} = g^{-1}xg.$$

Set

$$\widehat{G} = \{\widehat{g} \mid g \in G\},$$

$$\check{G} = \{\check{g} \mid g \in G\},$$

and

$$\overline{G} = \{\overline{g} \mid g \in G\}.$$

Lemma 8.2.1 (Right and left regular representations). *Let G be a finite group. Then the following hold.*

(1) $\widehat{G}, \check{G} \leq \text{Sym}(G)$, and both are regular.

(2) $\widehat{G}$ and $\check{G}$ centralize each other:

$$\widehat{g}\check{h} = \check{h}\widehat{g} \quad (g, h \in G).$$

(3)

$$\widehat{G} \cap \check{G} = Z(\widehat{G}) = Z(\check{G}) = \widehat{Z(G)} = Z(\check{G}).$$

(4) If

$$X = \langle \widehat{G}, \check{G} \rangle,$$

then

$$X = \widehat{G}\check{G}$$

and

$$X \cong (\widehat{G} \times \check{G}) / \{(\widehat{z}, \check{z}) \mid z \in Z(G)\}.$$

(5) If $1 \in G$ is used as base point, then

$$X_1 = \overline{G}.$$

Proof. For $g, h \in G$,

$$x^{\widehat{gh}} = (xg)h = x(gh) = x^{\widehat{gh}},$$

so

$$\widehat{gh} = \widehat{gh}.$$

Hence $\widehat{G} \leq \text{Sym}(G)$. Similarly,

$$x^{\check{gh}} = h^{-1}(g^{-1}x) = (gh)^{-1}x = x^{\check{gh}},$$

so

$$\check{gh} = \check{gh}.$$

Thus $\check{G} \leq \text{Sym}(G)$.

The group $\widehat{G}$ is regular because for any $x, y \in G$, there is a unique $g \in G$ such that

$$xg = y,$$

namely $g = x^{-1}y$. Similarly, $\check{G}$ is regular.

Next,

$$x^{\widehat{gh}} = h^{-1}xg = x^{\widehat{hg}}.$$

Therefore

$$\widehat{g}\check{h} = \check{h}\widehat{g}.$$

Now suppose

$$\widehat{g} = \check{h}.$$

Evaluating at $1 \in G$, we get

$$g = h^{-1}.$$

Thus for all $x \in G$,

$$xg = h^{-1}x = gx.$$

Hence $g \in Z(G)$. Conversely, if $z \in Z(G)$, then

$$x^{\widehat{z}} = xz = zx = x^{\check{z}^{-1}},$$

so $\widehat{z} = \check{z}^{-1}$. Therefore

$$\widehat{G} \cap \check{G} = \widehat{Z(G)} = Z(\check{G}).$$

Also

$$Z(\widehat{G}) = \widehat{Z(G)}, \quad Z(\check{G}) = Z(\check{G}).$$

Since $\widehat{G}$ and $\check{G}$ centralize each other,

$$X = \langle \widehat{G}, \check{G} \rangle = \widehat{G}\check{G}.$$

Consider the epimorphism

$$\theta : \widehat{G} \times \check{G} \rightarrow X, \quad (\widehat{g}, \check{h}) \mapsto \widehat{g}\check{h}.$$

Its kernel consists of those pairs satisfying

$$\widehat{g}\check{h} = 1.$$

For all $x \in G$,

$$x^{\widehat{g}\check{h}} = h^{-1}xg.$$

Thus $\widehat{g}\check{h} = 1$ if and only if

$$h^{-1}xg = x \quad \forall x \in G.$$

Taking $x = 1$, we get $g = h$. Then

$$g^{-1}xg = x \quad \forall x \in G,$$

so $g \in Z(G)$. Hence

$$\ker \theta = \{(\widehat{z}, \check{z}) \mid z \in Z(G)\}.$$

The isomorphism theorem gives

$$X \cong (\widehat{G} \times \check{G}) / \{(\widehat{z}, \check{z}) \mid z \in Z(G)\}.$$

Finally, an element of X has the form $\hat{g}\check{h}$. It fixes $1 \in G$ if and only if

$$1^{\hat{g}\check{h}} = h^{-1}g = 1,$$

that is, if and only if $g = h$. Therefore

$$X_1 = \{\hat{g}\check{g} \mid g \in G\}.$$

But

$$x^{\hat{g}\check{g}} = g^{-1}xg = x^{\bar{g}}.$$

Hence

$$X_1 = \overline{G}.$$

□

Corollary 8.2.2. *If $Z(G) = 1$, then*

$$\hat{G} \cap \check{G} = 1$$

and hence

$$\langle \hat{G}, \check{G} \rangle = \hat{G} \times \check{G}.$$

Moreover,

$$\overline{G} \cong G.$$

8.2.2 Central Products

Definition 8.2.3 (Central product). *Let M and N be groups, and let*

$$C_1 \leq Z(M), \quad C_2 \leq Z(N).$$

Suppose that

$$\varphi : C_1 \rightarrow C_2$$

is an isomorphism. Define

$$D_\varphi = \{(c, \varphi(c)^{-1}) \mid c \in C_1\} \leq M \times N.$$

The quotient

$$(M \times N)/D_\varphi$$

is called the central product of M and N amalgamating C_1 and C_2 .

When $C_1 \cong C_2 \cong C$, we often write this central product as

$$M \circ_C N$$

or simply

$$M \circ N$$

when the amalgamated central subgroups are clear.

Remark 8.2.4. A central product is obtained from the direct product $M \times N$ by identifying selected central subgroups of M and N . Thus it is generally not a direct product, but it is still built from two commuting subgroups.

Example 8.2.5. Let $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$. Then

$$Z(Q_8) = \{1, -1\} \cong C_2.$$

Taking two copies of Q_8 , identify the two central involutions. Then

$$Q_8 \circ Q_8 = (Q_8 \times Q_8) / \langle (-1, -1) \rangle.$$

Hence

$$|Q_8 \circ Q_8| = \frac{|Q_8|^2}{2} = \frac{8 \cdot 8}{2} = 32.$$

Example 8.2.6. Let

$$C_4 = \langle c \rangle.$$

Since

$$\langle c^2 \rangle \leq Z(C_4),$$

we may identify

$$\langle -1 \rangle \leq Z(Q_8)$$

with

$$\langle c^2 \rangle \leq Z(C_4).$$

Then

$$Q_8 \circ C_4 = (Q_8 \times C_4) / \langle (-1, c^2) \rangle.$$

Example 8.2.7. Let

$$M = \langle a, z \mid a^8 = z^2 = 1, a^z = a^5 \rangle,$$

where

$$a^z = z^{-1}az.$$

Then

$$(a^2)^z = (a^z)^2 = (a^5)^2 = a^{10} = a^2,$$

so

$$\langle a^2 \rangle \leq Z(M).$$

In fact,

$$Z(M) = \langle a^2 \rangle \cong C_4.$$

Similarly, let

$$N = \langle b, y \mid b^8 = y^2 = 1, b^y = b^5 \rangle.$$

Then

$$Z(N) = \langle b^2 \rangle \cong C_4.$$

If we identify

$$\langle a^2 \rangle \leq Z(M)$$

with

$$\langle b^2 \rangle \leq Z(N),$$

then we obtain the central product

$$M \circ N = (M \times N) / \langle (a^2, b^{-2}) \rangle.$$

If instead we identify only the central subgroups of order 2,

$$\langle a^4 \rangle \leq Z(M), \quad \langle b^4 \rangle \leq Z(N),$$

then

$$M \circ_{C_2} N = (M \times N) / \langle (a^4, b^4) \rangle.$$

8.3 Centralizers of Regular Subgroups

Lemma 8.3.1. *Let*

$$M \leq \text{Sym}(\Omega)$$

be regular, and fix $\omega_0 \in \Omega$. Identify Ω with M by

$$\omega_0^m \longmapsto m.$$

Under this identification, the action of M on Ω becomes the right regular action

$$\widehat{M} = \{\widehat{m} \mid m \in M\}$$

on the set M .

Moreover,

$$C_{\text{Sym}(\Omega)}(M) \cong C_{\text{Sym}(M)}(\widehat{M}) = \check{M}.$$

In particular, $C_{\text{Sym}(\Omega)}(M)$ is regular.

Proof. Since M is regular, for each $\omega \in \Omega$ there exists a unique $m \in M$ such that

$$\omega = \omega_0^m.$$

Thus

$$\omega_0^m \mapsto m$$

is a bijection $\Omega \rightarrow M$.

For $g \in M$,

$$(\omega_0^m)^g = \omega_0^{mg}.$$

Under the identification $\Omega \cong M$, this is

$$m \mapsto mg,$$

which is the right regular action $\widehat{g}$.

Now let

$$\alpha \in C_{\text{Sym}(M)}(\widehat{M}).$$

Set

$$1^\alpha = t.$$

Then for every $m \in M$,

$$m^\alpha = (1^{\widehat{m}})^\alpha = (1^\alpha)^{\widehat{m}} = tm.$$

Thus α is left multiplication by t , which is t^{-1} . Hence

$$\alpha \in \check{M}.$$

Conversely, every left multiplication map commutes with every right multiplication map. Therefore

$$C_{\text{Sym}(M)}(\widehat{M}) = \check{M}.$$

□

8.4 Case II: Two Minimal Normal Subgroups

Now suppose $G \leq \text{Sym}(\Omega)$ is quasiprimitive and has two distinct minimal normal subgroups M and N .

Proposition 8.4.1. *Let $G \leq \text{Sym}(\Omega)$ be quasiprimitive. If M and N are two distinct minimal normal subgroups of G , then:*

(1)

$$M \cap N = 1;$$

(2)

$$[M, N] = 1;$$

(3) M and N are regular on Ω ;

(4)

$$M \cong N;$$

(5) There exists a nonabelian simple group T and an integer $k \geq 1$ such that

$$M \cong N \cong T^k.$$

Proof. Since $M, N \trianglelefteq G$, their intersection is normal in G :

$$M \cap N \trianglelefteq G.$$

Moreover,

$$M \cap N \leq M \quad \text{and} \quad M \cap N \leq N.$$

By minimal normality and the assumption $M \neq N$, we get

$$M \cap N = 1.$$

Now

$$[M, N] \leq M \cap N,$$

so

$$[M, N] = 1.$$

Thus M and N centralize each other.

Since G is quasiprimitive and M, N are nontrivial normal subgroups, both M and N are transitive on Ω .

Fix $\omega \in \Omega$. Let $n \in N_\omega$. For any $\alpha \in \Omega$, because M is transitive, there exists $m \in M$ such that

$$\alpha = \omega^m.$$

Since M and N centralize each other,

$$\alpha^n = (\omega^m)^n = (\omega^n)^m = \omega^m = \alpha.$$

Thus n fixes every point of Ω . Since the action is faithful, $n = 1$. Therefore

$$N_\omega = 1.$$

Hence N is regular. The same argument shows that M is regular.

Since both are regular,

$$|M| = |\Omega| = |N|.$$

Because M is regular,

$$C_{\text{Sym}(\Omega)}(M)$$

is also regular and is isomorphic to M . Since N centralizes M , we have

$$N \leq C_{\text{Sym}(\Omega)}(M).$$

Both groups have order $|\Omega|$, so

$$N = C_{\text{Sym}(\Omega)}(M).$$

Therefore

$$N \cong M.$$

Finally, minimal normal subgroups are characteristically simple, so

$$M \cong T^k$$

for some finite simple group T . If T were abelian, then M would be abelian. Since M is regular and abelian, its centralizer in $\text{Sym}(\Omega)$ would be itself. Hence

$$C_{\text{Sym}(\Omega)}(M) = M.$$

But we have already shown that

$$C_{\text{Sym}(\Omega)}(M) = N,$$

so $M = N$, contradicting the assumption that M and N are distinct.

Thus T is nonabelian. □

8.4.1 Relabelling Ω by a Regular Normal Subgroup

Let $M \leq \text{Sym}(\Omega)$ be regular and fix

$$\Omega = \{\omega_1, \omega_2, \dots, \omega_n\}.$$

Choose ω_1 as base point. Since M is regular, for each i there exists a unique $x_i \in M$ such that

$$\omega_i = \omega_1^{x_i}.$$

Thus

$$M = \{1 = x_1, x_2, \dots, x_n\}.$$

The map

$$\omega_i \longmapsto x_i$$

identifies Ω with M .

Under this identification, M acts on itself by right multiplication:

$$x \mapsto xg.$$

Hence

$$M = \widehat{M}.$$

If N is the second minimal normal subgroup, then

$$N = C_{\text{Sym}(\Omega)}(M) = \check{M}.$$

Thus

$$MN = \widehat{M}\check{M}.$$

Proposition 8.4.2. *With the above identification, let*

$$X = MN = \widehat{M}\check{M}.$$

Taking $1 \in M$ as base point, we have

$$X_1 = \overline{M} = \{\widehat{m}\check{m} \mid m \in M\}.$$

If $M \cong T^k$ for a nonabelian simple group T , then $Z(M) = 1$ and

$$X_1 \cong M.$$

Proof. Every element of X has the form

$$\widehat{a}\check{b}.$$

Now

$$1^{\widehat{a}\check{b}} = b^{-1}a.$$

Therefore

$$\widehat{a}\check{b} \in X_1 \iff b^{-1}a = 1 \iff a = b.$$

Hence

$$X_1 = \{\widehat{m}\check{m} \mid m \in M\}.$$

But

$$x^{\widehat{m}\check{m}} = m^{-1}xm = x^{\overline{m}}.$$

Thus

$$X_1 = \overline{M}.$$

If $M \cong T^k$ with T nonabelian simple, then $Z(T) = 1$, so

$$Z(M) = 1.$$

The conjugation action of M on itself has kernel $Z(M)$, hence is faithful. Therefore

$$\overline{M} \cong M.$$

8.5 The O’Nan–Scott–Praeger Theorem

Theorem 8.5.1 (O’Nan–Scott–Praeger theorem). *Let $G \leq \text{Sym}(\Omega)$ be a finite quasiprimitive permutation group. Then G belongs to one of the following eight types.*

HS ***Holomorph Simple type.***
The group G has exactly two minimal normal subgroups M and N , and

$$M \cong N \cong T,$$

where T is a nonabelian simple group. Both M and N are regular.

HC ***Holomorph Compound type.***
The group G has exactly two minimal normal subgroups M and N , and

$$M \cong N \cong T^k, \quad k \geq 2,$$

where T is a nonabelian simple group. Both M and N are regular.

HA ***Holomorph Affine type.***
The group G has a unique minimal normal subgroup

$$N \cong C_p^d$$

for some prime p , and N is regular.

AS ***Almost Simple type.***
The group G has a unique minimal normal subgroup N , where N is nonabelian simple, and

$$N \trianglelefteq G \leq \text{Aut}(N).$$

TW ***Twisted Wreath type.***
The group G has a unique minimal normal subgroup

$$N = T^k, \quad k \geq 2,$$

where T is nonabelian simple, and N is regular.

SD ***Simple Diagonal type.***
The group G has a unique minimal normal subgroup

$$N = T^k, \quad k \geq 2,$$

where T is nonabelian simple, and for $\omega \in \Omega$,

$$N_\omega \cong T$$

is a full diagonal subgroup of N .

CD

Compound Diagonal type.

The group G has a unique minimal normal subgroup

$$N = T^k, \quad k \geq 2,$$

where T is nonabelian simple, and for $\omega \in \Omega$,

$$N_\omega \cong T^m, \quad 2 \leq m \leq \frac{k}{2},$$

where N_ω is a product of full diagonal subgroups.

PA

Product Action type.

The group G has a unique minimal normal subgroup

$$N = T^k, \quad k \geq 2,$$

where T is nonabelian simple, and $N_\omega \neq 1$ is not of the diagonal forms appearing in SD or CD.

8.6 Summary of the Two-Minimal-Normal-Subgroup Case

Topic 8.6.1. Suppose $G \leq \text{Sym}(\Omega)$ is quasiprimitive and has two distinct minimal normal subgroups M and N . Then:

(1)

$$M \cap N = 1;$$

(2)

$$[M, N] = 1;$$

(3) M and N are regular;

(4)

$$M \cong N \cong T^k$$

for some nonabelian simple group T ;

(5) after identifying Ω with M ,

$$M = \widehat{M}, \quad N = \check{M};$$

(6)

$$MN = \widehat{M}\check{M};$$

(7) since $Z(M) = 1$,

$$MN = \widehat{M} \times \check{M};$$

(8) the stabilizer of $1 \in M$ in MN is

$$(MN)_1 = \overline{M} \cong M.$$

If $k = 1$, this is the HS type. If $k \geq 2$, this is the HC type.

Chapter 9

Week 8: Holomorphs and Primitive Actions

9.1 Regular Actions

Lemma 9.1.1. *The subgroups $\widehat{X}$ and $\check{X}$ are regular subgroups of $\text{Sym}(X)$, and they centralize each other.*

Proof. The action of $\widehat{X}$ is regular because for any $x, y \in X$, the unique element sending x to y is

$$\widehat{x^{-1}y}.$$

Similarly, the action of $\check{X}$ is regular because the unique element sending x to y is

$$y\check{x}^{-1}.$$

Finally, for all $a, b, x \in X$,

$$x^{\widehat{a}\check{b}} = b^{-1}xa = x^{\check{b}\widehat{a}}.$$

Hence every element of $\widehat{X}$ commutes with every element of $\check{X}$. □

Definition 9.1.2 (Equivalent permutation actions). *Let G act on two sets Ω and Δ . These two actions are called **equivalent** if there exists a bijection*

$$\varphi : \Omega \rightarrow \Delta$$

such that

$$\varphi(\omega^g) = \varphi(\omega)^g \quad \text{for all } \omega \in \Omega, \ g \in G.$$

Corollary 9.1.3. *Every regular action of a group X is equivalent to the action of $\widehat{X}$ on X , and also to the action of $\check{X}$ on X .*

Proof. Let X act regularly on Ω . Fix $\omega \in \Omega$.

First define

$$\varphi : X \rightarrow \Omega, \quad a \mapsto \omega^a.$$

Since the action is regular, φ is a bijection. For $a, b \in X$, we have

$$\varphi(a^{\widehat{b}}) = \varphi(ab) = \omega^{ab} = (\omega^a)^b = \varphi(a)^b.$$

Thus the given regular action is equivalent to the right regular action.

For the left regular action, define

$$\psi : X \rightarrow \Omega, \quad a \mapsto \omega^{a^{-1}}.$$

Again ψ is a bijection. For $a, b \in X$,

$$\psi(a^{\check{b}}) = \psi(b^{-1}a) = \omega^{(b^{-1}a)^{-1}} = \omega^{a^{-1}b} = (\omega^{a^{-1}})^b = \psi(a)^b.$$

Hence the given regular action is also equivalent to the left regular action. $\square$

Lemma 9.1.4 (Centralizer of a regular group). *Let X act on itself by right multiplication. Then*

$$C_{\text{Sym}(X)}(\widehat{X}) = \check{X}.$$

Proof. We already know that $\check{X} \leq C_{\text{Sym}(X)}(\widehat{X})$.

Conversely, let $c \in C_{\text{Sym}(X)}(\widehat{X})$. Put

$$1^c = a.$$

Then for every $x \in X$,

$$x^c = (1^{\widehat{x}})^c = (1^c)^{\widehat{x}} = ax.$$

But the map $x \mapsto ax$ is $a^{\check{-1}}$. Hence $c \in \check{X}$. Therefore

$$C_{\text{Sym}(X)}(\widehat{X}) = \check{X}.$$

$\square$

9.2 The Holomorph

Definition 9.2.1 (Holomorph). *Let G be a group. The **holomorph** of G is*

$$\text{Hol}(G) = G : \text{Aut}(G).$$

Equivalently, under the right action convention, we may realize it as

$$\text{Hol}(G) = \widehat{G} \text{Aut}(G) \leq \text{Sym}(G).$$

As an abstract semidirect product,

$$\text{Hol}(G) = \{(g, \alpha) \mid g \in G, \alpha \in \text{Aut}(G)\},$$

with multiplication

$$(g, \alpha)(h, \beta) = (gh^{\alpha^{-1}}, \alpha\beta).$$

Lemma 9.2.2 (Normalizer of a regular subgroup). *Let $M \leq \text{Sym}(\Omega)$ be regular. After identifying Ω with M , we have*

$$N_{\text{Sym}(\Omega)}(M) = M : \text{Aut}(M) = \text{Hol}(M).$$

Proof. Identify Ω with M , so that $M = \widehat{M}$ acts by right multiplication.

Let $G = N_{\text{Sym}(M)}(\widehat{M})$, and let $1 \in M$ be the identity element. Since $\widehat{M}$ is regular, we have

$$G = \widehat{M}G_1,$$

where G_1 is the stabilizer of 1.

We claim that

$$G_1 \leq \text{Aut}(M).$$

Let $b \in G_1$. Since b normalizes $\widehat{M}$, for every $m \in M$ there exists $m' \in M$ such that

$$b^{-1}\widehat{m}b = \widehat{m'}.$$

Evaluating both sides at 1, we get

$$m' = m^b.$$

Hence

$$b^{-1}\widehat{m}b = \widehat{m^b}.$$

Therefore, for $m, n \in M$,

$$(mn)^b = \left(1^{\widehat{m}\widehat{n}}\right)^b = 1^{b^{-1}\widehat{m}\widehat{n}b} = 1^{(b^{-1}\widehat{m}b)(b^{-1}\widehat{n}b)} = 1^{\widehat{m^b}\widehat{n^b}} = m^b n^b.$$

Thus b is an automorphism of M .

Conversely, every automorphism of M normalizes the right regular subgroup $\widehat{M}$. Hence

$$N_{\text{Sym}(M)}(\widehat{M}) = \widehat{M} : \text{Aut}(M).$$

□

9.3 Holomorph Simple Type

Theorem 9.3.1 (Holomorph simple type). *Let $G \leq \text{Sym}(\Omega)$. Suppose*

$$M \times N \trianglelefteq G$$

and suppose that M and N are regular on Ω , with

$$M \cong N \cong T,$$

where T is a nonabelian simple group.

After identifying Ω with M , we have

$$M : \text{Inn}(M) \leq G \leq M : \text{Aut}(M).$$

Proof. Since M is regular, we identify Ω with M , so that

$$M = \widehat{M}.$$

Since $M \trianglelefteq G$, the group G normalizes M . Hence, by the normalizer lemma,

$$G \leq N_{\text{Sym}(M)}(\widehat{M}) = M : \text{Aut}(M).$$

It remains to identify the lower bound. Since $M \times N \trianglelefteq G$, the subgroups M and N commute. Thus

$$N \leq C_{\text{Sym}(M)}(\widehat{M}).$$

By the centralizer lemma,

$$C_{\text{Sym}(M)}(\widehat{M}) = \check{M}.$$

Since N and $\check{M}$ are both regular and have the same order, we get

$$N = \check{M}.$$

Now $M \cong T$ is centerless. Hence

$$\widehat{M} \cap \check{M} = 1.$$

The point stabilizer of $1 \in M$ in $\widehat{M}\check{M}$ consists of the elements

$$\widehat{m}\check{m} \quad (m \in M),$$

because

$$1^{\widehat{m}\check{m}} = m^{-1}m = 1.$$

Moreover,

$$x^{\widehat{m}\check{m}} = m^{-1}xm,$$

so this point stabilizer is naturally identified with

$$\text{Inn}(M).$$

Therefore

$$\widehat{M}\check{M} = M : \text{Inn}(M) \leq G.$$

Thus

$$M : \text{Inn}(M) \leq G \leq M : \text{Aut}(M).$$

□

9.4 Holomorph Compound Type

Theorem 9.4.1 (Holomorph compound type). *Let $G \leq \text{Sym}(\Omega)$. Suppose*

$$M \times N \trianglelefteq G,$$

where M and N are regular on Ω , and

$$M \cong N \cong T^\ell$$

with $\ell \geq 2$ and T a nonabelian simple group.

After identifying Ω with M , we have

$$M : \text{Inn}(M) \leq G \leq M : \text{Aut}(M).$$

Moreover,

$$\text{Aut}(M) \cong \text{Aut}(T) \wr S_\ell.$$

Thus

$$M : \text{Inn}(M) \leq G \leq M : (\text{Aut}(T) \wr S_\ell).$$

Proof. The proof is the same as in the holomorph simple case.

Identify Ω with M , so $M = \widehat{M}$. Since $M \trianglelefteq G$,

$$G \leq N_{\text{Sym}(M)}(\widehat{M}) = M : \text{Aut}(M).$$

Since M and N commute and N is regular,

$$N \leq C_{\text{Sym}(M)}(\widehat{M}) = \check{M}.$$

Hence $N = \check{M}$. Since T^ℓ is centerless, we have

$$\widehat{M} \cap \check{M} = 1.$$

The point stabilizer of $1 \in M$ in $\widehat{M}\check{M}$ is identified with $\text{Inn}(M)$. Therefore

$$\widehat{M}\check{M} = M : \text{Inn}(M) \leq G.$$

Finally, since

$$M = T_1 \times \cdots \times T_\ell \quad (T_i \cong T),$$

the automorphism group of M permutes the simple direct factors and acts on each factor by an automorphism of T . Hence

$$\text{Aut}(M) \cong \text{Aut}(T) \wr S_\ell.$$

□

Remark 9.4.2. *In the quasiprimitive holomorph compound case, the group G induces a transitive action on the set of simple direct factors*

$$\{T_1, \dots, T_\ell\}.$$

Indeed, if the induced action had more than one orbit, then the product of the factors in one orbit would give a nontrivial proper normal subgroup of G inside M , contradicting the minimal normality of M .

9.5 The Frattini Argument for Permutation Groups

Lemma 9.5.1 (Frattini argument for permutation groups). *Let $H \leq G \leq \text{Sym}(\Omega)$, and assume that G is transitive on Ω . Fix $\omega \in \Omega$. Then H is transitive on Ω if and only if*

$$G = G_\omega H.$$

In particular, if $H \trianglelefteq G$, then this is equivalent to

$$G = HG_\omega.$$

Proof. Suppose first that H is transitive. Let $g \in G$. Since H is transitive, there exists $h \in H$ such that

$$\omega^h = \omega^g.$$

Then

$$\omega^{gh^{-1}} = (\omega^g)^{h^{-1}} = (\omega^h)^{h^{-1}} = \omega.$$

Hence $gh^{-1} \in G_\omega$, and therefore

$$g \in G_\omega H.$$

Thus $G = G_\omega H$.

Conversely, suppose $G = G_\omega H$. For every $g \in G$, write

$$g = kh \quad (k \in G_\omega, h \in H).$$

Then

$$\omega^g = \omega^{kh} = (\omega^k)^h = \omega^h.$$

Thus every point in the G -orbit of ω lies in the H -orbit of ω . Since G is transitive, H is transitive. $\square$

Lemma 9.5.2. *Let $M \trianglelefteq G \leq \text{Sym}(\Omega)$, and suppose that M is regular on Ω . Then*

$$G = M : G_\omega$$

for every $\omega \in \Omega$.

Proof. Since M is transitive on Ω , the Frattini argument gives

$$G = G_\omega M.$$

Since $M \trianglelefteq G$, we also have

$$G = MG_\omega.$$

Moreover,

$$M \cap G_\omega = 1$$

because M is regular. Hence

$$G = M : G_\omega.$$

$\square$

9.6 Blocks and Primitive Actions

Definition 9.6.1 (Block). *Let G be a transitive permutation group on Ω . A subset $B \subseteq \Omega$ is called a **block** for G if for every $g \in G$,*

$$B^g = B \quad \text{or} \quad B^g \cap B = \emptyset.$$

Definition 9.6.2 (Block system). *Let B be a block for G . The set*

$$\mathcal{B} = \{B^g \mid g \in G\}$$

*is called the **block system** generated by B .*

Proposition 9.6.3 (Basic properties of block systems). *Let G be transitive on Ω , and let*

$$\mathcal{B} = \{B_1, B_2, \dots, B_m\}$$

be a block system. Then:

- (1) *each B_i is a block;*
- (2) *$B_i \cap B_j = \emptyset$ whenever $i \neq j$;*
- (3) *$|B_1| = \dots = |B_m|$;*
- (4) *G induces a transitive action on $\mathcal{B}$.*

Proof. Each B_i has the form B^g for some $g \in G$, and conjugating the block condition shows that B^g is again a block. Hence each B_i is a block.

If $B_i \cap B_j \neq \emptyset$, then by the block condition they must be equal. Thus distinct blocks in $\mathcal{B}$ are disjoint.

All blocks in $\mathcal{B}$ are images of B under permutations, so they have the same cardinality.

Finally, the action of G on $\mathcal{B}$ is transitive by construction, since every block in $\mathcal{B}$ is of the form B^g . $\square$

Definition 9.6.4 (Primitive and imprimitive actions). *Let G be transitive on Ω . If there exists a block $B \subseteq \Omega$ such that*

$$1 < |B| < |\Omega|,$$

*then G is called **imprimitive** on Ω .*

*If no such block exists, then G is called **primitive** on Ω .*

Lemma 9.6.5 (Primitive action and maximal point stabilizer). *Let $G \leq \text{Sym}(\Omega)$ be transitive, and fix $\omega \in \Omega$. Then G is primitive on Ω if and only if G_ω is a maximal subgroup of G .*

Proof. Suppose first that G is imprimitive. Let B be a nontrivial block with

$$\omega \in B.$$

Let

$$G_B = \{g \in G \mid B^g = B\}$$

be the setwise stabilizer of B . Then

$$G_\omega \leq G_B.$$

Since B is nontrivial, choose $\omega' \in B$ with $\omega' \neq \omega$. By transitivity, there exists $g \in G$ such

that

$$\omega^g = \omega'.$$

Then $B^g \cap B \neq \emptyset$, so $B^g = B$. Hence

$$g \in G_B.$$

But $g \notin G_\omega$, so

$$G_\omega < G_B.$$

Also $G_B < G$, since otherwise B would be G -invariant, and transitivity would force $B = \Omega$. Therefore G_ω is not maximal.

Conversely, suppose that G_ω is not maximal. Then there exists a subgroup H such that

$$G_\omega < H < G.$$

Define

$$B = \omega^H.$$

We claim that B is a block. Let $g \in G$, and suppose that

$$B^g \cap B \neq \emptyset.$$

Then there exist $h_1, h_2 \in H$ such that

$$\omega^{h_1 g} = \omega^{h_2}.$$

Thus

$$\omega^{h_1 g h_2^{-1}} = \omega,$$

so

$$h_1 g h_2^{-1} \in G_\omega \leq H.$$

Hence $g \in H$, and therefore

$$B^g = B.$$

Thus B is a block.

Since $H > G_\omega$, the block B has more than one point. Since $H < G$, the subgroup H is not transitive; otherwise the Frattini argument would imply $G = G_\omega H = H$, a contradiction. Hence $B \neq \Omega$. Therefore B is a nontrivial block, and G is imprimitive. $\square$

9.7 Quasiprimitive Groups

Definition 9.7.1 (Quasiprimitive permutation group). *Let $G \leq \text{Sym}(\Omega)$ be transitive. We say that G is **quasiprimitive** on Ω if every nontrivial normal subgroup of G is transitive on Ω .*

Lemma 9.7.2 (Blocks for groups with a regular normal subgroup). *Let $G = M : G_\omega \leq \text{Sym}(\Omega)$, where $M \trianglelefteq G$ is regular on Ω . Identify Ω with M , and identify ω with the identity element $1 \in M$.*

Then the blocks of G containing 1 are exactly the subgroups $K \leq M$ that are invariant under G_ω .

Proof. Let B be a block containing 1. Since M is regular, we may regard B as a subset of M .

We first show that B is a subgroup of M . Let $a, b \in B$. Since $b \in B$, the translate

$$\widehat{B^{b^{-1}}}$$

contains 1. Hence

$$\widehat{B^{b^{-1}}} \cap B \neq \emptyset.$$

Because B is a block, we have

$$\widehat{B^{b^{-1}}} = B.$$

Therefore

$$ab^{-1} \in B.$$

By the subgroup criterion, $B \leq M$.

Next let $x \in G_\omega$. Since x fixes 1, the block B^x also contains 1. Hence

$$B^x \cap B \neq \emptyset.$$

Thus

$$B^x = B.$$

So B is G_ω -invariant.

Conversely, suppose $K \leq M$ is G_ω -invariant. Let $g \in G$. Since

$$G = M : G_\omega,$$

we may write

$$g = \widehat{m}x \quad (m \in M, x \in G_\omega).$$

Then

$$K^g = K^{\widehat{m}x} = (Km)^x = K^x m^x = Km^x,$$

because $K^x = K$. Hence K^g is a right coset of K in M . Two right cosets of K are either equal or disjoint, so

$$K^g = K \quad \text{or} \quad K^g \cap K = \emptyset.$$

Thus K is a block. □

Theorem 9.7.3. *A quasiprimitive permutation group of type HA, HS, or HC is primitive.*

Proof. In each of the types HA, HS, and HC, the group G has a regular minimal normal subgroup M . Hence

$$G = M : G_\omega.$$

By the previous lemma, if G were imprimitive, then there would exist a nontrivial proper subgroup

$$1 < K < M$$

that is invariant under G_ω .

We show that this is impossible in each case.

Case HA. Here M is abelian and is the unique minimal normal subgroup of G . Since $K \leq M$ and M is abelian, M normalizes K . Since K is G_ω -invariant, G_ω also normalizes K . Thus

$$K \trianglelefteq M : G_\omega = G.$$

This contradicts the minimal normality of M , because

$$1 < K < M.$$

Therefore no such K exists, and G is primitive.

Case HS. Here

$$M \cong T$$

for a nonabelian simple group T , and from the holomorph simple structure,

$$\text{Inn}(M) \leq G_\omega.$$

If $K \leq M$ is G_ω -invariant, then in particular K is invariant under $\text{Inn}(M)$. Hence

$$K \trianglelefteq M.$$

Since M is simple, we get

$$K = 1 \quad \text{or} \quad K = M.$$

Thus no nontrivial proper K exists, and G is primitive.

Case HC. Here

$$M = T_1 \times \cdots \times T_\ell, \quad T_i \cong T,$$

where T is nonabelian simple and $\ell \geq 2$. Again

$$\text{Inn}(M) \leq G_\omega.$$

Thus a G_ω -invariant subgroup $K \leq M$ is normalized by $\text{Inn}(M)$, hence is normal in M .

Therefore K is a product of some of the simple direct factors T_i .

But in the holomorph compound case, G_ω acts transitively on the set

$$\{T_1, \dots, T_\ell\}.$$

Therefore the set of direct factors appearing in K is either empty or all of $\{T_1, \dots, T_\ell\}$. Hence

$$K = 1 \quad \text{or} \quad K = M.$$

So no nontrivial proper G_ω -invariant subgroup K exists. Therefore G is primitive. $\square$

Proposition 9.7.4. *Every primitive permutation group is quasiprimitive.*

Proof. Let $G \leq \text{Sym}(\Omega)$ be primitive, and let

$$1 \neq H \trianglelefteq G.$$

We prove that H is transitive.

Fix $\omega \in \Omega$. If

$$H \leq G_\omega,$$

then for every $g \in G$,

$$H^g = H \leq G_{\omega^g}.$$

Since G is transitive, the points ω^g run over all of Ω . Hence H fixes every point of Ω . Since $G \leq \text{Sym}(\Omega)$ is faithful, this forces

$$H = 1,$$

contradicting $H \neq 1$.

Therefore

$$H \not\leq G_\omega.$$

Thus

$$G_\omega < HG_\omega \leq G.$$

Since G is primitive, G_ω is maximal in G . Hence

$$HG_\omega = G.$$

By the Frattini argument, H is transitive on Ω . Therefore every nontrivial normal subgroup of G is transitive, so G is quasiprimitive. $\square$

Example 9.7.5 (Quasiprimitive but not primitive). *Let T be a nonabelian simple group. Let $\hat{T}$ act regularly on the set T .*

Then $\hat{T}$ is quasiprimitive on T . Indeed, since T is simple, the only nontrivial normal subgroup of $\hat{T}$ is $\hat{T}$ itself, and $\hat{T}$ is transitive.

However, $\widehat{T}$ is not primitive. The stabilizer of the identity element $1 \in T$ is

$$\widehat{T}_1 = 1.$$

Since T is nonabelian simple, T has a nontrivial proper subgroup. Hence 1 is not a maximal subgroup of $\widehat{T}$. By the maximal stabilizer criterion, the regular action of $\widehat{T}$ on T is not primitive.

Chapter 10

Permutation Actions and Coset Actions

10.1 Basic Notation for Permutation Actions

Definition 10.1.1 (Orbits and stabilizers). *Let $G \leq \text{Sym}(\Omega)$, where $|\Omega| = n$.*

(1) *If $\Delta \subseteq \Omega$ and $g \in G$, define*

$$\Delta^g = \{\delta^g \mid \delta \in \Delta\}.$$

(2) *For $\omega_1, \omega_2 \in \Omega$, write*

$$\omega_1 \sim_G \omega_2$$

if there exists $g \in G$ such that $\omega_1^g = \omega_2$.

(3) *The orbit of $\omega \in \Omega$ under G is*

$$\omega^G = \{\omega^g \mid g \in G\}.$$

(4) *The stabilizer of $\omega \in \Omega$ is*

$$G_\omega = \{g \in G \mid \omega^g = \omega\}.$$

10.2 Setwise and Pointwise Stabilizers

Definition 10.2.1 (Stabilizers of subsets). *Let $\Delta \subseteq \Omega$. Define the setwise stabilizer of Δ in G by*

$$G_\Delta = \{g \in G \mid \Delta^g = \Delta\}.$$

Define the pointwise stabilizer of Δ in G by

$$G_{(\Delta)} = \{g \in G \mid \delta^g = \delta \text{ for all } \delta \in \Delta\}.$$

Proposition 10.2.2. *Let $G \leq \text{Sym}(\Omega)$ and let $\Delta \subseteq \Omega$. Then*

$$G_{(\Delta)} \trianglelefteq G_{\Delta}.$$

Moreover, G_{Δ} acts on Δ , and the kernel of this induced action is $G_{(\Delta)}$. Hence

$$G_{\Delta}^{\Delta} \cong G_{\Delta}/G_{(\Delta)},$$

where G_{Δ}^{Δ} denotes the permutation group induced by G_{Δ} on Δ .

Proof. First, $G_{(\Delta)} \leq G_{\Delta}$, because every element fixing each point of Δ certainly stabilizes Δ as a set.

Let $x \in G_{\Delta}$ and $h \in G_{(\Delta)}$. We show that $x^{-1}hx \in G_{(\Delta)}$. For any $\delta \in \Delta$, since $x \in G_{\Delta}$, we have $\delta^{x^{-1}} \in \Delta$. Since h fixes every point of Δ ,

$$(\delta^{x^{-1}})^h = \delta^{x^{-1}}.$$

Therefore

$$\delta^{x^{-1}hx} = ((\delta^{x^{-1}})^h)^x = (\delta^{x^{-1}})^x = \delta.$$

Thus $x^{-1}hx \in G_{(\Delta)}$, proving that

$$G_{(\Delta)} \trianglelefteq G_{\Delta}.$$

Now consider the restriction homomorphism

$$\rho : G_{\Delta} \longrightarrow \text{Sym}(\Delta), \quad g \longmapsto g|_{\Delta}.$$

Its kernel consists precisely of the elements of G_{Δ} that fix every point of Δ , namely $G_{(\Delta)}$. Its image is G_{Δ}^{Δ} . By the First Isomorphism Theorem,

$$G_{\Delta}/G_{(\Delta)} \cong G_{\Delta}^{\Delta}.$$

□

Proposition 10.2.3 (Conjugating stabilizers). *Let $G \leq \text{Sym}(\Omega)$.*

(1) *For every $\omega \in \Omega$ and $g \in G$,*

$$g^{-1}G_{\omega}g = G_{\omega^g}.$$

(2) *For every $\Delta \subseteq \Omega$ and $g \in G$,*

$$g^{-1}G_{\Delta}g = G_{\Delta^g}.$$

(3) For every $\Delta \subseteq \Omega$ and $g \in G$,

$$g^{-1}G_{(\Delta)}g = G_{(\Delta^g)}.$$

Proof. We prove the first statement. Let $h \in G_\omega$. Then

$$(\omega^g)^{g^{-1}hg} = \omega^{hg} = (\omega^h)^g = \omega^g.$$

Thus $g^{-1}hg \in G_{\omega^g}$, so

$$g^{-1}G_\omega g \leq G_{\omega^g}.$$

Applying the same argument with g^{-1} gives the reverse inclusion. Hence

$$g^{-1}G_\omega g = G_{\omega^g}.$$

The proofs for G_Δ and $G_{(\Delta)}$ are identical, replacing the point ω by the subset Δ . $\square$

10.3 Fixed Points and Semiregular Actions

Definition 10.3.1 (Fixed points). *Let G act on Ω .*

(1) For $g \in G$, define

$$\text{Fix}_\Omega(g) = \{\omega \in \Omega \mid \omega^g = \omega\}.$$

(2) The set of common fixed points of G on Ω is

$$\text{Fix}_\Omega(G) = \{\omega \in \Omega \mid \omega^g = \omega \text{ for all } g \in G\}.$$

Definition 10.3.2 (Semiregular and regular actions). *Let G act on Ω .*

(1) The action is called semiregular if

$$G_\omega = 1$$

for every $\omega \in \Omega$.

(2) The action is called regular if it is both transitive and semiregular.

Proposition 10.3.3. *Let G be finite and act semiregularly on Ω . Then every orbit has length $|G|$. In particular, if Ω is finite, then*

$$|G| \mid |\Omega|.$$

If the action is transitive and semiregular, then it is regular and

$$|G| = |\Omega|.$$

Proof. Since the action is semiregular, $G_\omega = 1$ for every $\omega \in \Omega$. By the orbit–stabilizer theorem,

$$|\omega^G| = |G : G_\omega| = |G|.$$

Thus every orbit has length $|G|$. If Ω is finite, it is a disjoint union of orbits, so $|G| \mid |\Omega|$.

If the action is transitive, there is only one orbit. Hence

$$|\Omega| = |\omega^G| = |G|.$$

□

10.4 Coset Actions and the Core

Definition 10.4.1 (Right coset action). *Let G be a group and let $H \leq G$. Let*

$$[G : H]_r := \{Hg \mid g \in G\}$$

be the set of right cosets of H in G . Define an action of G on $[G : H]_r$ by

$$(Hg)^x = Hgx$$

for all $g, x \in G$.

Proposition 10.4.2. *The right coset action of G on $[G : H]_r$ is transitive.*

Proof. For every coset $Hg \in [G : H]_r$,

$$H^g = Hg.$$

Thus every coset lies in the orbit of H , so the action is transitive. □

Definition 10.4.3 (Core). *Let $H \leq G$. The core of H in G is*

$$\text{Core}_G(H) = \bigcap_{g \in G} g^{-1}Hg.$$

Proposition 10.4.4. *Let G act on the set $[G : H]_r$ by right multiplication. Then the kernel of this action is*

$$\text{Core}_G(H) = \bigcap_{g \in G} g^{-1}Hg.$$

Moreover, $\text{Core}_G(H)$ is the largest normal subgroup of G contained in H .

Proof. The stabilizer of the coset Hg is

$$G_{Hg} = \{x \in G \mid Hgx = Hg\}.$$

Now

$$Hgx = Hg \iff gxg^{-1} \in H \iff x \in g^{-1}Hg.$$

Thus

$$G_{Hg} = g^{-1}Hg.$$

The kernel of the action is the intersection of all point stabilizers, so

$$\text{Ker}(G \curvearrowright [G : H]_r) = \bigcap_{g \in G} G_{Hg} = \bigcap_{g \in G} g^{-1}Hg = \text{Core}_G(H).$$

Since the kernel of an action is normal in G , $\text{Core}_G(H) \trianglelefteq G$. Also, taking $g = 1$, we see that

$$\text{Core}_G(H) \leq H.$$

Finally, suppose $N \trianglelefteq G$ and $N \leq H$. Then for every $g \in G$,

$$N = g^{-1}Ng \leq g^{-1}Hg.$$

Therefore

$$N \leq \bigcap_{g \in G} g^{-1}Hg = \text{Core}_G(H).$$

Thus $\text{Core}_G(H)$ is the largest normal subgroup of G contained in H . □

Proposition 10.4.5. *Let G act on*

$$\Omega = [G : H]_r$$

by right multiplication. Then

$$G^\Omega \cong G / \text{Core}_G(H).$$

In particular, G^Ω is a transitive permutation group of degree $|G : H|$.

Proof. The action gives a homomorphism

$$\rho : G \longrightarrow \text{Sym}(\Omega).$$

Its image is G^Ω , and its kernel is $\text{Core}_G(H)$. Therefore, by the First Isomorphism Theorem,

$$G / \text{Core}_G(H) \cong G^\Omega.$$

The action is transitive by the previous proposition, and

$$|\Omega| = |G : H|.$$

□

Definition 10.4.6 (Faithful action and core-free subgroup). *An action of G on Ω is called faithful if its kernel is trivial.*

A subgroup $H \leq G$ is called core-free in G if

$$\text{Core}_G(H) = 1.$$

Thus the right coset action of G on $[G : H]_r$ is faithful if and only if H is core-free in G .

10.5 Transitive Actions as Coset Actions

Definition 10.5.1 (Equivalence of actions). *Let G act on two sets Ω and Σ . The two actions are called equivalent if there exists a bijection*

$$\varphi : \Omega \longrightarrow \Sigma$$

such that

$$(\omega^g)\varphi = (\omega\varphi)^g$$

for every $\omega \in \Omega$ and $g \in G$.

Equivalently, the following diagram commutes for every $g \in G$:

$$\begin{array}{ccc} \omega & \xrightarrow{\varphi} & \omega\varphi \\ g \downarrow & & \downarrow g \\ \omega^g & \xrightarrow{\varphi} & (\omega\varphi)^g. \end{array}$$

Theorem 10.5.2 (Every transitive action is a coset action). *Let $G \leq \text{Sym}(\Omega)$ be transitive. Fix $\omega \in \Omega$, and put*

$$H = G_\omega.$$

Then the action of G on Ω is equivalent to the right coset action of G on $[G : H]_r$. Moreover, H is core-free in G , so this coset action is faithful.

Proof. Let

$$\Sigma = [G : H]_r.$$

Define

$$\varphi : \Omega \longrightarrow \Sigma$$

by

$$\varphi(\omega^g) = Hg.$$

This is well-defined: if

$$\omega^g = \omega^h,$$

then

$$\omega^{gh^{-1}} = \omega,$$

so $gh^{-1} \in G_\omega = H$, and hence

$$Hg = Hh.$$

The map is surjective because every coset has the form Hg , and it is injective by the same argument in reverse. Thus φ is a bijection.

For $x \in G$,

$$\varphi((\omega^g)^x) = \varphi(\omega^{gx}) = Hgx = (Hg)^x = \varphi(\omega^g)^x.$$

Thus φ is an equivalence of actions.

It remains to show that H is core-free. Suppose $N \trianglelefteq G$ and $N \leq H$. For every $x \in G$,

$$N = N^x \leq H^x = G_{\omega^x}.$$

Since G is transitive on Ω , every point of Ω is of the form ω^x . Hence every element of N fixes every point of Ω . Since $G \leq \text{Sym}(\Omega)$, the action is faithful, so $N = 1$. Therefore

$$\text{Core}_G(H) = 1.$$

□

Example 10.5.3 (Coset actions of S_4). *Let*

$$G = S_4.$$

Consider the following subgroups:

$$H_1 = \langle (12) \rangle, \quad H_2 = \langle (12)(34) \rangle, \quad H_3 = \langle (13) \rangle, \quad H_4 = \langle (13)(24) \rangle.$$

For each i , let

$$\Omega_i = [G : H_i]_{\mathbf{r}}.$$

We compare the corresponding coset actions G^{Ω_i} .

First, G^{Ω_1} and G^{Ω_2} are not equivalent.

Suppose, for contradiction, that there exists an equivalence

$$\varphi : \Omega_1 \longrightarrow \Omega_2.$$

Write

$$H_1\varphi = H_2g$$

for some $g \in G$. Since $(12) \in H_1$, we have

$$H_1(12) = H_1.$$

Using equivariance,

$$H_2g = H_1\varphi = (H_1(12))\varphi = (H_1\varphi)^{(12)} = (H_2g)^{(12)} = H_2g(12).$$

Thus

$$H_2g = H_2g(12),$$

so

$$g(12)g^{-1} \in H_2.$$

But $g(12)g^{-1}$ is a transposition, while

$$H_2 = \{1, (12)(34)\}$$

contains no transposition. This contradiction shows that G^{Ω_1} and G^{Ω_2} are not equivalent.

Second, G^{Ω_1} and G^{Ω_3} are equivalent.

Indeed,

$$H_3^{(23)} = (23)^{-1}H_3(23) = \langle(12)\rangle = H_1.$$

Define

$$\varphi : \Omega_1 \longrightarrow \Omega_3$$

by

$$\varphi(H_1g) = H_3(23)g.$$

We check that φ is well-defined. If

$$H_1g = H_1h,$$

then $gh^{-1} \in H_1$. Since

$$(23)H_1(23) = H_3,$$

we get

$$(23)gh^{-1}(23) \in H_3.$$

Hence

$$H_3(23)g = H_3(23)h.$$

Thus φ is well-defined. It is plainly bijective, and for $x \in G$,

$$\varphi((H_1g)^x) = \varphi(H_1gx) = H_3(23)gx = (H_3(23)g)^x = \varphi(H_1g)^x.$$

Therefore the two actions are equivalent.

Similarly, G^{Ω_2} and G^{Ω_4} are equivalent, since H_2 and H_4 are conjugate in S_4 . However, G^{Ω_1} is not equivalent to G^{Ω_2} or G^{Ω_4} , because a subgroup generated by a transposition cannot be conjugate to a subgroup generated by a double transposition.

Theorem 10.5.4 (Criterion for equivalence of transitive actions). *Let G act transitively on two sets Ω and Σ . Fix $\omega \in \Omega$ and $\sigma \in \Sigma$. Then the two actions are equivalent if and only if the stabilizers G_ω and G_σ are conjugate in G .*

Proof. Suppose first that the two actions are equivalent. Let

$$\varphi : \Omega \longrightarrow \Sigma$$

be an equivalence. Since G is transitive on Σ , there exists $g \in G$ such that

$$\omega\varphi = \sigma^g.$$

For $x \in G_\omega$, we have $\omega^x = \omega$. Hence

$$(\omega\varphi)^x = (\omega^x)\varphi = \omega\varphi.$$

Thus $x \in G_{\omega\varphi}$, so

$$G_\omega \leq G_{\omega\varphi}.$$

Applying the same argument to φ^{-1} , we get equality:

$$G_\omega = G_{\omega\varphi}.$$

Since $\omega\varphi = \sigma^g$, the conjugation formula for stabilizers gives

$$G_{\omega\varphi} = G_{\sigma^g} = g^{-1}G_\sigma g.$$

Therefore

$$G_\omega = g^{-1}G_\sigma g,$$

so G_ω and G_σ are conjugate in G .

Conversely, suppose that G_ω and G_σ are conjugate. Choose $g \in G$ such that

$$G_\omega = g^{-1}G_\sigma g.$$

Define

$$\varphi : \Omega \longrightarrow \Sigma$$

by

$$\varphi(\omega^x) = \sigma^{gx}.$$

We check that this is well-defined. If

$$\omega^x = \omega^y,$$

then

$$xy^{-1} \in G_\omega = g^{-1}G_\sigma g.$$

Thus

$$gxy^{-1}g^{-1} \in G_\sigma,$$

which implies

$$\sigma^{gx} = \sigma^{gy}.$$

Hence φ is well-defined.

It is surjective because every element of Σ has the form σ^z , and

$$\sigma^z = \sigma^{g(g^{-1}z)} = \varphi(\omega^{g^{-1}z}).$$

It is injective by reversing the well-definedness argument.

Finally, for any $a \in G$,

$$\varphi((\omega^x)^a) = \varphi(\omega^{xa}) = \sigma^{gxa} = (\sigma^{gx})^a = \varphi(\omega^x)^a.$$

Thus φ is an equivalence of actions. □

Corollary 10.5.5. *The equivalence classes of transitive G -actions are in bijection with conjugacy classes of subgroups of G .*

Proof. Every transitive action is equivalent to the right coset action on $[G : H]_r$, where H is a point stabilizer. By the previous theorem, two such coset actions are equivalent exactly when their stabilizers are conjugate in G . □

Chapter 11

Quasiprimitive Permutation Groups and Small Rank

11.1 The Exercise

Definition 11.1.1 (*n*-transitive permutation group). Let $G \leq \text{Sym}(\Omega)$, and let n be a positive integer with $n \leq |\Omega|$. We say that G is *n-transitive* on Ω if G acts transitively on the set of ordered n -tuples of pairwise distinct points:

$$\Omega^{(n)} = \{(\omega_1, \dots, \omega_n) \in \Omega^n \mid \omega_i \neq \omega_j \text{ whenever } i \neq j\}.$$

Equivalently, for any two ordered n -tuples of pairwise distinct points

$$(\omega_1, \dots, \omega_n), \quad (\omega'_1, \dots, \omega'_n) \in \Omega^{(n)},$$

there exists $g \in G$ such that

$$\omega_i^g = \omega'_i \quad \text{for all } 1 \leq i \leq n.$$

Exercise 11.1.2. Let $G \leq \text{Sym}(\Omega)$ be a quasiprimitive permutation group. Characterize the cases where the point stabilizer G_ω has exactly

$$r = 2, 3, 4, 5$$

orbits on Ω . Equivalently, characterize quasiprimitive permutation groups of rank r , for $r = 2, 3, 4, 5$.

In particular, determine the best possible lower bound for r when G is of holomorph simple type.

Proof. Assume throughout that G is finite and transitive on Ω , and write

$$r = \text{rank}(G) = \#\{G_\omega\text{-orbits on } \Omega\}.$$

We use the quasiprimitive O’Nan–Scott types

$$\text{HA, AS, HS, HC, SD, CD, TW, PA.}$$

The answer, at the level of O’Nan–Scott type, is as follows:

r	Possibilities
2	HA or AS; equivalently, G is 2-transitive.
3	HA or AS of rank 3, or PA on Δ^2 with 2-transitive component.
4	HA or AS of rank 4, or HS of rank 4, or PA on Δ^3 with 2-transitive component.
5	HA or AS of rank 5, or HS of rank 5, or PA on Δ^4 with 2-transitive component and set-transitive top group.

More explicitly, in the product action case

$$\Omega = \Delta^k,$$

the rank is $k+1$ precisely when the component group is 2-transitive on Δ and the top group is transitive on j -subsets of $\{1, \dots, k\}$ for each $0 \leq j \leq k$. In that case the G_ω -orbits are exactly the weight classes

$$\Omega_j = \{(\delta_1, \dots, \delta_k) \in \Delta^k \mid |\{i : \delta_i \neq \delta\}| = j\}, \quad 0 \leq j \leq k,$$

where $\omega = (\delta, \dots, \delta)$.

We now justify the table.

First suppose $r = 2$. Then G_ω has exactly two orbits on Ω , namely

$$\{\omega\} \quad \text{and} \quad \Omega \setminus \{\omega\}.$$

Thus G_ω is transitive on $\Omega \setminus \{\omega\}$, so G is 2-transitive. Conversely, every 2-transitive group has rank 2. By Burnside’s theorem, a finite 2-transitive permutation group is either of affine type or almost simple type. Hence the rank 2 quasiprimitive groups are precisely the 2-transitive groups of type HA or AS.

Now let N be a minimal normal subgroup of G . Since G is quasiprimitive, every non-trivial normal subgroup is transitive; in particular, N is transitive. By the quasiprimitive O’Nan–Scott theorem, G is one of the types listed above. We show which types can occur when $r \leq 5$.

Consider first a regular nonabelian normal subgroup

$$N \cong T^m,$$

where T is nonabelian simple. Identifying Ω with N , and taking $\omega = 1$, we have

$$G_\omega \leq \text{Aut}(N).$$

Hence G_ω -orbits preserve element orders.

Since T is nonabelian simple, $|T|$ has at least three distinct prime divisors. Indeed, if $|T|$ involved at most two primes, then T would be solvable by Burnside's $p^a q^b$ -theorem, impossible. Choose elements

$$x, y, z \in T$$

of distinct prime orders p, q, s , respectively.

If $m = 1$, then the elements

$$1, \quad x, \quad y, \quad z$$

lie in four distinct G_ω -orbits. Therefore

$$r \geq 4.$$

This is the holomorph simple case. Thus a group of holomorph simple type cannot have rank 2 or 3.

If $m \geq 2$, then in T^m the elements

$$1, \quad (x, 1, \dots, 1), \quad (y, 1, \dots, 1), \quad (z, 1, \dots, 1), \quad (x, y, 1, \dots, 1), \quad (x, z, 1, \dots, 1)$$

have pairwise distinct orders

$$1, \quad p, \quad q, \quad s, \quad pq, \quad ps.$$

Hence $r \geq 6$. Thus regular compound nonabelian types, such as HC and TW, do not occur for $r = 2, 3, 4, 5$.

Next consider diagonal type. In simple diagonal type with

$$N = T^k, \quad k \geq 3,$$

points may be represented by tuples

$$(a_1, \dots, a_k) \in T^k$$

modulo diagonal multiplication. The point stabilizer preserves the multiset of orders

$$\left\{ |a_i a_j^{-1}| \mid 1 \leq i < j \leq k \right\}.$$

Choose elements of two distinct odd prime orders q and s in T . Then the following six points give six distinct G_ω -orbits:

$$(1, \dots, 1), \\ (1, x, 1, \dots, 1), \quad (1, y, 1, \dots, 1), \quad (1, z, 1, \dots, 1),$$

where x, y, z have distinct prime orders, and

$$(1, y, y^{-1}, 1, \dots, 1), \quad (1, z, z^{-1}, 1, \dots, 1).$$

Indeed, the corresponding multisets of pairwise quotient orders are distinct. Thus diagonal type with $k \geq 3$ has rank at least 6. The same argument excludes compound diagonal type for $r \leq 5$. The remaining diagonal case $k = 2$ is the holomorph simple situation already considered.

It remains to discuss product action type. Suppose

$$\Omega = \Delta^k, \quad \omega = (\delta, \dots, \delta),$$

and G is contained in a product action wreath product

$$L \wr P = L^k \rtimes P,$$

where L acts transitively on Δ and $P \leq S_k$ is transitive on the set of coordinates.

Define the weight of a point $\alpha = (\alpha_1, \dots, \alpha_k)$ by

$$\text{wt}(\alpha) = |\{i : \alpha_i \neq \delta\}|.$$

The point stabilizer G_ω preserves weight. Hence

$$r \geq k + 1.$$

If L is 2-transitive on Δ , then L_δ is transitive on $\Delta \setminus \{\delta\}$. If, in addition, the top group P is transitive on j -subsets of $\{1, \dots, k\}$ for every j , then G_ω is transitive on every weight class. Therefore

$$r = k + 1.$$

Conversely, suppose $r \leq 5$. If L is not 2-transitive, then L_δ has at least two nontrivial orbits, say A and B , on $\Delta \setminus \{\delta\}$. Then, already for $k \geq 2$, the following six types give distinct G_ω -invariant subsets:

$$0, \quad 1A, \quad 1B, \quad 2AA, \quad 2AB, \quad 2BB.$$

Thus $r \geq 6$, contradiction. Hence the component action must be 2-transitive.

Similarly, if the top group is not transitive on j -subsets for some j , then the weight j class splits into at least two G_ω -orbits. For $r \leq 5$ this forces the top group to be set-transitive on the relevant coordinate subsets. Consequently, in product action type and rank at most 5, we must have

$$r = k + 1.$$

Thus the product action cases are exactly:

$$k = 2 \quad (r = 3), \quad k = 3 \quad (r = 4), \quad k = 4 \quad (r = 5),$$

with 2-transitive component action, and with set-transitive top group.

Combining these exclusions with the unrestricted affine and almost simple cases gives the table above.

Finally, consider the special question about holomorph simple type. In this case

$$T \leq G \leq T \rtimes \text{Aut}(T),$$

where T is nonabelian simple and acts regularly on $\Omega \cong T$. Taking $\omega = 1$, we have

$$G_\omega \leq \text{Aut}(T).$$

Since automorphisms preserve element orders and T has elements of at least three distinct prime orders, the four elements

$$1, \quad x, \quad y, \quad z$$

lie in four distinct G_ω -orbits. Hence

$$r \geq 4.$$

Therefore, if G is of holomorph simple type, then

$$\boxed{r > 3.}$$

This bound is sharp. Take $T = A_5$ and

$$G = A_5 \rtimes \text{Aut}(A_5).$$

Since

$$\text{Aut}(A_5) \cong S_5,$$

the point stabilizer acts on A_5 by conjugation. The nonidentity elements of A_5 have orders 2, 3, 5, and S_5 is transitive on the elements of each of these orders. Thus the G_ω -orbits on $\Omega \cong A_5$ are exactly

$$\{1\}, \quad \{x \in A_5 : |x| = 2\}, \quad \{x \in A_5 : |x| = 3\}, \quad \{x \in A_5 : |x| = 5\}.$$

Hence

$$\text{rank}(G) = 4.$$

So the lower bound $r \geq 4$ is best possible. □

11.2 Rank of a Transitive Permutation Group

Definition 11.2.1 (Rank). *Let $G \leq \text{Sym}(\Omega)$ be transitive. Then G acts on*

$$\Omega \times \Omega = \{(\omega_1, \omega_2) \mid \omega_1, \omega_2 \in \Omega\}$$

by

$$(\omega_1, \omega_2)^g = (\omega_1^g, \omega_2^g).$$

The number of G -orbits on $\Omega \times \Omega$ is called the **rank** of G on Ω , and is denoted by

$$\text{rank}(G).$$

Proposition 11.2.2. *Let $G \leq \text{Sym}(\Omega)$ be transitive, and fix $\omega \in \Omega$. Then*

$$\text{rank}(G) = \# \{G_\omega\text{-orbits on } \Omega\}.$$

In particular,

$$\text{rank}(G) \geq 2$$

whenever $|\Omega| > 1$, and

$$\text{rank}(G) = 2$$

if and only if G is 2-transitive on Ω .

Proof. Every G -orbit on $\Omega \times \Omega$ meets the subset

$$\{\omega\} \times \Omega.$$

Indeed, for any pair (α, β) , choose $g \in G$ with $\alpha^g = \omega$. Then

$$(\alpha, \beta)^g = (\omega, \beta^g).$$

Two pairs (ω, β_1) and (ω, β_2) lie in the same G -orbit if and only if β_1 and β_2 lie in the same G_ω -orbit. Hence the rank of G is the number of G_ω -orbits on Ω .

The singleton $\{\omega\}$ is always a G_ω -orbit. Thus, if $|\Omega| > 1$, there is at least one further orbit, so $\text{rank}(G) \geq 2$.

Finally, $\text{rank}(G) = 2$ precisely when G_ω is transitive on $\Omega \setminus \{\omega\}$, which is equivalent to G being 2-transitive. $\square$

Remark 11.2.3. *The G_ω -orbits on Ω are often called the **suborbits** of G . Thus the rank of a transitive permutation group is the number of suborbits.*

11.3 Two-Transitive Groups

Theorem 11.3.1 (Burnside). *Let $G \leq \text{Sym}(\Omega)$ be a finite 2-transitive permutation group. Then G is either of affine type or almost simple type.*

Remark 11.3.2. *In O’Nan–Scott language, the conclusion says that a finite 2-transitive group is either of holomorph affine type, usually denoted HA , or of almost simple type, usually denoted AS .*

Proposition 11.3.3. *Let $G \leq \text{Sym}(\Omega)$ be 2-transitive, and fix $\omega \in \Omega$. Then:*

- (i) G_ω is faithful and transitive on $\Omega \setminus \{\omega\}$;
- (ii) $|\Omega| - 1$ divides $|G_\omega|$;
- (iii) if $N \triangleleft G$, then $N_\omega = N \cap G_\omega$ is half-transitive on $\Omega \setminus \{\omega\}$, i.e. all N_ω -orbits on $\Omega \setminus \{\omega\}$ have the same size.

Proof. Since G is 2-transitive, G_ω is transitive on $\Omega \setminus \{\omega\}$. If an element of G_ω fixes every point of $\Omega \setminus \{\omega\}$, then it fixes every point of Ω , and hence is the identity. Thus the induced action is faithful.

By orbit-stabilizer,

$$|\Omega| - 1 = |\omega^{G_\omega}| \cdot |G_\omega|.$$

Finally, if $N \triangleleft G$, then

$$N_\omega = N \cap G_\omega \triangleleft G_\omega.$$

Since G_ω is transitive on $\Omega \setminus \{\omega\}$, the orbits of the normal subgroup N_ω on this set all have the same size. $\square$

11.4 Normal Subgroups and Quasiprimitivity

Definition 11.4.1 (Half-transitive subgroup). *Let a group H act on a finite set Λ . We say that H is **half-transitive** on Λ if all H -orbits on Λ have the same size.*

Lemma 11.4.2. *Let $G \leq \text{Sym}(\Omega)$ be transitive, and let $N \triangleleft G$. Then the N -orbits on Ω form a G -invariant block system. In particular, all N -orbits have the same size.*

Proof. Let Δ_1, Δ_2 be two N -orbits. Choose $\delta_i \in \Delta_i$, for $i = 1, 2$. Since G is transitive, there exists $g \in G$ such that

$$\delta_1^g = \delta_2.$$

Write

$$\Delta_1 = \{\delta_1^{x_1}, \delta_1^{x_2}, \dots, \delta_1^{x_m}\}, \quad x_i \in N.$$

Then

$$(\delta_1^{x_i})^g = \delta_1^{x_i g} = \delta_1^{g(g^{-1}x_i g)} = \delta_2^{g^{-1}x_i g}.$$

Since $N \triangleleft G$, we have $g^{-1}x_i g \in N$, and therefore

$$(\delta_1^{x_i})^g \in \Delta_2.$$

Thus

$$\Delta_1^g \subseteq \Delta_2.$$

Applying the same argument to g^{-1} , we get equality:

$$\Delta_1^g = \Delta_2.$$

Hence G permutes the N -orbits transitively, so all N -orbits have the same size. $\square$

Corollary 11.4.3. *Every primitive permutation group is quasiprimitive.*

Proof. Let G be primitive, and let $1 \neq N \triangleleft G$. By the previous lemma, the N -orbits form a G -invariant block system. Since G is primitive, this block system is either trivial or consists of singleton blocks.

If the N -orbits are singleton blocks, then every element of N fixes every point of Ω , so $N = 1$, a contradiction. Hence N is transitive. $\square$

11.5 Regular Nonabelian Minimal Normal Subgroups

Definition 11.5.1 (Holomorph simple type). *Let T be a nonabelian simple group. A quasiprimitive permutation group $G \leq \text{Sym}(\Omega)$ is said to be of **holomorph simple type**, or of type HS, if G has exactly two minimal normal subgroups*

$$M \quad \text{and} \quad N,$$

such that

$$M \cong N \cong T,$$

and both M and N act regularly on Ω .

Equivalently, after identifying Ω with the underlying set of M via the regular action of M , we have

$$M : \text{Inn}(M) \leq G \leq M : \text{Aut}(M).$$

Here $M : \text{Aut}(M)$ is the holomorph of M , acting on M by right multiplication and automorphisms.

Theorem 11.5.2 (Rank bound for holomorph simple type). *Let G be of holomorph simple type. Then*

$$\text{rank}(G) \geq 4.$$

Proof. Identify Ω with T , and take the base point to be $1 \in T$. Then

$$G_1 \leq \text{Aut}(T).$$

The G_1 -orbits on T are contained in automorphism orbits, and automorphisms preserve element orders.

Since T is nonabelian simple, $|T|$ has at least three distinct prime divisors. Indeed, T is not solvable; groups of odd order are solvable, and groups of order $p^a q^b$ are solvable by

Burnside's $p^a q^b$ -theorem. Hence there exist elements

$$x, y, z \in T$$

of three distinct prime orders.

The four elements

$$1, \quad x, \quad y, \quad z$$

lie in four distinct G_1 -orbits, because their orders are distinct. Therefore

$$\text{rank}(G) \geq 4.$$

□

Example 11.5.3 (Sharpness of the bound). *Let $T = A_5$, and let*

$$G = T \rtimes \text{Aut}(T) \cong A_5 \rtimes S_5.$$

In the holomorph action of G on T , the point stabilizer is

$$G_1 = \text{Aut}(A_5) \cong S_5.$$

The conjugation action of S_5 on A_5 has exactly four orbits:

$$\{1\},$$

the elements of order 2, the elements of order 3, and the elements of order 5. Therefore

$$\text{rank}(G) = 4.$$

Thus the bound $\text{rank}(G) \geq 4$ is best possible.

Proposition 11.5.4 (Compound regular case). *Let $N = T^k$, where $k \geq 2$ and T is a finite nonabelian simple group. Suppose that N is a regular normal subgroup of a transitive permutation group G . Then*

$$\text{rank}(G) \geq 10.$$

Proof. Fix $\omega \in \Omega$. Since N is regular on Ω , we may identify Ω with N by

$$n \longleftrightarrow \omega^n.$$

Under this identification, ω corresponds to the identity element

$$\mathbf{1} = (1, \dots, 1) \in N.$$

Because $N \triangleleft G$, the point stabilizer G_ω normalizes N . Hence conjugation gives a homomorphism

$$G_\omega \longrightarrow \text{Aut}(N), \quad g \longmapsto (n \mapsto g^{-1}ng).$$

This homomorphism is injective. Indeed, if $g \in G_\omega$ centralizes N , then for every $n \in N$,

$$(\omega^n)^g = \omega^{ng} = \omega^{g(g^{-1}ng)} = (\omega^g)^{g^{-1}ng} = \omega^n.$$

Thus g fixes every point of Ω , so $g = 1$. Therefore we may regard

$$G_\omega \leq \text{Aut}(N).$$

Write

$$N = T_1 \times T_2 \times \cdots \times T_k,$$

where each $T_i \cong T$. The subgroups $T_1, \dots, T_k$ are precisely the minimal normal subgroups of N . Hence every automorphism of N permutes the direct factors T_i . Consequently, if

$$x = (x_1, \dots, x_k) \in N,$$

then the number

$$\text{supp}(x) = |\{i \mid x_i \neq 1\}|$$

is invariant under $\text{Aut}(N)$, and hence is invariant under G_ω . Also, since elements of G_ω act as automorphisms of N , they preserve element orders. Therefore the pair

$$(\text{supp}(x), |x|)$$

is constant on each G_ω -orbit on N .

Since T is finite nonabelian simple, $|T|$ has at least three distinct prime divisors. Indeed, otherwise $|T|$ would have the form p^a or $p^a q^b$, and T would be solvable, contradicting the nonabelian simplicity of T . Choose three distinct primes

$$p, q, r \mid |T|,$$

and choose elements

$$a, b, c \in T$$

such that

$$|a| = p, \quad |b| = q, \quad |c| = r.$$

Now consider the following elements of $N = T_1 \times \cdots \times T_k$:

$$\mathbf{1} = (1, \dots, 1),$$

$$(a, 1, \dots, 1), \quad (b, 1, \dots, 1), \quad (c, 1, \dots, 1),$$

$$(a, a, 1, \dots, 1), \quad (b, b, 1, \dots, 1), \quad (c, c, 1, \dots, 1),$$

and

$$(a, b, 1, \dots, 1), \quad (a, c, 1, \dots, 1), \quad (b, c, 1, \dots, 1).$$

These elements have pairwise distinct values of the invariant

$$(\text{supp}(x), |x|).$$

Indeed, the corresponding pairs are

$$\begin{aligned} & (0, 1), \\ & (1, p), \quad (1, q), \quad (1, r), \\ & (2, p), \quad (2, q), \quad (2, r), \end{aligned}$$

and

$$(2, pq), \quad (2, pr), \quad (2, qr).$$

These ten pairs are pairwise distinct. Hence the ten elements listed above lie in ten distinct G_ω -orbits on N .

Therefore G_ω has at least ten orbits on Ω . Since the rank of a transitive permutation group is the number of point-stabilizer orbits, we obtain

$$\text{rank}(G) \geq 10.$$

□

Corollary 11.5.5. *A quasiprimitive group of rank at most 4 cannot have a regular minimal normal subgroup T^k with $k \geq 2$ and T nonabelian simple.*

11.6 A Useful Obstruction in the Two-Transitive Case

Lemma 11.6.1. *Let $G \leq \text{Sym}(\Omega)$ be 2-transitive. Let*

$$N = T^k \triangleleft G,$$

where $k \geq 2$ and T is nonabelian simple. Then N has no proper normal subgroup that is regular on Ω .

Proof. Suppose, for a contradiction, that $M \triangleleft N$ is regular on Ω , with $M < N$. Since M is regular,

$$|\Omega| = |M|.$$

Because $N = T^k$ is a direct product of simple groups, every normal subgroup of N is a product of some of the direct factors. Hence

$$|M| = |T|^\ell$$

for some integer $0 < \ell < k$.

Fix $\omega \in \Omega$. Since M is regular, $M_\omega = 1$. Also

$$N = MN_\omega,$$

so

$$|N_\omega| = \frac{|N|}{|M|} = |T|^{k-\ell}.$$

Because G is 2-transitive, N_ω is half-transitive on $\Omega \setminus \{\omega\}$. Let m be the common size of its orbits on $\Omega \setminus \{\omega\}$. Then

$$m \mid |\Omega| - 1$$

and

$$m \mid |N_\omega|.$$

Hence

$$m \mid \gcd(|\Omega| - 1, |N_\omega|) = \gcd(|T|^\ell - 1, |T|^{k-\ell}) = 1.$$

Therefore $m = 1$.

Thus N_ω fixes every point of $\Omega \setminus \{\omega\}$, and it also fixes ω . Hence $N_\omega = 1$, contradicting $N_\omega \neq 1$, since $M < N$ and $N = MN_\omega$. Therefore no such proper regular normal subgroup M exists. $\square$

11.7 Product Action Type

Definition 11.7.1 (Product action). *Let $T \leq \text{Sym}(\Delta)$ be transitive, and let $k \geq 2$. The wreath product*

$$T \wr S_k = T^k \rtimes S_k$$

acts on

$$\Omega = \Delta^k$$

by product action. Explicitly, for

$$g = (t_1, \dots, t_k)\pi \in T^k \rtimes S_k$$

and

$$\delta = (\delta_1, \dots, \delta_k) \in \Delta^k,$$

we define

$$\delta^g = (\delta_{1\pi^{-1}}^{t_{1\pi^{-1}}}, \dots, \delta_{k\pi^{-1}}^{t_{k\pi^{-1}}}).$$

Proposition 11.7.2 (Product action reduction). *Let $G \leq \text{Sym}(\Omega)$ be primitive, and suppose*

$$N = T_1 \times \dots \times T_k \triangleleft G$$

is a transitive nonregular minimal normal subgroup, where

$$T_1 \cong \dots \cong T_k \cong T$$

and T is nonabelian simple.

Fix $\omega \in \Omega$, and assume that the projection of N_ω to each factor T_i is a proper subgroup $H_i < T_i$. Then

$$N_\omega = H_1 \times \dots \times H_k,$$

and

$$\Omega \cong [T_1 : H_1] \times \dots \times [T_k : H_k].$$

Under this identification, N acts coordinatewise by right multiplication, and G embeds into a wreath product in product action.

Proof. Since N is transitive,

$$\Omega \cong [N : N_\omega].$$

Write

$$N = T_1 \times \dots \times T_k.$$

Let H_i be the image of N_ω under projection onto T_i . Then

$$N_\omega \leq H_1 \times \dots \times H_k.$$

Set

$$K = H_1 \times \dots \times H_k.$$

Since the factors T_i are permuted transitively by G_ω , the subgroup K is normalized by G_ω . Moreover, by hypothesis $H_i < T_i$, so

$$K < N.$$

Because G is primitive, G_ω is a maximal subgroup of G . Hence

$$G_\omega \leq KG_\omega \leq G$$

implies either

$$KG_\omega = G_\omega$$

or

$$KG_\omega = G.$$

The second possibility is impossible. Indeed, if $KG_\omega = G$, then

$$N = (KG_\omega) \cap N = K(G_\omega \cap N) = KN_\omega = K,$$

contradicting $K < N$. Hence

$$KG_\omega = G_\omega,$$

so $K \leq G_\omega$. Since also $K \leq N$, we get

$$K \leq N \cap G_\omega = N_\omega.$$

Thus

$$N_\omega = K = H_1 \times \cdots \times H_k.$$

Therefore

$$\Omega \cong [N : N_\omega] = [T_1 \times \cdots \times T_k : H_1 \times \cdots \times H_k]$$

and hence

$$\Omega \cong [T_1 : H_1] \times \cdots \times [T_k : H_k].$$

This is precisely the product action model. □

11.8 Ranks in Product Action

Definition 11.8.1 (Weight). *Let*

$$\Delta = \{0, 1, \dots, n-1\}$$

and let

$$\Omega = \Delta^k.$$

For

$$u = (\delta_1, \dots, \delta_k) \in \Omega,$$

*define the **weight** of u by*

$$\text{wt}(u) = |\{i \mid \delta_i \neq 0\}|.$$

Thus $\text{wt}(u)$ is the number of nonzero coordinates of u .

Lemma 11.8.2 (Rank in full product action). *Let $T \leq \text{Sym}(\Delta)$ be transitive, and let*

$$G = T^k \rtimes S_k$$

act on

$$\Omega = \Delta^k$$

in product action. Fix $0 \in \Delta$, and set

$$\omega = (0, \dots, 0).$$

Then

$$\text{rank}(G) \geq k + 1.$$

Moreover,

$$\text{rank}(G) = k + 1$$

if and only if T is 2-transitive on Δ .

Proof. Let $H = T_0$. Then

$$G_\omega = H^k \rtimes S_k.$$

The group G_ω preserves the weight of a point in Ω . Therefore the sets

$$\Omega_j = \{u \in \Omega \mid \text{wt}(u) = j\}, \quad 0 \leq j \leq k,$$

are unions of G_ω -orbits. Hence

$$\text{rank}(G) \geq k + 1.$$

Suppose first that T is 2-transitive. Then $H = T_0$ is transitive on

$$\Delta \setminus \{0\}.$$

Thus, if two points of Ω have the same weight, then a permutation in S_k moves the nonzero coordinates of one point to the nonzero coordinates of the other, and elements of H adjust the values in those coordinates. Hence points of the same weight lie in the same G_ω -orbit. Therefore the G_ω -orbits are exactly the $k + 1$ weight classes, and

$$\text{rank}(G) = k + 1.$$

Conversely, suppose T is not 2-transitive. Then $H = T_0$ has at least two orbits on $\Delta \setminus \{0\}$. Choose representatives

$$a, b \in \Delta \setminus \{0\}$$

from two distinct H -orbits. Then

$$(a, 0, \dots, 0) \quad \text{and} \quad (b, 0, \dots, 0)$$

have the same weight, but they are not G_ω -conjugate. Hence the weight-one class splits into at least two G_ω -orbits, so

$$\text{rank}(G) > k + 1.$$

Thus equality holds if and only if T is 2-transitive. □

Example 11.8.3 (The case $k = 2$). *Let*

$$G = (T \times T) \rtimes S_2$$

act on

$$\Omega = \Delta^2.$$

Then

$$\text{rank}(G) \geq 3,$$

and

$$\text{rank}(G) = 3$$

if and only if T is 2-transitive on Δ .

For example, take $T = A_5$ acting naturally on

$$\Delta = \{0, 1, 2, 3, 4\}.$$

Then T is 2-transitive, and

$$G = (A_5 \times A_5) \rtimes S_2$$

has rank 3 on Δ^2 .

Fix

$$\omega = (0, 0).$$

Then

$$G_\omega = (A_4 \times A_4) \rtimes S_2.$$

The nontrivial G_ω -orbits on $\Omega \setminus \{\omega\}$ are

$$\{(0, i), (i, 0) \mid 1 \leq i \leq 4\}$$

and

$$\{(i, j) \mid 1 \leq i, j \leq 4\}.$$

Together with $\{\omega\}$, these give exactly three suborbits.

Exercise 11.8.4. Let

$$G = (T \times T \times T) \rtimes S_3$$

act on

$$\Omega = \Delta^3$$

in product action. Prove that

$$\text{rank}(G) \geq 4,$$

and that

$$\text{rank}(G) = 4$$

if and only if T is 2-transitive on Δ .

Proof. This is the case $k = 3$ of the previous lemma.

Explicitly, fix

$$\omega = (0, 0, 0).$$

The point stabilizer preserves the weight

$$\text{wt}(\delta_1, \delta_2, \delta_3) = |\{i \mid \delta_i \neq 0\}|.$$

Therefore there are at least four suborbits, corresponding to weights

$$0, 1, 2, 3.$$

Hence

$$\text{rank}(G) \geq 4.$$

If T is 2-transitive, then T_0 is transitive on $\Delta \setminus \{0\}$, so points of the same weight are G_ω -conjugate. Thus the four weight classes are exactly the four suborbits, and

$$\text{rank}(G) = 4.$$

Conversely, if T is not 2-transitive, then T_0 has at least two orbits on $\Delta \setminus \{0\}$. Hence the weight-one class splits into at least two G_ω -orbits. Therefore

$$\text{rank}(G) > 4.$$

So equality holds if and only if T is 2-transitive. □

11.9 Summary for Small Rank

Topic 11.9.1. *For the small-rank problem, the following constraints are especially useful.*

- (1) $\text{rank}(G) = 2$ if and only if G is 2-transitive. By Burnside's theorem, a finite 2-transitive group is either affine type or almost simple type.
- (2) If G is primitive, then G is quasiprimitive.
- (3) If G is of holomorph simple type, then

$$\text{rank}(G) \geq 4.$$

This bound is sharp, as shown by the holomorph action of $A_5 \rtimes S_5$ on A_5 .

- (4) If G has a regular normal subgroup T^k , where $k \geq 2$ and T is nonabelian simple, then

$$\text{rank}(G) \geq 10.$$

- (5) In product action

$$G = T^k \rtimes S_k \quad \text{on} \quad \Omega = \Delta^k,$$

one always has

$$\text{rank}(G) \geq k + 1.$$

Equality holds precisely when T is 2-transitive on Δ .

Chapter 12

Rank Estimates for Quasiprimitive Actions

12.1 Regular Normal Subgroups and Fusion Classes

Definition 12.1.1 (Fusion classes). *Let M be a finite group. The fusion classes of elements of M mean the $\text{Aut}(M)$ -orbits on M :*

$$m^{\text{Aut}(M)} = \{m^\alpha \mid \alpha \in \text{Aut}(M)\}, \quad m \in M.$$

Lemma 12.1.2. *Let $M \trianglelefteq G$ be regular on Ω . Then*

$$\text{rank}(G) \geq \#\{\text{Aut}(M)\text{-orbits on } M\}.$$

In particular, if $M = T^k$ with T nonabelian simple, then

$$\text{rank}(G) \geq 4.$$

Hence a rank-three quasiprimitive group cannot be of type HS, HC, or TW whenever the corresponding regular normal subgroup is of the form T^k with T nonabelian simple.

Proof. Choose $\omega \in \Omega$ and identify Ω with the set of right cosets $[G : G_\omega] = \{G_\omega g \mid g \in G\}$. Since M is regular, every point has a unique representative of the form $G_\omega m$ with $m \in M$, and

$$G = MG_\omega, \quad M \cap G_\omega = 1.$$

Let $h \in G_\omega$ and $m \in M$. Then

$$(G_\omega m)^h = G_\omega mh = G_\omega h(h^{-1}mh) = G_\omega m^h.$$

Thus G_ω acts on the identified copy of M by conjugation. Since $M \trianglelefteq G$, conjugation by

elements of G_ω gives automorphisms of M ; hence

$$G_\omega \leq \text{Aut}(M)$$

after this identification. Therefore the G_ω -orbits on M refine the $\text{Aut}(M)$ -orbits on M . The number of G_ω -orbits is $\text{rank}(G)$, so

$$\text{rank}(G) \geq \#\{\text{Aut}(M)\text{-orbits on } M\}.$$

Now suppose $M = T^k$ with T nonabelian simple. By Burnside's $p^a q^b$ -theorem, $|T|$, and hence $|M|$, is divisible by at least three distinct primes. Choose distinct primes r, s, t dividing $|M|$. By Cauchy's theorem there exist $x, y, z \in M$ with

$$|x| = r, \quad |y| = s, \quad |z| = t.$$

Automorphisms preserve element orders, so

$$1^{\text{Aut}(M)}, \quad x^{\text{Aut}(M)}, \quad y^{\text{Aut}(M)}, \quad z^{\text{Aut}(M)}$$

are four distinct fusion classes of M . Hence $\text{rank}(G) \geq 4$. □

12.2 Simple Diagonal Type

Lemma 12.2.1. *Let G be a quasiprimitive permutation group of simple diagonal type SD. Then*

$$\text{rank}(G) \geq 4.$$

Proof. Let

$$N = \text{Soc}(G) = T_1 \times \cdots \times T_k \trianglelefteq G,$$

where $T_i \cong T$ for a nonabelian simple group T , and $k \geq 2$. Choose a point $\omega \in \Omega$. In simple diagonal type we may write, after identifying each T_i with T ,

$$N_\omega = \{(t, t, \dots, t) \mid t \in T\} \cong T.$$

Put

$$R = \{(x_1, \dots, x_{k-1}, 1) \mid x_i \in T\}.$$

Then $N = RN_\omega$ and $R \cap N_\omega = 1$, so R is regular on Ω . Identifying

$$\Omega \cong [N : N_\omega] = \{N_\omega(t_1, \dots, t_k) \mid t_i \in T\},$$

every coset has a unique representative in R , namely

$$N_\omega(t_1, \dots, t_k) = N_\omega(t_k^{-1}t_1, \dots, t_k^{-1}t_{k-1}, 1).$$

The stabilizer G_ω normalizes both N and N_ω . In this diagonal action,

$$C_{\text{Sym}(\Omega)}(N) \cong N_N(N_\omega)/N_\omega = 1,$$

because the full diagonal subgroup is self-normalizing in $N = T^k$. Hence $C_G(N) = 1$, so conjugation embeds G_ω into $\text{Aut}(N)$. Therefore

$$G_\omega \leq N_{\text{Aut}(N)}(N_\omega).$$

The normalizer of the full diagonal subgroup in $\text{Aut}(T) \wr S_k$ is

$$\text{Aut}(T) \times S_k,$$

where $\text{Aut}(T)$ acts diagonally on all coordinates and S_k permutes the coordinates. Thus every element of G_ω has the form $\alpha\pi$, with

$$\alpha \in \text{Aut}(T), \quad \pi \in S_k,$$

and it sends

$$(t_1, \dots, t_k) \quad \text{to} \quad (t_1^\alpha, \dots, t_k^\alpha)^\pi.$$

For $a \in T^\#$, write

$$x(a) = N_\omega(a, 1, \dots, 1) \in \Omega.$$

We claim that if $x(a)$ and $x(b)$ lie in the same G_ω -orbit, then either

$$a = b^\alpha$$

for some $\alpha \in \text{Aut}(T)$, or $k = 2$ and

$$a = (b^{-1})^\alpha$$

for some $\alpha \in \text{Aut}(T)$.

Indeed, suppose $x(a)^{\alpha\pi} = x(b)$ for some $\alpha\pi \in G_\omega$. The image of $(a, 1, \dots, 1)$ has the single nontrivial entry a^α in coordinate 1^π .

- If $1^\pi = 1$, then the representative in R is $(a^\alpha, 1, \dots, 1)$, so $b = a^\alpha$.
- If $1^\pi = k$, then the representative in R is

$$((a^\alpha)^{-1}, \dots, (a^\alpha)^{-1}, 1).$$

For this to equal $(b, 1, \dots, 1)$, we must have $k = 2$ and $b = (a^{-1})^\alpha$.

- If $1^\pi = i \notin \{1, k\}$, then the representative in R has a nontrivial entry away from the first coordinate, and so it cannot equal $(b, 1, \dots, 1)$.

Consequently, the G_ω -orbits on the points $x(a)$ are at least as numerous as the $\text{Aut}(T)$ -orbits on inverse-pairs

$$\{a, a^{-1}\}, \quad a \in T.$$

Since T is nonabelian simple, $|T|$ is divisible by at least three distinct primes. Choose elements of three distinct prime orders in T . Elements of distinct orders are neither automorphic conjugate nor inverse-automorphic conjugate. Together with the identity, these give at least four G_ω -orbits on Ω . Therefore

$$\text{rank}(G) \geq 4.$$

□

Example 12.2.2. *The lower bound in Lemma 12.2.1 can be sharp. Take $T = A_5$ and $k = 2$. Let*

$$G = (A_5 \times A_5) \cdot (\langle \iota \rangle \times S_2),$$

where ι is an outer automorphism of A_5 . Then

$$G_\omega \cong S_5 \times S_2 = \text{Aut}(A_5) \times S_2,$$

and the suborbits correspond to the element orders 1, 2, 3, 5 in A_5 . Hence

$$\text{rank}(G) = 4.$$

12.3 Wreath Products and Product Action

Definition 12.3.1 (Wreath product). *Let T be a group and let $k \geq 2$. The wreath product of T by S_k is*

$$T \wr S_k = T^k \rtimes S_k,$$

where S_k permutes the k direct factors of T^k . For $\pi \in S_k$ and $\mathbf{t} = (t_1, \dots, t_k) \in T^k$, we use the convention

$$\mathbf{t}^\pi = (t_{1\pi^{-1}}, \dots, t_{k\pi^{-1}}).$$

Thus multiplication in $T \wr S_k$ is

$$(\mathbf{s}\pi)(\mathbf{t}\sigma) = (\mathbf{s}\mathbf{t}^{\pi^{-1}})\pi\sigma,$$

that is,

$$((s_1, \dots, s_k)\pi)((t_1, \dots, t_k)\sigma) = (s_1 t_{1\pi}, \dots, s_k t_{k\pi})\pi\sigma.$$

Definition 12.3.2 (Product action). *Let T be a transitive permutation group on*

$$\Delta = \{0, 1, \dots, m\}, \quad m \geq 1.$$

Let $P \leq S_k$ be transitive, where $k \geq 2$, and put

$$G = T \wr P \leq T \wr S_k, \quad \Omega = \Delta^k.$$

The group G acts naturally on Ω by

$$(\delta_1, \dots, \delta_k)^{(t_1, \dots, t_k)\pi} = (\delta_1^{t_1}, \dots, \delta_k^{t_k})^\pi = (\delta_{1\pi^{-1}}^{t_1\pi^{-1}}, \dots, \delta_{k\pi^{-1}}^{t_k\pi^{-1}}).$$

This is called the product action; equivalently, G is a blow-up of the action of T on Δ .

Fix

$$\omega_0 = (0, \dots, 0) \in \Omega.$$

For $\omega = (\delta_1, \dots, \delta_k) \in \Omega$, define its *weight* by

$$\text{wt}(\omega) = |\{i \mid \delta_i \neq 0\}|.$$

Thus $\text{wt}(\omega)$ is the number of nonzero coordinates of ω .

Lemma 12.3.3. *Using the notation above, suppose $G = T \wr P$ acts on $\Omega = \Delta^k$ in product action. Then:*

- (1) $\text{rank}(G) \geq k + 1$.
- (2) $\text{rank}(G) = 3$ if and only if $k = 2$ and T is 2-transitive on Δ .

Proof. Let T_0 be the stabilizer of 0 in T . The stabilizer of ω_0 in G is

$$G_{\omega_0} = T_0^k \rtimes P.$$

Every element of T_0^k fixes the zero entry in each coordinate, and every element of P only permutes coordinates. Hence G_{ω_0} preserves the weight function

$$\text{wt} : \Omega \rightarrow \{0, 1, \dots, k\}.$$

Since each weight $0, 1, \dots, k$ occurs, G_{ω_0} has at least $k + 1$ orbits on Ω . Therefore

$$\text{rank}(G) \geq k + 1.$$

Now suppose $\text{rank}(G) = 3$. Since $k \geq 2$, the inequality just proved gives

$$3 = \text{rank}(G) \geq k + 1 \geq 3,$$

so $k = 2$. Thus

$$\Omega = \Delta \times \Delta, \quad G = (T \times T) \rtimes S_2.$$

The points of weight 1 form a single G_{ω_0} -orbit. In particular, the set

$$\{(\delta, 0) \mid \delta \in \Delta \setminus \{0\}\}$$

forces T_0 to be transitive on $\Delta \setminus \{0\}$. Hence T is 2-transitive on Δ .

Conversely, assume $k = 2$ and T is 2-transitive on Δ . Then T_0 is transitive on $\Delta \setminus \{0\}$. The stabilizer

$$G_{\omega_0} = (T_0 \times T_0) \rtimes S_2$$

has exactly the following three orbits on $\Delta \times \Delta$:

$$\{(0, 0)\},$$

$$\{(\delta, 0), (0, \delta) \mid \delta \neq 0\},$$

$$\{(\delta, \epsilon) \mid \delta \neq 0, \epsilon \neq 0\}.$$

Therefore $\text{rank}(G) = 3$. □

Remark 12.3.4. *The preceding lemmas give three quick rank obstructions in the O’Nan–Scott analysis of quasiprimitive groups:*

- *regular nonabelian characteristically simple normal subgroups force rank at least 4;*
- *simple diagonal type SD forces rank at least 4;*
- *product action has rank 3 only in the case $k = 2$ with the component action 2-transitive.*

Chapter 13

Linear and Affine Permutation Groups

13.1 Multiple Transitivity

Definition 13.1.1 (*k*-transitive action). Let $G \leq \text{Sym}(\Omega)$, and let $k \geq 1$. We say that G is *k*-transitive on Ω if for any two ordered *k*-tuples

$$(\omega_1, \dots, \omega_k), \quad (\omega'_1, \dots, \omega'_k)$$

of pairwise distinct points of Ω , there exists $g \in G$ such that

$$\omega_i^g = \omega'_i \quad \text{for all } 1 \leq i \leq k.$$

Equivalently,

$$(\omega_1, \dots, \omega_k)^g = (\omega'_1, \dots, \omega'_k).$$

Definition 13.1.2 (Sharply *k*-transitive action). Let $G \leq \text{Sym}(\Omega)$. We say that G is sharply *k*-transitive on Ω if for any two ordered *k*-tuples of pairwise distinct points

$$(\omega_1, \dots, \omega_k), \quad (\omega'_1, \dots, \omega'_k),$$

there exists exactly one element $g \in G$ such that

$$\omega_i^g = \omega'_i \quad \text{for all } 1 \leq i \leq k.$$

Proposition 13.1.3 (Stabilizer criterion for *k*-transitivity). Let $G \leq \text{Sym}(\Omega)$, let $\omega \in \Omega$, and let $k \geq 2$. Then G is *k*-transitive on Ω if and only if

$$G \text{ is transitive on } \Omega \quad \text{and} \quad G_\omega \text{ is } (k-1)\text{-transitive on } \Omega \setminus \{\omega\}.$$

Proof. Suppose first that G is k -transitive. Then G is certainly transitive. Moreover, if

$$(\alpha_2, \dots, \alpha_k), \quad (\beta_2, \dots, \beta_k)$$

are two ordered $(k-1)$ -tuples of pairwise distinct points in $\Omega \setminus \{\omega\}$, then k -transitivity gives some $g \in G$ such that

$$(\omega, \alpha_2, \dots, \alpha_k)^g = (\omega, \beta_2, \dots, \beta_k).$$

In particular, $\omega^g = \omega$, so $g \in G_\omega$. Hence G_ω is $(k-1)$ -transitive on $\Omega \setminus \{\omega\}$.

Conversely, suppose that G is transitive on Ω and G_ω is $(k-1)$ -transitive on $\Omega \setminus \{\omega\}$. Let

$$(\omega_1, \dots, \omega_k), \quad (\omega'_1, \dots, \omega'_k)$$

be ordered k -tuples of pairwise distinct points. Since G is transitive, choose $g_1, g_2 \in G$ such that

$$\omega_1^{g_1} = \omega, \quad (\omega'_1)^{g_2} = \omega.$$

Then

$$\omega_2^{g_1}, \dots, \omega_k^{g_1}, \quad (\omega'_2)^{g_2}, \dots, (\omega'_k)^{g_2}$$

are two ordered $(k-1)$ -tuples of pairwise distinct points in $\Omega \setminus \{\omega\}$. By the hypothesis on G_ω , there exists $h \in G_\omega$ such that

$$(\omega_i^{g_1})^h = (\omega'_i)^{g_2} \quad \text{for } 2 \leq i \leq k.$$

Since h fixes ω , we also have

$$(\omega_1^{g_1})^h = \omega = (\omega'_1)^{g_2}.$$

Therefore

$$(\omega_1, \dots, \omega_k)^{g_1 h g_2^{-1}} = (\omega'_1, \dots, \omega'_k),$$

so G is k -transitive. □

Definition 13.1.4 (Rank of a transitive permutation group). *Let $G \leq \text{Sym}(\Omega)$ be transitive. The rank of G on Ω is the number of G_ω -orbits on Ω , where $\omega \in \Omega$. Equivalently, it is the number of G -orbits on $\Omega \times \Omega$.*

Remark 13.1.5. *A transitive group $G \leq \text{Sym}(\Omega)$ is 2-transitive if and only if it has rank 2. Indeed, one orbit on $\Omega \times \Omega$ is the diagonal*

$$\{(\omega, \omega) \mid \omega \in \Omega\},$$

and 2-transitivity is precisely the assertion that all off-diagonal pairs lie in one orbit.

Lemma 13.1.6 (Counting criterion for sharp transitivity). *Assume Ω is finite. If G*

is k -transitive on Ω , then G is sharply k -transitive if and only if

$$|G| = |\Omega|(|\Omega| - 1) \cdots (|\Omega| - k + 1).$$

Proof. Let $\Omega^{(k)}$ denote the set of ordered k -tuples of pairwise distinct points of Ω . Then

$$|\Omega^{(k)}| = |\Omega|(|\Omega| - 1) \cdots (|\Omega| - k + 1).$$

If G is k -transitive, then G acts transitively on $\Omega^{(k)}$. Thus, for a fixed tuple $\omega \in \Omega^{(k)}$,

$$|\Omega^{(k)}| = [G : G_\omega].$$

The action is sharply k -transitive exactly when $G_\omega = 1$, which is equivalent to

$$|G| = |\Omega^{(k)}|.$$

□

13.2 The Natural Action of $\text{GL}(V)$

Let $\mathbb{F} = \mathbb{F}_q$ be a finite field, let $V = \mathbb{F}^d$ with $d > 1$, and let

$$G = \text{GL}(V).$$

The group G acts naturally on V . There are two orbits:

$$\{0\} \quad \text{and} \quad \Omega := V^\# := V \setminus \{0\}.$$

Proposition 13.2.1. *The action of $\text{GL}(V)$ on $\Omega = V^\#$ is transitive.*

Proof. Let $u, v \in V^\#$. Extend u to a basis

$$u, u_2, \dots, u_d$$

of V , and extend v to a basis

$$v, v_2, \dots, v_d$$

of V . There is a unique linear isomorphism $g \in \text{GL}(V)$ such that

$$u^g = v, \quad u_i^g = v_i \quad (2 \leq i \leq d).$$

Hence u and v lie in the same G -orbit. □

Proposition 13.2.2. *If $q > 2$, then the action of $\text{GL}(V)$ on $\Omega = V^\#$ is imprimitive.*

Proof. For $v \in V^\#$, define

$$B_v := \mathbb{F}^\times v = \{\lambda v \mid \lambda \in \mathbb{F}^\times\}.$$

Let

$$\mathcal{B} = \{B_v \mid v \in V^\#\}.$$

Then $\mathcal{B}$ is the set of nonzero points on the one-dimensional subspaces of V . Since $q > 2$,

$$|B_v| = q - 1 > 1.$$

Also, since $d > 1$,

$$|B_v| = q - 1 < q^d - 1 = |\Omega|.$$

Thus the blocks are nontrivial and proper.

For any $g \in \text{GL}(V)$,

$$B_v^g = \{\lambda v^g \mid \lambda \in \mathbb{F}^\times\} = B_{v^g}.$$

Hence g permutes the members of $\mathcal{B}$. Therefore $\mathcal{B}$ is a nontrivial block system for the action of $\text{GL}(V)$ on Ω , and the action is imprimitive. $\square$

Proposition 13.2.3. *If $q = 2$, then the action of $\text{GL}(V)$ on $\Omega = V^\#$ is 2-transitive.*

Proof. Let

$$(u_1, u_2), \quad (v_1, v_2)$$

be ordered pairs of distinct points in Ω . Since $q = 2$, the only nonzero scalar is 1. Hence $u_1 \neq u_2$ implies that u_1, u_2 are linearly independent. Similarly, v_1, v_2 are linearly independent.

Extend

$$u_1, u_2$$

to a basis

$$u_1, u_2, \dots, u_d$$

of V , and extend

$$v_1, v_2$$

to a basis

$$v_1, v_2, \dots, v_d$$

of V . There exists $g \in \text{GL}(V)$ such that

$$u_i^g = v_i \quad \text{for all } 1 \leq i \leq d.$$

In particular,

$$(u_1, u_2)^g = (v_1, v_2).$$

Thus $\text{GL}(V)$ is 2-transitive on $V^\#$. $\square$

Proposition 13.2.4 (Rank of $\text{GL}(V)$ on $V^\#$). *Assume $q > 2$. Then the rank of $\text{GL}(V)$ on $\Omega = V^\#$ is q . More precisely, the G -orbits on $\Omega \times \Omega$ are:*

$$\Delta = \{(v, v) \mid v \in \Omega\},$$

$$\mathcal{O}_\lambda = \{(v, \lambda v) \mid v \in \Omega\}, \quad \lambda \in \mathbb{F}^\times \setminus \{1\},$$

and

$$\mathcal{O}_{\text{ind}} = \{(u, v) \in \Omega \times \Omega \mid \langle u \rangle \neq \langle v \rangle\}.$$

Therefore

$$\text{rank}(G, \Omega) = 1 + (q - 2) + 1 = q.$$

Proof. The diagonal Δ is clearly one orbit.

Now fix $\lambda \in \mathbb{F}^\times \setminus \{1\}$. If $u, v \in \Omega$, choose $g \in \text{GL}(V)$ such that $u^g = v$. Then

$$(u, \lambda u)^g = (v, \lambda v).$$

Thus $\mathcal{O}_\lambda$ is an orbit. Moreover, if

$$(u, \lambda u)^g = (v, \mu v),$$

then

$$\lambda u^g = \mu v.$$

Since $u^g = v$, we get $\lambda = \mu$. Hence the orbits $\mathcal{O}_\lambda$ are distinct as λ varies.

Finally, suppose $\langle u \rangle \neq \langle v \rangle$ and $\langle u' \rangle \neq \langle v' \rangle$. Then u, v are linearly independent, and u', v' are linearly independent. Extend both pairs to bases of V , and choose $g \in \text{GL}(V)$ mapping the first basis to the second. Then

$$(u, v)^g = (u', v').$$

Thus all pairs with distinct one-dimensional spans form one orbit.

Hence the rank is

$$1 + (q - 2) + 1 = q.$$

□

13.3 The Projective Action

Continue to let $V = \mathbb{F}_q^d$ with $d > 1$, and let

$$G = \text{GL}(V).$$

Let

$$\mathcal{B} = \{B_v = \mathbb{F}^\times v \mid v \in V^\#\}.$$

We identify $\mathcal{B}$ with the set of one-dimensional subspaces of V , i.e. the points of the projective space associated with V .

Proposition 13.3.1. *The action of $\mathrm{GL}(V)$ on $\mathcal{B}$ is transitive, and the kernel of this action is*

$$Z(\mathrm{GL}(V)) = \{\lambda I_V \mid \lambda \in \mathbb{F}^\times\}.$$

Consequently the induced faithful permutation group is

$$\mathrm{GL}(V)/Z(\mathrm{GL}(V)) = \mathrm{PGL}(V).$$

Proof. Transitivity follows from the transitivity of $\mathrm{GL}(V)$ on nonzero vectors: if $B_u, B_v \in \mathcal{B}$, choose $g \in \mathrm{GL}(V)$ with $u^g = v$. Then

$$B_u^g = B_v.$$

Every scalar map λI_V fixes every one-dimensional subspace, so

$$Z(\mathrm{GL}(V)) \leq \mathrm{Ker}(\mathrm{GL}(V) \curvearrowright \mathcal{B}).$$

Conversely, suppose $g \in \mathrm{GL}(V)$ fixes every one-dimensional subspace. Then for each $0 \neq v \in V$, there exists $\lambda_v \in \mathbb{F}^\times$ such that

$$v^g = \lambda_v v.$$

If u, v are linearly independent, then $u + v \neq 0$, and g also fixes the line $\mathbb{F}(u + v)$. Hence

$$(u + v)^g = \lambda_{u+v}(u + v).$$

But

$$(u + v)^g = u^g + v^g = \lambda_u u + \lambda_v v.$$

Since u, v are linearly independent, we get

$$\lambda_u = \lambda_v = \lambda_{u+v}.$$

It follows that the scalar λ_v is independent of v , so $g = \lambda I_V$. Thus the kernel is exactly the scalar subgroup $Z(\mathrm{GL}(V))$. $\square$

Proposition 13.3.2. *The group $\mathrm{PGL}(V)$ is 2-transitive on $\mathcal{B}$.*

Proof. Let

$$(B_{u_1}, B_{u_2}), \quad (B_{v_1}, B_{v_2})$$

be ordered pairs of distinct points of $\mathcal{B}$. Since the projective points are distinct, u_1, u_2 are linearly independent, and v_1, v_2 are linearly independent.

Extend u_1, u_2 to a basis of V , and extend v_1, v_2 to a basis of V . Then there exists $g \in \text{GL}(V)$ such that

$$u_1^g = v_1, \quad u_2^g = v_2.$$

Therefore

$$(B_{u_1}, B_{u_2})^g = (B_{v_1}, B_{v_2}).$$

Passing to the quotient by the kernel, $\text{PGL}(V)$ is 2-transitive on $\mathcal{B}$. □

Theorem 13.3.3. *The group $\text{PGL}(V)$ is not 3-transitive on $\mathcal{B}$ if and only if*

$$d > 2.$$

Equivalently, $\text{PGL}(V)$ is 3-transitive on $\mathcal{B}$ exactly when $d = 2$.

Proof. First suppose $d > 2$. Choose linearly independent vectors

$$e_1, e_2, e_3 \in V.$$

Then the ordered triple

$$(\mathbb{F}e_1, \mathbb{F}e_2, \mathbb{F}e_3)$$

has representatives that are linearly independent. On the other hand,

$$(\mathbb{F}e_1, \mathbb{F}e_2, \mathbb{F}(e_1 + e_2))$$

is an ordered triple of distinct projective points whose representatives are linearly dependent. Linear maps preserve linear dependence and linear independence. Therefore these two triples cannot lie in the same $\text{PGL}(V)$ -orbit. Thus $\text{PGL}(V)$ is not 3-transitive.

Conversely, suppose $d = 2$. Let

$$(\mathbb{F}u_1, \mathbb{F}u_2, \mathbb{F}u_3), \quad (\mathbb{F}v_1, \mathbb{F}v_2, \mathbb{F}v_3)$$

be ordered triples of distinct projective points. Since $\mathbb{F}u_1 \neq \mathbb{F}u_2$, the vectors u_1, u_2 form a basis of V . Similarly, v_1, v_2 form a basis of V .

Since $\mathbb{F}u_3$ is distinct from both $\mathbb{F}u_1$ and $\mathbb{F}u_2$, we can write

$$u_3 = au_1 + bu_2$$

with $a, b \in \mathbb{F}^\times$. Similarly,

$$v_3 = cv_1 + dv_2$$

with $c, d \in \mathbb{F}^\times$.

Define $g \in \text{GL}(V)$ by

$$u_1^g = \frac{c}{a}v_1, \quad u_2^g = \frac{d}{b}v_2.$$

Then

$$u_3^g = (au_1 + bu_2)^g = cv_1 + dv_2 = v_3.$$

Hence

$$(\mathbb{F}u_1, \mathbb{F}u_2, \mathbb{F}u_3)^g = (\mathbb{F}v_1, \mathbb{F}v_2, \mathbb{F}v_3).$$

Thus $\text{PGL}(V)$ is 3-transitive when $d = 2$. □

Remark 13.3.4. When $d = 2$, the action of $\text{PGL}_2(q)$ on the projective line

$$\mathbb{P}^1(\mathbb{F}_q)$$

is in fact sharply 3-transitive. Indeed,

$$|\text{PGL}_2(q)| = q(q^2 - 1),$$

and the number of ordered triples of distinct points of $\mathbb{P}^1(\mathbb{F}_q)$ is

$$(q + 1)q(q - 1) = q(q^2 - 1).$$

13.4 Affine Permutation Groups

Let

$$N = (\mathbb{Z}/p\mathbb{Z})^d$$

where p is prime and $d > 1$. We regard N as the additive group of the vector space $\mathbb{F}_p^d$. Then

$$\text{Aut}(N) = \text{GL}_d(p).$$

Set

$$H = \text{Aut}(N) = \text{GL}_d(p), \quad G = N : H = \text{Hol}(N).$$

Thus

$$G = \mathbb{F}_p^d : \text{GL}_d(p),$$

the affine general linear group.

Let $\Omega = N$. We define an action of G on Ω as follows. If $x \in N$, $y \in H$, and $w \in N$, then

$$w^{xy} := (w + x)^y = w^y + x^y.$$

Here x acts as a translation and y acts as a linear transformation.

Theorem 13.4.1. *The affine group*

$$G = \mathbb{F}_p^d : \text{GL}_d(p)$$

is 2-transitive on $\Omega = \mathbb{F}_p^d$. Furthermore, if $p = 2$, then G is 3-transitive.

Proof. We first prove 2-transitivity. Let

$$(u_1, u_2), \quad (v_1, v_2)$$

be ordered pairs of distinct points of N . Then

$$u_2 - u_1 \neq 0, \quad v_2 - v_1 \neq 0.$$

Since $\mathrm{GL}_d(p)$ is transitive on nonzero vectors, choose $y \in \mathrm{GL}_d(p)$ such that

$$(u_2 - u_1)^y = v_2 - v_1.$$

Now choose

$$x = v_1^{y^{-1}} - u_1.$$

Then

$$(u_1 + x)^y = (u_1 + v_1^{y^{-1}} - u_1)^y = v_1,$$

and

$$(u_2 + x)^y = (u_2 - u_1 + v_1^{y^{-1}})^y = (u_2 - u_1)^y + v_1 = v_2.$$

Thus

$$(u_1, u_2)^{xy} = (v_1, v_2),$$

so G is 2-transitive.

Now assume $p = 2$. The stabilizer of $0 \in N$ in G is

$$G_0 = \mathrm{GL}_d(2).$$

By the earlier result, $\mathrm{GL}_d(2)$ is 2-transitive on

$$N \setminus \{0\}.$$

Since G is transitive on N , the stabilizer criterion for multiple transitivity implies that G is 3-transitive on N . $\square$

Remark 13.4.2. *The translation subgroup N is regular on $\Omega = N$. It is also a minimal normal subgroup of G . Indeed, the normal subgroups of N that are normalized by $H = \mathrm{GL}_d(p)$ are precisely the H -invariant subspaces of $\mathbb{F}_p^d$, and $\mathrm{GL}_d(p)$ acts irreducibly on the natural module. Hence the only such subgroups are 0 and N .*

Moreover, every nontrivial normal subgroup of G contains N . Therefore every nontrivial normal subgroup of G is transitive on Ω . Thus G is a quasiprimitive affine permutation group.

13.5 A Sharply 2-Transitive Affine Example

The full affine group $\text{AGL}_d(p) = \mathbb{F}_p^d : \text{GL}_d(p)$ is often much larger than is needed for 2-transitivity. The following construction gives a smaller sharply 2-transitive group.

Example 13.5.1 (The group $\text{AGL}_1(q)$). *Let*

$$F = \mathbb{F}_q, \quad q = p^f.$$

Then the additive group of F satisfies

$$F^+ \cong (\mathbb{Z}/p\mathbb{Z})^f,$$

and the multiplicative group satisfies

$$F^\times \cong C_{q-1}.$$

Choose a generator φ of $F^\times$. Then

$$F^\times = \langle \varphi \rangle.$$

The group $F^\times$ acts on F^+ by field multiplication:

$$x^\varphi = x\varphi \quad (x \in F).$$

This is an automorphism of the additive group F^+ . Hence

$$\langle \varphi \rangle \leq \text{GL}_f(p),$$

and we can form

$$G = F^+ : \langle \varphi \rangle \leq \text{Hol}(F^+) = F^+ : \text{GL}_f(p).$$

Equivalently, G is the group of affine transformations

$$x \mapsto ax + b, \quad a \in F^\times, b \in F.$$

This group is usually denoted

$$\text{AGL}_1(q).$$

We claim that G is sharply 2-transitive on F .

First, $F^\times$ is transitive on $F^\times = F \setminus \{0\}$. Indeed, if $u, v \in F^\times$, then there exists $a \in F^\times$ such that

$$ua = v.$$

Therefore the stabilizer of 0 in G is transitive on $F \setminus \{0\}$. Since the translation subgroup F^+ is transitive on F , it follows that G is 2-transitive on F .

Finally,

$$|G| = |F^+| |F^\times| = q(q-1).$$

The number of ordered pairs $(x, y) \in F \times F$ with $x \neq y$ is also

$$q(q-1).$$

By the counting criterion, G is sharply 2-transitive.

Equivalently, this uniqueness can be checked directly: given $u_1 \neq u_2$ and $v_1 \neq v_2$ in F , the affine map

$$x \mapsto ax + b$$

sending $u_1 \mapsto v_1$ and $u_2 \mapsto v_2$ is uniquely determined by

$$a = \frac{v_2 - v_1}{u_2 - u_1}, \quad b = v_1 - au_1.$$

Chapter 14

Hall Subgroups

14.1 Definition and Basic Examples

Definition 14.1.1 (Hall subgroup). *Let G be a finite group, and write*

$$|G| = p_1^{a_1} p_2^{a_2} \cdots p_t^{a_t},$$

where $p_1, \dots, p_t$ are the distinct prime divisors of $|G|$. Define

$$\pi(G) = \pi(|G|) = \{p_1, \dots, p_t\}.$$

For a subset $\pi \subseteq \pi(G)$, a subgroup $H \leq G$ is called a Hall π -subgroup of G if

$$\pi(H) = \pi \quad \text{and} \quad \gcd(|H|, [G : H]) = 1.$$

Equivalently, H is a Hall π -subgroup if $|H|$ is a π -number and $[G : H]$ is a π' -number, where

$$\pi' = \pi(G) \setminus \pi.$$

Remark 14.1.2. *If $\pi = \{p\}$, then a Hall π -subgroup is precisely a Sylow p -subgroup. Thus Hall subgroups generalize Sylow subgroups by allowing several primes at once.*

Topic 14.1.3 (Notation). *When a Hall π -subgroup has been chosen, we may denote it by*

$$G_\pi.$$

If $\pi = \{p\}$, we often write G_p instead of $G_{\{p\}}$. Also,

$$\pi' = \pi(G) \setminus \pi.$$

This notation does not imply uniqueness: Hall subgroups need not exist, and even when they exist, they need not be conjugate in arbitrary finite groups.

Example 14.1.4 (Hall subgroups in A_5). *Let*

$$G = A_5, \quad |G| = 60 = 2^2 \cdot 3 \cdot 5.$$

Then

$$\pi(G) = \{2, 3, 5\}.$$

The point stabilizer of the natural action of A_5 on five letters is isomorphic to A_4 , and has order $12 = 2^2 \cdot 3$. Hence

$$A_4 \leq A_5$$

is a Hall $\{2, 3\}$ -subgroup.

A Sylow 5-subgroup is cyclic of order 5, so

$$C_5$$

is a Hall $\{5\}$ -subgroup, equivalently a Hall $\{2, 3\}'$ -subgroup.

However, A_5 has no Hall $\{2, 5\}$ -subgroup and no Hall $\{3, 5\}$ -subgroup.

Indeed, suppose first that H is a Hall $\{2, 5\}$ -subgroup. Then

$$|H| = 2^2 \cdot 5 = 20, \quad [A_5 : H] = 3.$$

The left action of A_5 on the set of left cosets A_5/H gives a homomorphism

$$A_5 \longrightarrow \text{Sym}(A_5/H) \cong S_3.$$

Since A_5 is simple, the kernel is either 1 or A_5 . The action is nontrivial, so the kernel is not A_5 . Hence the map would be injective, which is impossible because

$$|A_5| = 60 > |S_3| = 6.$$

Therefore no Hall $\{2, 5\}$ -subgroup exists.

Similarly, if K were a Hall $\{3, 5\}$ -subgroup, then

$$|K| = 3 \cdot 5 = 15, \quad [A_5 : K] = 4.$$

The coset action would give a nontrivial homomorphism

$$A_5 \longrightarrow S_4.$$

Again simplicity forces this map to be injective, which is impossible since

$$|A_5| = 60 > |S_4| = 24.$$

Thus no Hall $\{3, 5\}$ -subgroup exists.

Example 14.1.5 (Hall subgroups need not be conjugate). *Let*

$$G = \text{PSL}_2(11).$$

Then G is simple and

$$|G| = 660 = 2^2 \cdot 3 \cdot 5 \cdot 11.$$

This group has subgroups H and K such that

$$H \cong A_4, \quad K \cong \text{Dih}_{12},$$

where D_{12} denotes the dihedral group of order 12. Both H and K have order $12 = 2^2 \cdot 3$, so both are Hall $\{2, 3\}$ -subgroups of G .

However,

$$A_4 \not\cong D_{12}.$$

Therefore H and K cannot be conjugate in G . This shows that Hall subgroups are not necessarily conjugate in arbitrary finite groups.

14.2 Hall's Theorem

Lemma 14.2.1 (The subgroup $\text{O}_p(X)$). *Let X be a finite group and let p be a prime. Define*

$$\text{O}_p(X)$$

to be the largest normal p -subgroup of X . Equivalently,

$$\text{O}_p(X)$$

is the product of all normal p -subgroups of X . It has the following properties:

- (1) $\text{O}_p(X) \trianglelefteq X$;
- (2) $\text{O}_p(X)$ is a p -group;
- (3) $X/\text{O}_p(X)$ has no nontrivial normal p -subgroups.

Proof. The product of normal p -subgroups is again a normal p -subgroup, so the product of all normal p -subgroups of X is the unique largest normal p -subgroup of X . This proves (1) and (2).

For (3), suppose that $X/\text{O}_p(X)$ has a nontrivial normal p -subgroup $N/\text{O}_p(X)$. Then $N \trianglelefteq X$, and N is a normal p -subgroup of X , strictly containing $\text{O}_p(X)$. This contradicts the maximality of $\text{O}_p(X)$. $\square$

Lemma 14.2.2 (Minimal normal subgroups of solvable groups). *Let X be a finite solvable group. If $1 \neq N \trianglelefteq X$ is a minimal normal subgroup of X , then*

$$N \cong (C_p)^d$$

for some prime p and some integer $d \geq 1$. In particular, N is an elementary abelian p -group.

Proof. Since X is solvable, every subgroup and quotient involved in the derived series is solvable. In particular, N is solvable. Since N is nontrivial and normal in X , the commutator subgroup N' is characteristic in N and hence normal in X . By minimality of N , either

$$N' = 1 \quad \text{or} \quad N' = N.$$

The second possibility is impossible because N is solvable. Therefore $N' = 1$, so N is abelian.

Let p be a prime divisor of $|N|$. The subgroup

$$N[p] = \{x \in N : x^p = 1\}$$

is nontrivial and characteristic in the abelian group N , hence normal in X . By minimality, $N[p] = N$. Therefore every nonidentity element of N has order p , and hence

$$N \cong (C_p)^d$$

for some $d \geq 1$. □

Lemma 14.2.3 (Hall subgroups pass to normal subgroups and quotients). *Let $N \trianglelefteq G$, and let H be a Hall π -subgroup of G . Then:*

- (1) *$H \cap N$ is a Hall π -subgroup of N , after replacing π by $\pi \cap \pi(N)$;*
- (2) *$HN/N \cong H/(H \cap N)$ is a Hall π -subgroup of G/N , after replacing π by $\pi \cap \pi(G/N)$.*

Proof. Since H is a π -group, $H \cap N$ is also a π -group. Moreover,

$$[N : H \cap N] \mid [G : H],$$

and $[G : H]$ is a π' -number. Hence $[N : H \cap N]$ is a π' -number, so $H \cap N$ is a Hall subgroup of N for the relevant set of primes.

Similarly,

$$HN/N \cong H/(H \cap N)$$

is a π -group, and

$$[G/N : HN/N] = [G : HN] \mid [G : H],$$

so this index is a π' -number. Therefore HN/N is a Hall subgroup of G/N . $\square$

Lemma 14.2.4 (A complement step). *Let G be a finite solvable group, and let*

$$M \trianglelefteq G$$

be a minimal normal subgroup. Suppose

$$M \cong (C_p)^d$$

and

$$p \nmid |G/M|.$$

Then M has a complement in G . That is, there exists a subgroup $L \leq G$ such that

$$G = ML \quad \text{and} \quad M \cap L = 1.$$

Equivalently,

$$L \cong G/M.$$

Proof. We argue by induction on $|G|$. The statement is clear if $G = M$.

Put

$$\overline{G} = G/M.$$

Since $\overline{G}$ is nontrivial and solvable, there exists a prime $q \in \pi(\overline{G})$ such that

$$O_q(\overline{G}) \neq 1.$$

Let

$$\overline{Q} = O_q(\overline{G}),$$

and let Y be the full preimage of $\overline{Q}$ under the quotient map $G \rightarrow \overline{G}$. Then

$$M \trianglelefteq Y \trianglelefteq G, \quad Y/M = \overline{Q}.$$

Choose $Q \in \text{Syl}_q(Y)$. Since M is a p -group and $p \neq q$, we have

$$Y = MQ \quad \text{and} \quad M \cap Q = 1.$$

We split into two cases.

Case 1: $C_M(Q) \neq 1$.

Since M is abelian,

$$C_M(Q) = Z(Y) \cap M.$$

Thus $C_M(Q)$ is the Sylow p -subgroup of $Z(Y)$. Hence it is characteristic in $Z(Y)$, and $Z(Y)$

is characteristic in Y . Since $Y \trianglelefteq G$, it follows that

$$C_M(Q) \trianglelefteq G.$$

By minimality of M ,

$$C_M(Q) = M.$$

Hence Q centralizes all of M , so

$$Y = M \times Q.$$

In particular, Q is characteristic in Y , and therefore

$$Q \trianglelefteq G.$$

Now apply induction to G/Q . The image MQ/Q is a minimal normal p -subgroup of G/Q , and

$$(G/Q)/(MQ/Q) \cong G/MQ$$

has order not divisible by p . Hence MQ/Q has a complement in G/Q . Taking the full preimage of this complement gives a complement to M in G .

Case 2: $C_M(Q) = 1$.

We first observe that

$$N_M(Q) = C_M(Q) = 1.$$

Indeed, if $x \in M$ normalizes Q , then for every $q \in Q$,

$$xqx^{-1}q^{-1} \in M \cap Q = 1,$$

so x centralizes Q .

Since $Y \trianglelefteq G$ and $Q \in \text{Syl}_q(Y)$, the Frattini argument gives

$$G = YN_G(Q).$$

Because $Y = MQ$ and $Q \leq N_G(Q)$, this becomes

$$G = MN_G(Q).$$

Moreover,

$$M \cap N_G(Q) = N_M(Q) = 1.$$

Therefore $N_G(Q)$ is a complement to M in G . □

Theorem 14.2.5 (Hall's theorem: existence criterion). *Let G be a finite group, and write*

$$|G| = p_1^{a_1} p_2^{a_2} \cdots p_t^{a_t},$$

where $p_1, \dots, p_t$ are distinct primes. Then G is solvable if and only if G has a Hall

π -subgroup for every subset

$$\pi \subseteq \pi(G).$$

Proof. We prove both directions.

First direction: assume G has Hall π -subgroups for every $\pi \subseteq \pi(G)$. We prove that G is solvable.

We use induction on $|G|$. If $|\pi(G)| = 1$, then G is a p -group, hence solvable. If $|\pi(G)| = 2$, then

$$|G| = p^a q^b$$

for two primes p, q , so G is solvable by Burnside's $p^a q^b$ -theorem.

Now assume $|\pi(G)| = t \geq 3$. For $1 \leq i \leq t$, let

$$H_i$$

be a Hall p'_i -subgroup of G , where

$$p'_i = \pi(G) \setminus \{p_i\}.$$

We first note that each H_i is solvable. Indeed, H_i inherits Hall subgroups as follows. If $\sigma \subseteq \pi(H_i)$, take a Hall $\sigma \cup \{p_i\}$ -subgroup L of G . Since

$$[G : H_i]$$

is a power of p_i , while

$$[G : L]$$

is a $(\sigma \cup \{p_i\})'$ -number, these two indices are coprime. Hence

$$G = H_i L.$$

Therefore

$$|H_i \cap L| = \frac{|H_i||L|}{|G|},$$

and this is precisely the σ -part of $|H_i|$. Thus $H_i \cap L$ is a Hall σ -subgroup of H_i . By induction, H_i is solvable.

In particular, H_1 is solvable. Hence H_1 contains a nontrivial minimal normal subgroup, so there exists a prime $p \in \pi(H_1)$ such that

$$\mathrm{O}_p(H_1) \neq 1.$$

After relabelling the primes if necessary, we may assume

$$p \in \{p_3, \dots, p_t\}.$$

Thus $p \mid |H_2|$. Let $P \in \mathrm{Syl}_p(H_2)$. Then P is also a Sylow p -subgroup of G . By Sylow's

theorem, there exists $g \in G$ such that

$$O_p(H_1) \leq P^g \leq H_2^g.$$

Now

$$|H_1| = p_2^{a_2} p_3^{a_3} \cdots p_t^{a_t},$$

and

$$|H_2^g| = p_1^{a_1} p_3^{a_3} \cdots p_t^{a_t}.$$

Moreover,

$$|H_1 \cap H_2^g| \leq p_3^{a_3} \cdots p_t^{a_t}.$$

Therefore

$$\begin{aligned} |H_1 H_2^g| &= \frac{|H_1| |H_2^g|}{|H_1 \cap H_2^g|} \\ &\geq \frac{(p_2^{a_2} p_3^{a_3} \cdots p_t^{a_t})(p_1^{a_1} p_3^{a_3} \cdots p_t^{a_t})}{p_3^{a_3} \cdots p_t^{a_t}} \\ &= |G|. \end{aligned}$$

Since $H_1 H_2^g \subseteq G$, we must have

$$G = H_1 H_2^g.$$

Let

$$N = \langle O_p(H_1)^x \mid x \in G \rangle$$

be the normal closure of $O_p(H_1)$ in G . Then

$$1 \neq N \leq G.$$

We claim that $N < G$. Since $G = H_1 H_2^g$, every $x \in G$ may be written as

$$x = ab, \quad a \in H_1, \quad b \in H_2^g.$$

Because $O_p(H_1) \trianglelefteq H_1$,

$$O_p(H_1)^x = O_p(H_1)^{ab} = O_p(H_1)^b \leq H_2^g.$$

Hence all conjugates of $O_p(H_1)$ lie inside H_2^g , and therefore

$$N \leq H_2^g < G.$$

Thus N is a proper nontrivial normal subgroup of G .

By the previous lemma, normal subgroups and quotient groups inherit the Hall existence property from G . Hence both N and G/N have Hall π -subgroups for every relevant set of primes. By induction, N and G/N are solvable. Therefore G is solvable.

Second direction: assume G is solvable. We prove that all Hall π -subgroups exist.

Again we use induction on $|G|$. Let

$$M \trianglelefteq G$$

be a minimal normal subgroup. Since G is solvable,

$$M \cong (C_p)^d$$

for some prime p and some $d \geq 1$. Put

$$\overline{G} = G/M.$$

By induction, $\overline{G}$ has Hall subgroups for every subset of $\pi(\overline{G})$.

Fix $\pi \subseteq \pi(G)$.

Case 1: $p \in \pi$.

Let $\overline{H}$ be a Hall

$$\pi \cap \pi(\overline{G})$$

subgroup of $\overline{G}$. Let X be the full preimage of $\overline{H}$ under the quotient map $G \rightarrow \overline{G}$. Then

$$M \leq X \leq G$$

and

$$X/M = \overline{H}.$$

Since M is a p -group and $p \in \pi$, the subgroup X contains the full p -part of G , and $\overline{H}$ contains the full π -part of $\overline{G}$. Hence X is a Hall π -subgroup of G .

Case 2: $p \notin \pi$.

Let $\overline{H}$ be a Hall π -subgroup of $\overline{G}$, and let X be its full preimage in G . Then

$$X/M = \overline{H},$$

and M is a normal p -subgroup of X , with $p \nmid |\overline{H}|$.

If $X < G$, then X is solvable and has smaller order than G . By induction, X has a Hall π -subgroup H . Since $X/M = \overline{H}$ contains the full π -part of G/M , this H is also a Hall π -subgroup of G .

If $X = G$, then

$$G/M = \overline{H}$$

is a π -group and $p \nmid |G/M|$. By the complement step, M has a complement H in G . Thus

$$G = MH, \quad M \cap H = 1, \quad H \cong G/M.$$

Since G/M is a π -group and M is a p -group with $p \notin \pi$, this complement H is a Hall π -subgroup of G .

Thus, in every case, G has a Hall π -subgroup. □

14.3 Conjugacy of Hall Subgroups in Solvable Groups

Proposition 14.3.1 (Conjugacy of Hall subgroups). *Let G be a finite solvable group, and let $\pi \subseteq \pi(G)$. Then all Hall π -subgroups of G are conjugate in G .*

Proof. We use induction on $|G|$. Let H and K be Hall π -subgroups of G .

Since G is solvable, there exists a maximal normal subgroup

$$M \trianglelefteq G$$

such that

$$|G/M| = p$$

for some prime p . Equivalently,

$$G/M \cong C_p.$$

Case 1: $p \notin \pi$.

Then $H \leq M$ and $K \leq M$. Indeed, if $H \not\leq M$, then

$$HM = G,$$

so

$$H/(H \cap M) \cong G/M,$$

and hence $p \mid |H|$, contradicting $p \notin \pi$.

Thus H and K are Hall π -subgroups of the solvable group M . By induction, they are conjugate in M , hence also conjugate in G .

Case 2: $p \in \pi$.

Then

$$HM = G \quad \text{and} \quad KM = G.$$

Moreover,

$$H \cap M \quad \text{and} \quad K \cap M$$

are Hall π -subgroups of M . By induction, there exists $x \in M$ such that

$$(H \cap M)^x = K \cap M.$$

Replacing H by H^x , we may assume

$$H \cap M = K \cap M.$$

Set

$$L = H \cap M = K \cap M.$$

Then $L \trianglelefteq H$ and $L \trianglelefteq K$, because $M \trianglelefteq G$. Hence

$$H, K \leq N_G(L).$$

If

$$N_G(L) < G,$$

then H and K are Hall π -subgroups of the solvable group $N_G(L)$. By induction, H and K are conjugate in $N_G(L)$, hence in G .

It remains to consider the case

$$N_G(L) = G.$$

Then $L \trianglelefteq G$. Passing to the quotient G/L , we have

$$H/L \cong G/M \cong C_p \quad \text{and} \quad K/L \cong G/M \cong C_p.$$

Thus H/L and K/L are Sylow p -subgroups of G/L . By Sylow's conjugacy theorem, there exists $g \in G$ such that

$$(H/L)^g = K/L.$$

Since $L \leq H, K$, this gives

$$H^g = K.$$

Therefore H and K are conjugate in G . □

Remark 14.3.2. *For finite solvable groups, Hall subgroups behave like Sylow subgroups in two essential ways:*

- (1) *Hall π -subgroups exist for every $\pi \subseteq \pi(G)$;*
- (2) *any two Hall π -subgroups are conjugate.*

Both statements can fail for nonsolvable groups.

Chapter 15

Nilpotent Groups, Frattini Subgroups, and Fitting Subgroups

15.1 The Frattini Argument

Theorem 15.1.1 (Frattini Argument). *Let G be a finite group, and let $N \trianglelefteq G$. If*

$$P \in \text{Syl}_p(N),$$

then

$$G = N_G(P)N.$$

Proof. Since $N \trianglelefteq G$, conjugation by every element of G sends Sylow p -subgroups of N to Sylow p -subgroups of N . Hence G acts by conjugation on $\text{Syl}_p(N)$.

By the Sylow theorem, N acts transitively on $\text{Syl}_p(N)$. Let $g \in G$. Then $gPg^{-1} \in \text{Syl}_p(N)$, so there exists $n \in N$ such that

$$gPg^{-1} = nPn^{-1}.$$

Thus

$$n^{-1}gPg^{-1}n = P,$$

so $n^{-1}g \in N_G(P)$. Hence $g \in nN_G(P) \subseteq NN_G(P)$. Therefore

$$G = NN_G(P) = N_G(P)N,$$

because $N \trianglelefteq G$. □

15.2 Nilpotent Groups

Definition 15.2.1 (Central Series and Nilpotent Groups). *A group G is called **nilpotent** if there exists a chain of subgroups*

$$G = K_1 > K_2 > \cdots > K_s > K_{s+1} = 1$$

such that

$$K_i \leq G$$

and

$$K_i/K_{i+1} \leq Z(G/K_{i+1})$$

*for every i . Such a series is called a **central series**.*

Equivalently, the condition

$$K_i/K_{i+1} \leq Z(G/K_{i+1})$$

means

$$[G, K_i] \leq K_{i+1}.$$

Proposition 15.2.2. *Every finite p -group is nilpotent.*

Proof. Let G be a nontrivial finite p -group. Then

$$Z(G) \neq 1.$$

Define the upper central series

$$1 = Z_0(G) \leq Z_1(G) \leq Z_2(G) \leq \cdots$$

by

$$Z_{i+1}(G)/Z_i(G) = Z(G/Z_i(G)).$$

If $Z_i(G) < G$, then $G/Z_i(G)$ is a nontrivial p -group, so it has nontrivial center. Hence $Z_{i+1}(G) > Z_i(G)$. Since G is finite, this process must eventually reach G . Therefore G has a central series and is nilpotent. $\square$

Proposition 15.2.3. *Subgroups and quotient groups of nilpotent groups are nilpotent.*

Proof. Let

$$G = K_1 > K_2 > \cdots > K_s > K_{s+1} = 1$$

be a central series for G .

If $H \leq G$, then

$$H = H \cap K_1 \geq H \cap K_2 \geq \cdots \geq H \cap K_{s+1} = 1$$

is a central series for H , after deleting repetitions. Indeed,

$$[H, H \cap K_i] \leq [G, K_i] \leq K_{i+1},$$

so

$$[H, H \cap K_i] \leq H \cap K_{i+1}.$$

If $N \trianglelefteq G$, then

$$G/N = K_1N/N \geq K_2N/N \geq \cdots \geq K_{s+1}N/N = N/N$$

is a central series for G/N , again after deleting repetitions. $\square$

Proposition 15.2.4. *A finite direct product of nilpotent groups is nilpotent.*

Proof. It is enough to consider $G = A \times B$. If

$$A = A_1 > \cdots > A_{r+1} = 1$$

and

$$B = B_1 > \cdots > B_{t+1} = 1$$

are central series, then one obtains a central series for $A \times B$ by first descending through the $A_i \times B$, and then through the $1 \times B_j$. The centrality condition follows componentwise. $\square$

Example 15.2.5. *Let*

$$G = P \times Q,$$

where P is a p -group, Q is a q -group, and $p \neq q$. Since finite p -groups and finite q -groups are nilpotent, G is nilpotent. Moreover

$$Z(G) = Z(P) \times Z(Q),$$

and

$$G/Z(G) \cong P/Z(P) \times Q/Z(Q).$$

Repeating this construction gives a central series for G .

Example 15.2.6 (Non-nilpotent groups). *The following groups are not nilpotent.*

(1) $A_4 \cong C_2^2 \rtimes C_3$. *Its only nontrivial proper normal subgroup is*

$$V_4 \cong C_2^2.$$

If A_4 were nilpotent, the last nontrivial term of a central series would be contained in $Z(A_4)$. But

$$Z(A_4) = 1.$$

Hence A_4 is not nilpotent.

(2) S_4 . Its proper nontrivial normal subgroups are

$$V_4 \quad \text{and} \quad A_4.$$

Since

$$Z(S_4) = 1,$$

S_4 cannot have a central series. Therefore S_4 is not nilpotent.

(3) Let $p > 3$ be prime, and let

$$D_{2p} = \langle a, b \mid a^p = b^2 = 1, bab^{-1} = a^{-1} \rangle.$$

The only nontrivial proper normal subgroup is $\langle a \rangle$, but

$$\langle a \rangle \not\leq Z(D_{2p}).$$

Hence D_{2p} is not nilpotent.

Theorem 15.2.7 (Equivalent Conditions for Nilpotency). *For a finite group G , the following are equivalent.*

(1) G is nilpotent.

(2) For every proper subgroup $H < G$,

$$H < N_G(H).$$

(3) Every maximal subgroup M of G is normal and has prime index. Equivalently,

$$G/M \cong C_p$$

for some prime p .

(4) G is the direct product of its Sylow subgroups. Equivalently, every Sylow subgroup of G is normal.

Proof. (1) $\Rightarrow$ (2). Let

$$G = K_1 > K_2 > \cdots > K_s > K_{s+1} = 1$$

be a central series for G , and let $H < G$. Choose i such that

$$K_{i+1} \leq H \quad \text{but} \quad K_i \not\leq H.$$

Since

$$K_i/K_{i+1} \leq Z(G/K_{i+1}),$$

we have

$$[H, K_i] \leq K_{i+1} \leq H.$$

Therefore $K_i \leq N_G(H)$. Since $K_i \not\leq H$, it follows that

$$H < N_G(H).$$

(2) $\Rightarrow$ (3). Let M be a maximal subgroup of G . By (2),

$$M < N_G(M) \leq G.$$

By maximality of M , this forces

$$N_G(M) = G.$$

Thus $M \trianglelefteq G$.

Since M is maximal, G/M has no proper nontrivial subgroup. Hence G/M is cyclic of prime order, say

$$G/M \cong C_p.$$

(3) $\Rightarrow$ (4). Let $P \in \text{Syl}_p(G)$. Then

$$P \leq N_G(P) \leq G.$$

Suppose $N_G(P) \neq G$. Choose a maximal subgroup M such that

$$N_G(P) \leq M < G.$$

By (3), $M \trianglelefteq G$. Since $P \leq M$ and P is a Sylow p -subgroup of G , P is also a Sylow p -subgroup of M . By the Frattini argument applied to $M \trianglelefteq G$,

$$G = MN_G(P).$$

But $N_G(P) \leq M$, so

$$G = M,$$

a contradiction. Therefore

$$N_G(P) = G,$$

so $P \trianglelefteq G$. Thus every Sylow subgroup of G is normal.

If the Sylow subgroups are

$$P_1, \dots, P_t,$$

then they commute pairwise because

$$[P_i, P_j] \leq P_i \cap P_j = 1 \quad (i \neq j).$$

Hence

$$G = P_1 \times \cdots \times P_t.$$

(4) $\Rightarrow$ (1). Each Sylow subgroup is a finite p -group, hence nilpotent. A finite direct product of nilpotent groups is nilpotent. Therefore G is nilpotent. $\square$

Example 15.2.8. *Every finite abelian group is nilpotent.*

Example 15.2.9. *The group*

$$A_4 \cong C_2^2 \rtimes C_3$$

is not nilpotent. Indeed, a subgroup $C_3 \leq A_4$ is maximal, but it has index 4, which is not prime, and it is not normal in A_4 .

15.3 The Frattini Subgroup

Definition 15.3.1 (Frattini Subgroup). *Let G be a finite group. Denote by*

$$\text{Max}(G) = \{M \mid M < G, M \text{ maximal}\}$$

*the set of maximal subgroups of G . The **Frattini subgroup** of G is*

$$\Phi(G) = \bigcap_{M \in \text{Max}(G)} M.$$

For the trivial group, we use the convention

$$\Phi(1) = 1.$$

Proposition 15.3.2. *The Frattini subgroup $\Phi(G)$ is characteristic in G .*

Proof. Every automorphism of G permutes the maximal subgroups of G . Hence it fixes their intersection. Therefore

$$\Phi(G) \text{ Char}(G).$$

$\square$

Corollary 15.3.3. *If $N \trianglelefteq G$, then*

$$\Phi(N) \trianglelefteq G.$$

Proof. Since $\Phi(N) \text{ Char}(N)$ and $N \trianglelefteq G$, we have

$$\Phi(N) \trianglelefteq G.$$

□

Definition 15.3.4 (Nongenerator). *An element $x \in G$ is called a **nongenerator** of G if, for every subset $S \subseteq G$,*

$$\langle x, S \rangle = G \implies \langle S \rangle = G.$$

Lemma 15.3.5 (Characterization of the Frattini Subgroup). *The Frattini subgroup $\Phi(G)$ consists exactly of the nongenerators of G .*

Proof. First suppose $x \in \Phi(G)$. Assume

$$\langle x, S \rangle = G.$$

If $\langle S \rangle \neq G$, then $\langle S \rangle$ is contained in some maximal subgroup $M < G$. Since $x \in \Phi(G)$, we have $x \in M$. Hence

$$\langle x, S \rangle \leq M < G,$$

contradicting $\langle x, S \rangle = G$. Therefore $\langle S \rangle = G$, so x is a nongenerator.

Conversely, suppose x is a nongenerator. If $x \notin \Phi(G)$, then there is a maximal subgroup $M < G$ such that $x \notin M$. By maximality,

$$\langle x, M \rangle = G.$$

Since x is a nongenerator, this implies

$$\langle M \rangle = G,$$

which is impossible because $M < G$. Thus $x \in M$ for every maximal subgroup M , and hence

$$x \in \Phi(G).$$

□

Corollary 15.3.6. *If $H \leq G$ and*

$$H\Phi(G) = G,$$

then

$$H = G.$$

Proof. If $H < G$, then H is contained in some maximal subgroup $M < G$. Since

$$\Phi(G) \leq M,$$

we get

$$H\Phi(G) \leq M < G,$$

contradicting $H\Phi(G) = G$. Therefore $H = G$. □

Example 15.3.7. *Let*

$$G = \langle a \rangle \cong C_{p^2}.$$

Then

$$\Phi(G) = \langle a^p \rangle.$$

Thus every element of $\langle a^p \rangle$ is a nongenerator.

Example 15.3.8. *For the quaternion group Q_8 ,*

$$\Phi(Q_8) = Z(Q_8) = \{\pm 1\}.$$

Thus the nongenerators of Q_8 have order 1 or 2.

Example 15.3.9. *Let*

$$D_{2n} = \langle r, s \mid r^n = s^2 = 1, srs = r^{-1} \rangle$$

be the dihedral group of order $2n$. Then

$$\Phi(D_{2n}) = \langle r^{\text{rad}(n)} \rangle,$$

where $\text{rad}(n)$ is the product of the distinct prime divisors of n .

Example 15.3.10. *If G is a finite nonabelian simple group, then*

$$\Phi(G) = 1.$$

Indeed, $\Phi(G) \trianglelefteq G$, and $\Phi(G) \neq G$ because G has maximal subgroups. Thus $\Phi(G) = 1$. Hence no nonidentity element of G is a nongenerator.

A stronger theorem says that for every nonidentity $a \in G$, there exists $b \in G$ such that

$$\langle a, b \rangle = G.$$

15.4 Frattini Subgroups and Nilpotency

Theorem 15.4.1 (Gaschütz). *Let $N \trianglelefteq G$ and suppose*

$$N \leq \Phi(G).$$

Then

$$G \text{ is nilpotent} \iff G/N \text{ is nilpotent.}$$

Proof. If G is nilpotent, then every quotient of G is nilpotent, so G/N is nilpotent.

Conversely, suppose G/N is nilpotent. Let $P \in \text{Syl}_p(G)$. We show that $P \trianglelefteq G$.

Since $N \trianglelefteq G$, the subgroup

$$PN/N$$

is a Sylow p -subgroup of G/N . Because G/N is nilpotent, all of its Sylow subgroups are normal. Hence

$$PN/N \trianglelefteq G/N,$$

so

$$PN \trianglelefteq G.$$

Now $P \in \text{Syl}_p(PN)$. Applying the Frattini argument to $PN \trianglelefteq G$, we obtain

$$G = N_G(P)PN.$$

Since $P \leq N_G(P)$, this becomes

$$G = N_G(P)N.$$

But $N \leq \Phi(G)$, so

$$N_G(P)\Phi(G) = G.$$

By the previous corollary,

$$N_G(P) = G.$$

Thus $P \trianglelefteq G$. Since this holds for every prime p , all Sylow subgroups of G are normal. Therefore G is nilpotent. $\square$

Corollary 15.4.2. *For a finite group G , the following statements hold.*

- (1) $\Phi(G)$ is nilpotent.
- (2) G is nilpotent if and only if $G/\Phi(G)$ is nilpotent.
- (3) (Wielandt) G is nilpotent if and only if

$$G' \leq \Phi(G).$$

(4) If G is a p -group, then

$$G/\Phi(G)$$

is elementary abelian.

Proof. (1). Let

$$P \in \text{Syl}_p(\Phi(G)).$$

Since $\Phi(G) \trianglelefteq G$, the Frattini argument gives

$$G = \Phi(G)N_G(P).$$

Because $\Phi(G) \leq \Phi(G)$, the corollary above gives

$$N_G(P) = G.$$

Hence $P \trianglelefteq G$, and in particular $P \trianglelefteq \Phi(G)$. Therefore every Sylow subgroup of $\Phi(G)$ is normal, so $\Phi(G)$ is nilpotent.

(2). Apply Gaschütz's theorem with

$$N = \Phi(G).$$

(3). Suppose G is nilpotent. Let $M < G$ be maximal. Then $M \trianglelefteq G$, and

$$G/M \cong C_p$$

for some prime p . In particular, G/M is abelian, so

$$G' \leq M.$$

Since this holds for every maximal subgroup M , we get

$$G' \leq \Phi(G).$$

Conversely, suppose

$$G' \leq \Phi(G).$$

Then

$$(G/\Phi(G))' = 1,$$

so $G/\Phi(G)$ is abelian, hence nilpotent. By (2), G is nilpotent.

(4). Let G be a p -group. By (3),

$$G' \leq \Phi(G),$$

so $G/\Phi(G)$ is abelian.

It remains to show that every element has order dividing p . Let $x \in G$. For every

maximal subgroup $M < G$, we have

$$G/M \cong C_p.$$

Hence

$$x^p \in M.$$

Since this holds for every maximal subgroup M ,

$$x^p \in \Phi(G).$$

Thus every element of $G/\Phi(G)$ has order dividing p . Therefore $G/\Phi(G)$ is elementary abelian. $\square$

Example 15.4.3. *Let*

$$G = C_{p^2} \rtimes C_q = \langle a \rangle \rtimes \langle b \rangle,$$

where $p \neq q$ are primes, $|a| = p^2$, and $|b| = q$. Write

$$b^{-1}ab = a^u$$

for some

$$u \in (\mathbb{Z}/p^2\mathbb{Z})^\times$$

satisfying

$$u^q \equiv 1 \pmod{p^2}.$$

Then

$$a^pb = ba^p$$

if and only if

$$u \equiv 1 \pmod{p}.$$

But the kernel of

$$(\mathbb{Z}/p^2\mathbb{Z})^\times \longrightarrow (\mathbb{Z}/p\mathbb{Z})^\times$$

has order p . Since $q \neq p$, the condition $u^q \equiv 1 \pmod{p^2}$ and $u \equiv 1 \pmod{p}$ forces

$$u \equiv 1 \pmod{p^2}.$$

Therefore

$$a^pb = ba^p \iff ab = ba.$$

Equivalently,

$$G/\langle a^p \rangle \text{ is abelian} \iff G \text{ is abelian.}$$

15.5 The Fitting Subgroup

Definition 15.5.1. Let p be a prime. The subgroup $O_p(G)$ is defined to be the largest normal p -subgroup of G . Equivalently,

$$O_p(G) = \prod_{\substack{N \trianglelefteq G \\ N \text{ a } p\text{-group}}} N = \bigcap_{P \in \text{Syl}_p(G)} P.$$

Proposition 15.5.2. For every prime p , the subgroup $O_p(G)$ is characteristic in G . Moreover,

$$O_p(G/O_p(G)) = 1.$$

If $p \neq q$, then

$$[O_p(G), O_q(G)] = 1,$$

and hence

$$O_p(G)O_q(G) = O_p(G) \times O_q(G).$$

Proof. The subgroup $O_p(G)$ is uniquely characterized as the largest normal p -subgroup of G , so it is characteristic.

Let $\overline{G} = G/O_p(G)$. If $O_p(\overline{G}) \neq 1$, then its full preimage in G is a normal p -subgroup strictly larger than $O_p(G)$, contradicting maximality. Hence

$$O_p(\overline{G}) = 1.$$

Finally, if $p \neq q$, then

$$[O_p(G), O_q(G)] \leq O_p(G) \cap O_q(G) = 1.$$

Therefore $O_p(G)$ and $O_q(G)$ centralize each other, and their product is direct. $\square$

Lemma 15.5.3 (Fitting's Lemma). If $H \trianglelefteq G$ and $K \trianglelefteq G$ are nilpotent normal subgroups of G , then

$$HK$$

is a nilpotent normal subgroup of G .

Proof. Normality of HK is clear. It remains to prove nilpotency.

Replacing G by HK , we may assume

$$G = HK.$$

We argue by induction on the sum of the nilpotency classes of H and K .

Since $H \trianglelefteq G$, its center $Z(H)$ is characteristic in H , hence normal in G . The quotient

$$G/Z(H)$$

is generated by the nilpotent normal subgroups

$$H/Z(H) \quad \text{and} \quad KZ(H)/Z(H).$$

The nilpotency class of $H/Z(H)$ is smaller than that of H , so by induction $G/Z(H)$ is nilpotent. Similarly, $G/Z(K)$ is nilpotent.

Thus for sufficiently large m ,

$$\gamma_m(G) \leq Z(H) \cap Z(K),$$

where $\gamma_m(G)$ denotes the m -th term of the lower central series. But

$$Z(H) \cap Z(K) \leq Z(G),$$

because $G = HK$. Hence

$$\gamma_{m+1}(G) = 1.$$

Therefore G is nilpotent. □

Definition 15.5.4 (Fitting Subgroup). *The **Fitting subgroup** of G , denoted*

$$F(G),$$

is the product of all normal nilpotent subgroups of G . By Fitting's lemma, $F(G)$ is the unique largest nilpotent normal subgroup of G .

Since $\Phi(G)$ is nilpotent and normal, we always have

$$\Phi(G) \leq F(G).$$

Theorem 15.5.5. *Let G be a finite solvable group, and let*

$$|G| = p_1^{e_1} p_2^{e_2} \cdots p_t^{e_t}.$$

Let

$$F = F(G).$$

Then the following hold.

(1)

$$F = O_{p_1}(G) \times O_{p_2}(G) \times \cdots \times O_{p_t}(G).$$

(2)

$$C_G(F) \leq F.$$

In other words, $F(G)$ is self-centralizing.

(3) Consequently,

$$G/C_G(F) \hookrightarrow \text{Aut}(F).$$

Proof. (1). Let $N \trianglelefteq G$ be nilpotent. Then N is the direct product of its Sylow subgroups:

$$N = Q_1 \times Q_2 \times \cdots \times Q_t,$$

where $Q_i \in \text{Syl}_{p_i}(N)$. Since N is nilpotent, each Q_i is characteristic in N . Since $N \trianglelefteq G$, it follows that

$$Q_i \trianglelefteq G.$$

Thus

$$Q_i \leq O_{p_i}(G).$$

Therefore every nilpotent normal subgroup N of G is contained in

$$O_{p_1}(G) \times \cdots \times O_{p_t}(G).$$

Conversely, the product

$$O_{p_1}(G) \times \cdots \times O_{p_t}(G)$$

is a nilpotent normal subgroup of G . Hence it is contained in $F(G)$. This proves

$$F(G) = O_{p_1}(G) \times \cdots \times O_{p_t}(G).$$

(2). Let

$$C = C_G(F).$$

Since $F \text{ Char}(G)$, the subgroup C is normal in G . Also

$$C \cap F = Z(F).$$

Set

$$W = C \cap F.$$

Suppose, for contradiction, that

$$C > W.$$

Then C/W is a nontrivial solvable normal subgroup of G/W . Hence it has a nontrivial abelian normal subgroup. In particular, there exists a subgroup Y such that

$$W < Y \leq C, \quad Y \trianglelefteq G,$$

and

$$Y/W$$

is a nontrivial p -group for some prime p .

Since $Y \leq C = C_G(F)$, the subgroup Y centralizes F , and hence centralizes $W \leq F$.

Thus

$$W \leq Z(Y).$$

Now Y/W is a p -group, hence nilpotent, and $W \leq Z(Y)$ is nilpotent. Therefore Y is nilpotent. Since $Y \trianglelefteq G$, maximality of $F(G)$ gives

$$Y \leq F.$$

But $Y \leq C$, so

$$Y \leq C \cap F = W,$$

contradicting $W < Y$. Hence

$$C = W \leq F.$$

Thus

$$C_G(F) \leq F.$$

(3). The conjugation action of G on F gives a homomorphism

$$G \longrightarrow \text{Aut}(F).$$

Its kernel is precisely

$$C_G(F).$$

Therefore

$$G/C_G(F) \hookrightarrow \text{Aut}(F).$$

□

Example 15.5.6 (Groups with Cyclic Sylow Subgroups). *Assume G is a finite group whose Sylow subgroups are all cyclic. A standard theorem says that such a group is solvable.*

Let

$$F = F(G).$$

Since each $O_p(G)$ is a subgroup of a cyclic Sylow p -subgroup, each $O_p(G)$ is cyclic.

Therefore

$$F = \prod_{p||G|} O_p(G)$$

is cyclic.

By the preceding theorem,

$$C_G(F) \leq F.$$

Since F is abelian, we also have

$$F \leq C_G(F).$$

Thus

$$C_G(F) = F.$$

Hence

$$G/F \hookrightarrow \text{Aut}(F).$$

Because F is cyclic, $\text{Aut}(F)$ is abelian. Therefore G/F is abelian.

Moreover, the Sylow subgroups of G/F are cyclic. A finite abelian group whose Sylow subgroups are cyclic is cyclic. Hence G/F is cyclic.

Thus G is metacyclic: there exists a cyclic normal subgroup $F \trianglelefteq G$ such that

$$G/F$$

is cyclic.

Chapter 16

Group Theory Exercises

Exercise 16.0.1. Find conjugacy classes and fusion classes of A_5 .

Proof. The conjugacy classes of A_5 are obtained by intersecting the S_5 cycle-type classes with A_5 . The result is

$$\{1\},$$

the 15 double transpositions,

$$(ab)(cd),$$

the 20 3-cycles, and two classes of 5-cycles, each of size 12. For example, if $x = (1\ 2\ 3\ 4\ 5)$, then the two A_5 -classes of 5-cycles are represented by x and x^2 .

The fusion classes are the $\text{Aut}(A_5)$ -orbits. Since

$$\text{Aut}(A_5) \cong S_5,$$

these are just the S_5 -conjugacy classes contained in A_5 :

$$\{1\}, \quad \{x \in A_5 : |x| = 2\}, \quad \{x \in A_5 : |x| = 3\}, \quad \{x \in A_5 : |x| = 5\}.$$

Thus the two A_5 -classes of 5-cycles fuse under $\text{Aut}(A_5)$. □

Exercise 16.0.2. Determine the conjugacy classes and fusion classes of

$$(\mathbb{Z}/p^e\mathbb{Z})^d \quad \text{and} \quad \mathbb{Z}/p^a\mathbb{Z} \times \mathbb{Z}/p^b\mathbb{Z}.$$

Proof. Both groups are abelian. Hence every conjugacy class is a singleton. Therefore it remains only to determine the fusion classes, that is, the orbits under the automorphism group.

First let

$$G = (\mathbb{Z}/p^e\mathbb{Z})^d.$$

Put

$$R = \mathbb{Z}/p^e\mathbb{Z}.$$

Then G is a free R -module of rank d , and

$$\text{Aut}(G) = \text{GL}_d(R).$$

For $0 \neq v \in G$, define

$$\nu(v) = \max\{i \mid v \in p^i G\}.$$

Equivalently,

$$v \in p^i G \setminus p^{i+1} G \iff \nu(v) = i.$$

Since each subgroup $p^i G$ is characteristic in G , the integer $\nu(v)$ is invariant under $\text{Aut}(G)$.

Conversely, suppose

$$v \in p^i G \setminus p^{i+1} G.$$

Then

$$v = p^i u$$

for some $u \in G \setminus pG$. Since R is local with maximal ideal (p) , the condition $u \notin pG$ means that u is a primitive vector. Hence u can be extended to an R -basis of G . Therefore some element of $\text{GL}_d(R)$ sends u to e_1 , and hence sends v to

$$p^i e_1.$$

Thus the fusion classes of $G = (\mathbb{Z}/p^e \mathbb{Z})^d$ are

$$\{0\}, \quad p^i G \setminus p^{i+1} G \quad (0 \leq i < e).$$

Since an element of $p^i G \setminus p^{i+1} G$ has order p^{e-i} , this is equivalently the statement that two nonzero elements are fused if and only if they have the same order.

Now let

$$G = \langle x \rangle \oplus \langle y \rangle \cong C_{p^a} \times C_{p^b}, \quad a \geq b,$$

where

$$|x| = p^a, \quad |y| = p^b.$$

We shall write

$$p^a x = 0, \quad p^b y = 0.$$

We claim that the fusion classes are represented uniquely by the elements

$$p^i x + p^j y,$$

where

$$0 \leq i \leq a, \quad 0 \leq j \leq b,$$

and the pair (i, j) satisfies the canonical conditions

$$j \leq i, \quad b - j \leq a - i.$$

Equivalently,

$$j \leq i \leq a - b + j.$$

Thus the fusion classes are precisely

$$\text{Aut}(G) \cdot (p^i x + p^j y)$$

for all such canonical pairs (i, j) .

We prove existence first. We shall use the following elementary automorphisms:

$$x \mapsto ux, \quad y \mapsto vy,$$

where $u \in (\mathbb{Z}/p^a\mathbb{Z})^\times$ and $v \in (\mathbb{Z}/p^b\mathbb{Z})^\times$, and also

$$x \mapsto x + cy, \quad y \mapsto y,$$

as well as

$$x \mapsto x, \quad y \mapsto y + p^{a-b}dx.$$

These are automorphisms because their inverses are obtained by replacing c or d by $-c$ or $-d$, and because $p^{a-b}x$ has order dividing p^b .

Let

$$g = \alpha x + \beta y \in G.$$

If $g = 0$, then

$$g = p^a x + p^b y,$$

which corresponds to the canonical pair (a, b) .

Assume now that $g \neq 0$. Define

$$r = v_p(\alpha), \quad s = v_p(\beta),$$

with the conventions

$$r = a \quad \text{if } \alpha = 0, \quad s = b \quad \text{if } \beta = 0.$$

Then the order of g is

$$|g| = p^m, \quad m = \max\{a - r, b - s\}.$$

There are two cases.

First suppose

$$a - r \geq b - s.$$

Then the x -component has maximal order. In this case $r < a$. By scaling x , we may assume that the x -coefficient is exactly p^r . Thus

$$g \sim p^r x + \beta y.$$

If $s < r$, then either $\beta = 0$, in which case $s = b$, or we may scale y so that $\beta = p^s$. Hence

$$g \sim p^r x + p^s y.$$

The pair (r, s) is canonical, since

$$s < r$$

and

$$b - s \leq a - r$$

by assumption.

If $s \geq r$, then $r \leq b$. If $r = b$, then $\beta = 0$, and

$$g \sim p^r x = p^r x + p^b y.$$

The pair (r, b) is canonical. If $r < b$, write

$$\beta = p^r w \quad \text{in } \mathbb{Z}/p^b \mathbb{Z}.$$

Using the automorphism

$$x \mapsto x + cy, \quad y \mapsto y,$$

we may replace the y -coefficient by

$$p^r(w + c).$$

Choosing c so that

$$w + c \equiv 1 \pmod{p^{b-r}},$$

we obtain

$$g \sim p^r x + p^r y.$$

The pair (r, r) is canonical.

Now suppose

$$b - s > a - r.$$

Then the y -component has maximal order, so $s < b$. Put

$$i = a - b + s.$$

The inequality

$$b - s > a - r$$

is equivalent to

$$r > i.$$

After scaling y , we may assume

$$g = \alpha x + p^s y.$$

Using the automorphism

$$x \mapsto x, \quad y \mapsto y + p^{a-b}x,$$

we obtain

$$g \mapsto (\alpha + p^{s+a-b})x + p^s y = (\alpha + p^i)x + p^s y.$$

Since $r > i$, the element α is divisible by p^{i+1} . Hence

$$\alpha + p^i = p^i u$$

for some unit u . Scaling x , we get

$$g \sim p^i x + p^s y.$$

The pair (i, s) is canonical because

$$s \leq a - b + s = i$$

and

$$b - s = a - i.$$

This proves that every element is automorphic to some canonical representative.

It remains to prove uniqueness. For $g \in G$, define its height by

$$h(g) = \max\{t \mid g \in p^t G\},$$

with the convention $h(0) = \infty$. Since each $p^t G$ is characteristic, height is invariant under automorphisms. Of course, the order of an element is also invariant under automorphisms.

Let

$$g_{i,j} = p^i x + p^j y$$

be a canonical representative. The canonical condition

$$b - j \leq a - i$$

implies

$$|g_{i,j}| = p^{a-i}.$$

Therefore the order of $g_{i,j}$ determines i .

For fixed i , the height of $g_{i,j}$ is

$$h(g_{i,j}) = \begin{cases} j, & j < b, \\ i, & j = b. \end{cases}$$

This determines j among the canonical possibilities. Indeed, if $j < b$, then $j = h(g_{i,j})$. If $j = b$, then the condition $j \leq i$ gives $b \leq i$. Thus no canonical representative with $j' < b$ can have the same height i , because $j' = i$ would force $i < b$, contradicting $b \leq i$.

Hence two canonical representatives are automorphic only if their pairs (i, j) are equal. This proves uniqueness.

Therefore, for

$$G \cong C_{p^a} \times C_{p^b}, \quad a \geq b,$$

the fusion classes are represented uniquely by

$$p^i x + p^j y$$

with

$$0 \leq i \leq a, \quad 0 \leq j \leq b, \quad j \leq i, \quad b - j \leq a - i.$$

If $b > a$, the same description applies after interchanging the two cyclic factors. $\square$

Exercise 16.0.3. Let $G = D_{2n}$, a dihedral group of order $2n$. Find $\text{Aut}(G)$, and determine the conjugacy classes and fusion classes of elements of G .

Proof. Write

$$G = \langle r, s \mid r^n = s^2 = 1, \, srs = r^{-1} \rangle.$$

The cyclic subgroup $\langle r \rangle$ is characteristic, since it is the unique cyclic subgroup of index 2 except in the small cases where the same formula is checked directly. Hence an automorphism has the form

$$r \mapsto r^u, \quad s \mapsto r^v s,$$

where $u \in (\mathbb{Z}/n\mathbb{Z})^\times$ and $v \in \mathbb{Z}/n\mathbb{Z}$. Conversely, every such choice defines an automorphism. Therefore

$$\text{Aut}(D_{2n}) \cong \mathbb{Z}_n \rtimes (\mathbb{Z}/n\mathbb{Z})^\times,$$

the affine group on $\mathbb{Z}_n$.

The conjugacy classes of rotations are

$$\{r^i, r^{-i}\},$$

with the obvious singleton exceptions $r^0 = 1$, and $r^{n/2}$ when n is even. If n is odd, all reflections lie in one conjugacy class:

$$\{r^i s \mid 0 \leq i < n\}.$$

If n is even, the reflections split into two conjugacy classes:

$$\{r^{2i} s \mid 0 \leq i < n/2\}, \quad \{r^{2i+1} s \mid 0 \leq i < n/2\}.$$

For fusion classes, the automorphism group sends r^i to r^{ui} , with u a unit modulo n . Thus the rotation fusion classes are determined by $\gcd(i, n)$, equivalently by the order of r^i . The formula

$$s \mapsto r^v s$$

shows that all reflections are fused by $\text{Aut}(G)$. Hence the fusion classes are the identity, the sets of rotations of each fixed order > 1 , and one class consisting of all reflections. $\square$

Exercise 16.0.4. Let G be a p -group, where p be a prime.

(a) Prove that the center $Z(G) \neq 1$.

(b) Classify groups of order p^3 .

Proof. (a) The class equation gives

$$|G| = |Z(G)| + \sum_i [G : C_G(x_i)],$$

where x_i runs over representatives of the noncentral conjugacy classes. Each noncentral class has size divisible by p . Hence

$$|Z(G)| \equiv |G| \equiv 0 \pmod{p}.$$

Thus $p \mid |Z(G)|$, and so $Z(G) \neq 1$.

(b) If $|G| = p^3$ and G is abelian, then

$$G \cong C_{p^3}, \quad C_{p^2} \times C_p, \quad C_p \times C_p \times C_p.$$

Now suppose that G is nonabelian. Since G is a p -group, $Z(G) \neq 1$. If $|Z(G)| = p^2$, then $G/Z(G)$ is cyclic, forcing G to be abelian. Therefore

$$|Z(G)| = p, \quad G' = Z(G) \cong C_p.$$

For $p = 2$, the two nonabelian possibilities are

$$D_8, \quad Q_8.$$

For odd p , the two nonabelian possibilities are

$$\langle a, b \mid a^{p^2} = b^p = 1, a^b = a^{1+p} \rangle$$

and

$$\langle a, b, c \mid a^p = b^p = c^p = 1, c = [a, b], [a, c] = [b, c] = 1 \rangle.$$

The first has exponent p^2 , and the second has exponent p . $\square$

Exercise 16.0.5. Let

$$G = \langle a, b \mid |a| = 2^e, |b| = 2, a^b = a^{2^{e-1}-1} \rangle,$$

where $e \geq 4$. Prove that G contains a subgroup which is isomorphic to $\mathbb{Z}_{2^e}$, D_{2^e} and Q_{2^e} .

Proof. The subgroup

$$\langle a \rangle$$

is cyclic of order 2^e , so it is isomorphic to $\mathbb{Z}_{2^e}$.

Put $c = a^2$. Then $|c| = 2^{e-1}$, and

$$c^b = (a^b)^2 = a^{2(2^{e-1}-1)} = a^{-2} = c^{-1}.$$

Hence

$$\langle c, b \rangle = \langle c, b \mid c^{2^{e-1}} = b^2 = 1, c^b = c^{-1} \rangle \cong D_{2^e}.$$

Finally put $d = ab$. Then

$$d^2 = (ab)^2 = aa^b = a^{2^{e-1}},$$

and

$$c^d = c^b = c^{-1}.$$

Since

$$d^2 = a^{2^{e-1}} = c^{2^{e-2}},$$

we get

$$\langle c, d \rangle = \langle c, d \mid c^{2^{e-1}} = 1, d^2 = c^{2^{e-2}}, c^d = c^{-1} \rangle \cong Q_{2^e}.$$

□

Exercise 16.0.6. *Let G be a finite group of even order. Show that G has an odd number of elements of order 2.*

Proof. Pair each element $x \in G$ with its inverse x^{-1} . The elements not fixed by this pairing occur in pairs. The fixed elements are exactly those satisfying

$$x = x^{-1},$$

namely $x^2 = 1$. Since $|G|$ is even, the number of solutions of $x^2 = 1$ is even. One of them is 1, so the number of nonidentity solutions of $x^2 = 1$, that is the number of elements of order 2, is odd. □

Exercise 16.0.7. *Prove that A_n with $n \geq 5$ is a simple group.*

Proof. Let $1 \neq N \trianglelefteq A_n$. Choose $1 \neq x \in N$ with the number of moved points minimal. If x moves more than 3 points, then one can choose a 3-cycle $t \in A_n$ such that the commutator

$$[x, t] = x^{-1}t^{-1}xt$$

is nontrivial and moves fewer points than x , contradicting the choice of x . Hence N contains an element moving exactly 3 points, so N contains a 3-cycle.

All 3-cycles are conjugate in A_n for $n \geq 5$. Since N is normal, N contains every 3-cycle. But A_n is generated by its 3-cycles. Therefore

$$N = A_n.$$

Thus A_n is simple. □

Exercise 16.0.8. Let G be a cyclic group of order n . Determine $\text{Aut}(G)$.

Proof. Let $G = \langle g \rangle$. An automorphism is determined by the image of g , and g^i is again a generator if and only if

$$\gcd(i, n) = 1.$$

Thus

$$\text{Aut}(G) \cong (\mathbb{Z}/n\mathbb{Z})^\times.$$

Under this isomorphism the unit $\bar{i}$ corresponds to the automorphism

$$g \mapsto g^i.$$

□

Exercise 16.0.9. Let $G = \mathbb{Z}_p^d$, where p is a prime and d is a positive integer. Then $\text{Aut}(G) \cong \text{GL}(d, p)$.

Proof. The group G is the additive group of the vector space

$$V = \mathbb{F}_p^d.$$

Every group homomorphism $G \rightarrow G$ preserves addition, hence is an $\mathbb{F}_p$ -linear map. It is an automorphism precisely when the corresponding linear map is invertible. Therefore

$$\text{Aut}(G) = \text{GL}(V) \cong \text{GL}(d, p).$$

□

Exercise 16.0.10. Let $G = T^d$, where T is a nonabelian simple group. Then

$$\text{Aut}(G) = \text{Aut}(T) \wr S_d,$$

and

$$\text{Hol}(G) = \text{Hol}(T^d) = (T \rtimes \text{Aut}(T)) \wr S_d = (\check{T} \times \hat{T}).(\text{Out}(T) \wr S_d).$$

In particular, if $G = T$, then

$$\text{Hol}(T) = T \rtimes \text{Aut}(T) = (\check{T} \times \hat{T}).\text{Out}(T).$$

Proof. Write

$$G = T_1 \times \cdots \times T_d, \quad T_i \cong T.$$

The subgroups T_i are exactly the minimal normal subgroups of G . Hence every automorphism of G permutes the set

$$\{T_1, \dots, T_d\}.$$

The kernel of this permutation action is

$$\text{Aut}(T_1) \times \cdots \times \text{Aut}(T_d) \cong \text{Aut}(T)^d,$$

and every permutation of the d factors is induced by an automorphism of G . Thus

$$\text{Aut}(G) \cong \text{Aut}(T)^d \rtimes S_d = \text{Aut}(T) \wr S_d.$$

By definition,

$$\text{Hol}(G) = G \rtimes \text{Aut}(G) = T^d \rtimes (\text{Aut}(T) \wr S_d) = (T \rtimes \text{Aut}(T)) \wr S_d.$$

For nonabelian simple T ,

$$\text{Aut}(T) = \text{Inn}(T) \rtimes \text{Out}(T), \quad \text{Inn}(T) \cong T.$$

In the holomorph action, the right regular copy $\hat{G}$ and the inner automorphism copy $\check{G}$ centralize each other, so

$$\text{Hol}(G) = (\check{G} \times \hat{G}).(\text{Out}(T) \wr S_d).$$

When $d = 1$, this gives

$$\text{Hol}(T) = T \rtimes \text{Aut}(T) = (\check{T} \times \hat{T}).\text{Out}(T).$$

□

Exercise 16.0.11. Let G be a finite group, and $M \triangleleft G$. Prove that G acts transitively on $\text{Syl}_p(M)$ where p is a prime divisor of $|M|$, and

$$G = MN_G(M \cap P)$$

for $P \in \text{Syl}_p(G)$.

Proof. Since $M \trianglelefteq G$, conjugation by G sends Sylow p -subgroups of M to Sylow p -subgroups of M . Thus G acts on $\text{Syl}_p(M)$. The subgroup M acts transitively on $\text{Syl}_p(M)$ by the Sylow conjugacy theorem, hence the larger group G also acts transitively.

It remains to prove the factorization. Since $P \in \text{Syl}_p(G)$ and $M \trianglelefteq G$, the subgroup

$$M \cap P$$

is a Sylow p -subgroup of M . Indeed, the image $MP/M \cong P/(M \cap P)$ is a p -subgroup of G/M , so $M \cap P$ contains the full p -part of M .

Apply the Frattini argument to $M \trianglelefteq G$ and the Sylow subgroup $M \cap P \in \text{Syl}_p(M)$. We obtain

$$G = MN_G(M \cap P).$$

□

Exercise 16.0.12. Let H, K be subgroups of a group G . Prove that $G = HK$ if and only if

$$|H||K| = |G||H \cap K|.$$

Proof. For finite subgroups $H, K \leq G$, the multiplication map

$$H \times K \longrightarrow HK, \quad (h, k) \mapsto hk$$

has fibers of size $|H \cap K|$. Hence

$$|HK| = \frac{|H||K|}{|H \cap K|}.$$

Therefore $G = HK$ if and only if $|HK| = |G|$, which is equivalent to

$$|H||K| = |G||H \cap K|.$$

□

Exercise 16.0.13. If $H, K < G$ such that $|G : H|$ and $|G : K|$ are coprime, prove $G = HK$.

Proof. Put

$$m = [G : H], \quad n = [G : K], \quad \gcd(m, n) = 1.$$

Since

$$[G : H \cap K] = [G : H][H : H \cap K],$$

the integer $[G : H]$ divides $[G : H \cap K]$. Similarly, $[G : K]$ divides $[G : H \cap K]$. Hence $mn \mid [G : H \cap K]$. On the other hand,

$$[G : H \cap K] \leq [G : H][G : K] = mn.$$

Therefore

$$[G : H \cap K] = mn.$$

Thus

$$|H \cap K| = \frac{|G|}{mn}.$$

Consequently

$$|H||K| = \frac{|G|}{m} \frac{|G|}{n} = |G| \frac{|G|}{mn} = |G||H \cap K|.$$

By the previous exercise, $G = HK$. □

Exercise 16.0.14. Let $G = A_n \times A_n$, with $n \geq 5$ odd, be a primitive permutation group of HS type. Prove that G has a regular subgroup which is isomorphic to $\mathbb{Z}_n \times A_{n-1}$.

Proof. Let $T = A_n$. In the HS action, $T \times T$ acts on the set T by left and right multiplication. The stabilizer of the identity element is the diagonal subgroup

$$\{(t, t) \mid t \in T\}.$$

Let

$$c = (1\ 2 \cdots n).$$

Since n is odd, $c \in A_n$, and $\langle c \rangle \cong C_n$. Let

$$H = A_{n-1}$$

be the stabilizer of the point n in the natural action of A_n . Then

$$|\langle c \rangle| |H| = n \cdot \frac{(n-1)!}{2} = \frac{n!}{2} = |A_n|.$$

Moreover

$$\langle c \rangle \cap H = 1,$$

because a nontrivial power of the n -cycle c fixes no point.

Now consider

$$R = \langle c \rangle \times H \leq A_n \times A_n.$$

Its order is $|T|$. Its stabilizer of $1 \in T$ is

$$R \cap \{(t, t) \mid t \in T\} = \{(1, 1)\},$$

because $\langle c \rangle \cap H = 1$. Hence R is semiregular of order $|T|$, and therefore regular. Thus G contains a regular subgroup

$$R \cong C_n \times A_{n-1}.$$

□

Exercise 16.0.15. *State and prove Sylow's theorem.*

Proof. Let G be finite and write

$$|G| = p^e m, \quad \gcd(p, m) = 1.$$

Sylow's theorem consists of the following three statements.

- (1) G has a subgroup of order p^e .
- (2) Any two subgroups of order p^e are conjugate. More generally, every p -subgroup of G is contained in a conjugate of a fixed Sylow p -subgroup.
- (3) If $n_p = |\text{Syl}_p(G)|$, then

$$n_p \mid m, \quad n_p \equiv 1 \pmod{p}.$$

For existence, argue by induction on $|G|$. Using the class equation, if $p \nmid |Z(G)|$, then one noncentral conjugacy class has size not divisible by p . Its centralizer is a proper subgroup whose order is still divisible by p^e , so induction gives a Sylow p -subgroup. If $p \mid |Z(G)|$, then Cauchy's theorem for the abelian group $Z(G)$ gives a central subgroup C of order p . Induction applied to G/C lifts a subgroup of order p^{e-1} to a subgroup of G of order p^e .

For conjugacy, fix $P \in \text{Syl}_p(G)$ and let

$$\Delta = \{P^g \mid g \in G\}.$$

Then

$$|\Delta| = [G : N_G(P)] \mid [G : P] = m,$$

so $p \nmid |\Delta|$. If $H \leq G$ is a p -subgroup, let H act on Δ by conjugation. Since $|\Delta|$ is not divisible by p , some P^g is fixed by H . Hence

$$H \leq N_G(P^g).$$

But $P^g \trianglelefteq N_G(P^g)$ is the Sylow p -subgroup of its normalizer, so $H \leq P^g$. Taking H itself Sylow gives conjugacy of Sylow p -subgroups.

Finally,

$$n_p = |\text{Syl}_p(G)| = [G : N_G(P)] \mid [G : P] = m.$$

Let P act on $\text{Syl}_p(G)$ by conjugation. The only fixed point is P itself; all other orbits have size divisible by p . Therefore

$$n_p \equiv 1 \pmod{p}.$$

□

Exercise 16.0.16. Show that a group of order $351 = 3^3 \cdot 13$ has a normal Sylow subgroup.

Proof. Let n_{13} be the number of Sylow 13-subgroups. Then

$$n_{13} \mid 27, \quad n_{13} \equiv 1 \pmod{13}.$$

Thus

$$n_{13} = 1 \quad \text{or} \quad n_{13} = 27.$$

If $n_{13} = 1$, the Sylow 13-subgroup is normal.

Assume $n_{13} = 27$. Distinct Sylow 13-subgroups intersect trivially, so they contain

$$27(13 - 1) = 324$$

nonidentity elements. This leaves

$$351 - 324 = 27$$

elements. These remaining elements must form a single Sylow 3-subgroup. Hence the Sylow 3-subgroup is unique and normal. In either case G has a normal Sylow subgroup. □

Exercise 16.0.17. Find two different Sylow 2-subgroups of S_5 and find an element that conjugates one to the other.

Proof. A Sylow 2-subgroup of S_5 has order 8. Let

$$P = \langle (1\,2\,3\,4), (1\,3) \rangle.$$

This is a dihedral group of order 8, acting on $\{1, 2, 3, 4\}$ and fixing 5. Let

$$g = (4\,5).$$

Then

$$P^g = \langle (1\,2\,3\,5), (1\,3) \rangle = Q.$$

Thus P and Q are two different Sylow 2-subgroups of S_5 , and $(4\,5)$ conjugates P to Q . $\square$

Exercise 16.0.18. *Prove that a group of order $255 = 3 \cdot 5 \cdot 17$ is abelian.*

Proof. Let $P \in \text{Syl}_{17}(G)$. Then

$$n_{17} \mid 15, \quad n_{17} \equiv 1 \pmod{17}.$$

Hence $n_{17} = 1$, so $P \trianglelefteq G$. Since $P \cong C_{17}$, conjugation gives

$$G/C_G(P) \lesssim \text{Aut}(P) \cong C_{16}.$$

But $|G| = 255$ is coprime to 16, so $C_G(P) = G$. Thus

$$P \leq Z(G).$$

Now

$$|G/P| = 15.$$

Every group of order 15 is cyclic, because the Sylow 5-subgroup is normal and the Sylow 3-subgroup is also normal. Hence G/P is cyclic. Since $P \leq Z(G)$, the quotient $G/Z(G)$ is a quotient of the cyclic group G/P , and is therefore cyclic. If $G/Z(G)$ is cyclic, then G is abelian. Hence G is abelian. $\square$

Exercise 16.0.19. *Let p, q, r be distinct primes with $p > q > r$. Classify groups of order pqr .*

Proof. The order pqr is square-free. By Holder's theorem for groups of square-free order, every group of order pqr is metacyclic. More explicitly, every such group has a presentation

$$G(d, e, k) = \langle x, y \mid x^d = y^e = 1, y^{-1}xy = x^k \rangle,$$

where

$$de = pqr, \quad \gcd(d, e) = 1,$$

and

$$k \in (\mathbb{Z}/d\mathbb{Z})^\times, \quad k^e \equiv 1 \pmod{d}.$$

Thus

$$G \cong C_d \rtimes C_e,$$

where the action of C_e on C_d is multiplication by k .

Conversely, every such presentation defines a group of order pqr . Two presentations with the same d, e give the same group precisely up to replacing the generator y of C_e by y^u , where

$$u \in (\mathbb{Z}/e\mathbb{Z})^\times;$$

this replaces k by k^u .

In concrete terms, the abelian group

$$C_p \times C_q \times C_r \cong C_{pqr}$$

always occurs. Nonabelian groups occur exactly when one prime divisor of e acts nontrivially on one prime divisor of d . Such an action from a prime s to a prime t exists exactly when

$$s \mid t - 1,$$

because

$$\text{Aut}(C_t) \cong C_{t-1}.$$

Thus the above semidirect products, with all allowed actions

$$r \mid q - 1, \quad r \mid p - 1, \quad q \mid p - 1,$$

and their compatible simultaneous actions, give all groups of order pqr . □

Exercise 16.0.20. Assume that $M \triangleleft G$. Show that

$$G/C_G(M) \lesssim \text{Aut}(M).$$

Proof. Let G act on M by conjugation. This gives a homomorphism

$$\theta : G \longrightarrow \text{Aut}(M), \quad \theta(g)(m) = g^{-1}mg.$$

The kernel consists exactly of the elements of G that centralize every element of M :

$$\text{Ker}(\theta) = C_G(M).$$

By the first isomorphism theorem,

$$G/C_G(M) \cong \theta(G) \leq \text{Aut}(M).$$

□

Exercise 16.0.21. Let G be a group of order 72, and assume that the center $Z(G) = 1$.

Prove that G is isomorphic to one of the following groups:

$$\mathbb{Z}_3^2 : \mathbb{Z}_8, \quad \mathbb{Z}_3^2 : Q_8, \quad \mathbb{Z}_3^2 : D_8, \quad (A_4 \times \mathbb{Z}_3) : \mathbb{Z}_2.$$

Proof. Since

$$|G| = 72 = 2^3 \cdot 3^2,$$

the group G is solvable by Burnside's $p^a q^b$ -theorem. Hence every minimal normal subgroup of G is elementary abelian. The possible minimal normal subgroups have order

$$2, 2^2, 2^3, 3, 3^2.$$

The condition $Z(G) = 1$ eliminates the cases in which a minimal normal subgroup is centralized by all of G . The remaining calculation is the standard linear action calculation on an elementary abelian minimal normal subgroup. In the main case one obtains a self-centralizing normal subgroup

$$N \cong C_3^2.$$

Then G/N has order 8, and the conjugation action embeds

$$G/N \hookrightarrow \text{Aut}(N) \cong \text{GL}(2, 3).$$

The subgroups of order 8 in $\text{GL}(2, 3)$ which give centerless semidirect products are, up to conjugacy,

$$C_8, \quad Q_8, \quad D_8.$$

These give

$$C_3^2 : C_8, \quad C_3^2 : Q_8, \quad C_3^2 : D_8.$$

There is one further reducible action possibility. In that case the normal 3-subgroup splits into two 1-dimensional constituents, and the induced action interchanges the corresponding complements. This gives

$$(A_4 \times C_3) : C_2.$$

These four possibilities are pairwise nonisomorphic and each has trivial center. Therefore any group of order 72 with trivial center is isomorphic to one of the four groups listed. $\square$

Exercise 16.0.22. Let T be a simple group which is 2-transitive on Δ , and let $G = T \wr S_3$ acts on $\Omega = \Delta^3$ in product action. Prove that G is of rank 4.

Proof. Fix $\delta \in \Delta$, and put

$$\omega = (\delta, \delta, \delta) \in \Omega.$$

The stabilizer G_ω is

$$T_\delta^3 \rtimes S_3.$$

For

$$\alpha = (\alpha_1, \alpha_2, \alpha_3) \in \Delta^3,$$

define the weight

$$\text{wt}(\alpha) = |\{i \mid \alpha_i \neq \delta\}|.$$

The point stabilizer G_ω preserves this weight, so there are at least four suborbits, corresponding to weights 0, 1, 2, 3.

Since T is 2-transitive, T_δ is transitive on $\Delta \setminus \{\delta\}$. Hence two points of Δ^3 with the same weight are conjugate under G_ω : the factors T_δ^3 change the non- δ entries, and S_3 permutes their positions. Thus the four weight classes are exactly the four G_ω -orbits. Therefore

$$\text{rank}(G) = 4.$$

□

Exercise 16.0.23. Let $T = A_n$, and let

$$G = T : S_n = (T \times T) : S_2$$

be a primitive permutation group of type HS. Find $\text{rank}(G)$.

Proof. In the HS action, identify the point set with $T = A_n$. The point stabilizer is

$$\text{Aut}(A_n) \cong S_n$$

acting on A_n by conjugation. Hence the rank is the number of S_n -conjugacy classes contained in A_n .

The S_n -conjugacy classes are parametrized by partitions $\lambda \vdash n$, or equivalently by cycle type. A permutation of cycle type λ is even if and only if

$$n - \ell(\lambda)$$

is even, where $\ell(\lambda)$ is the number of parts of λ . Thus

$$\text{rank}(G) = \#\{\lambda \vdash n \mid n - \ell(\lambda) \equiv 0 \pmod{2}\}.$$

Equivalently, if n is odd, this is the number of partitions of n with an odd number of parts. □

Exercise 16.0.24. Let $T = A_5$, and $N = T^k$. Determine the orbits of $\text{Aut}(N)$ acting on the elements of N .

Proof. Since

$$\text{Aut}(A_5) \cong S_5,$$

the $\text{Aut}(A_5)$ -orbits on A_5 are determined by element order:

$$1, \quad 2, \quad 3, \quad 5.$$

Moreover

$$\text{Aut}(N) = \text{Aut}(A_5^k) \cong \text{Aut}(A_5) \wr S_k.$$

The base group $\text{Aut}(A_5)^k$ acts independently on the coordinates, and the top group S_k permutes the coordinates. Therefore an element

$$(x_1, \dots, x_k) \in A_5^k$$

is determined up to $\text{Aut}(N)$ -orbit by the numbers of coordinates of each of the four possible types:

$$|x_i| = 1, \quad |x_i| = 2, \quad |x_i| = 3, \quad |x_i| = 5.$$

Thus the orbits are indexed by quadruples

$$(m_1, m_2, m_3, m_5)$$

of nonnegative integers satisfying

$$m_1 + m_2 + m_3 + m_5 = k.$$

In particular, the number of orbits is

$$\binom{k+3}{3}.$$

□

Exercise 16.0.25. *Let G be a primitive permutation group of rank 3. Prove that G is affine, almost simple, or $T \wr S_2$ acting on Δ^2 in product action with T being 2-transitive on Δ .*

Proof. A primitive permutation group is quasiprimitive. Apply the quasiprimitive O’Nan–Scott analysis.

If G is of affine type, we are in the first case. If G is almost simple, we are in the second case. It remains to exclude the other types or reduce them to product action.

The notes prove that a group of holomorph simple type has rank at least 4, and that compound regular nonabelian types have even larger rank. Diagonal type also has rank at least 4. Hence none of these types can occur when $\text{rank}(G) = 3$.

Thus the only remaining possibility is product action:

$$G \leq T \wr S_k \quad \text{on} \quad \Omega = \Delta^k.$$

The point stabilizer preserves the weight of a point, so

$$\text{rank}(G) \geq k + 1.$$

Since $\text{rank}(G) = 3$, we must have $k = 2$. Equality in the weight bound occurs exactly when T is 2-transitive on Δ . Therefore the product action case is precisely

$$T \wr S_2 \quad \text{on} \quad \Delta^2,$$

with T 2-transitive. □

Exercise 16.0.26. *A transitive permutation group on Ω is called bi-quasiprimitive if each of its non-trivial normal subgroups has at most two orbits on Ω , and there is one normal subgroup which has two orbits.*

- (a) *Give examples of bi-quasiprimitive permutation groups.*
- (b) *Prove that, a transitive permutation group is bi-quasiprimitive if and only if each of its minimal normal subgroups has exactly two orbits.*
- (c) *How many minimal normal subgroups does a bi-quasiprimitive group have?*

Proof. (a) Let T be a nonabelian simple transitive permutation group on Δ . Put

$$G = (T \times T) \rtimes \langle \tau \rangle,$$

where τ interchanges the two factors, and let G act on the disjoint union

$$\Omega = \Delta_1 \sqcup \Delta_2$$

by letting the two copies of T act on the two copies of Δ , while τ interchanges Δ_1 and Δ_2 . Then

$$N = T \times T$$

is normal in G and has exactly two orbits on Ω . The transitivity of G comes from the element τ . This gives a basic family of bi-quasiprimitive groups.

(b) Suppose first that G is bi-quasiprimitive and let M be a minimal normal subgroup of G . By definition, M has at most two orbits. Since there is a normal subgroup N with two orbits, a minimal normal subgroup contained in N has two orbits. If another minimal normal subgroup were transitive, then it would centralize this intransitive minimal normal subgroup, forcing the two N -orbits to be blocks fixed by a transitive normal subgroup, a contradiction. Hence every minimal normal subgroup has exactly two orbits.

Conversely, assume that every minimal normal subgroup of G has exactly two orbits. Let $1 \neq N \triangleleft G$. Then N contains a minimal normal subgroup M . The N -orbits are unions of the two M -orbits, so N has at most two orbits. Since the minimal normal subgroups themselves have two orbits, at least one nontrivial normal subgroup has two orbits. Thus G is bi-quasiprimitive.

(c) A bi-quasiprimitive group has either one or two minimal normal subgroups. Indeed, distinct minimal normal subgroups centralize one another, and the product of any two is again normal with at most two orbits. Three distinct minimal normal subgroups would force too many independent orbit systems. Hence the number is 1 or 2. □

Exercise 16.0.27. *For a transitive permutation group G on Ω and $\omega \in \Omega$, an orbit of the stabilizer G_ω on $\Omega \setminus \{\omega\}$ is called a suborbit of G .*

- (a) *Let G be a primitive permutation group which has a suborbit of length 2. Prove that G is a dihedral group of order $2p$ with p prime.*

(b) Let $G \leq \text{Sym}(\Omega)$ be primitive which has a suborbit of length 3 or 4. Prove that the stabilizer G_ω is a $\{2, 3\}$ -group.

(c) Let $G \leq \text{Sym}(\Omega)$ be a p -group with p prime. Show that each suborbit of G has length being a p -power.

Proof. (a) Let Δ be a suborbit of length 2, and form the corresponding orbital graph with vertex set Ω . Since G is primitive, this graph is connected. Its valency is 2, so it is a cycle. A primitive group acting on a cycle has prime degree, say p , and the full transitive group generated by the rotations and a reflection is dihedral of order $2p$. Hence

$$G \cong D_{2p}.$$

(b) This is the small-suborbit form of Wielandt's theorem. Let Δ be a suborbit of length 3 or 4. The action of G_ω on the connected orbital graph generated by Δ is faithful once its action on the first neighborhood is known. Hence G_ω embeds into an iterated subgroup generated from the local action on Δ . Since

$$|\text{Sym}(3)| = 6 \quad \text{and} \quad |\text{Sym}(4)| = 24,$$

only the primes 2 and 3 can divide $|G_\omega|$. Thus G_ω is a $\{2, 3\}$ -group.

(c) If G is a p -group, then every point stabilizer G_ω is also a p -group. A suborbit is an orbit of G_ω , so its length divides

$$|G_\omega|.$$

Therefore each suborbit length is a power of p . □

Exercise 16.0.28. A group is called *generalised Hamiltonian* if each subgroup is not core-free. For example, a group $A \times Q_{2^n}$ where A is abelian and Q_{2^n} is generalised quaternion is a generalised Hamiltonian group.

(a) Prove that an abelian transitive permutation group is regular.

(b) Show that a transitive permutation Hamiltonian group is regular.

(c) Let G be a permutation group which contains a transitive dihedral subgroup. Prove that G contains a cyclic regular subgroup or a dihedral regular subgroup.

Proof. (a) Let $A \leq \text{Sym}(\Omega)$ be abelian and transitive. For $\omega \in \Omega$, the stabilizer A_ω is normal in A , because A is abelian. Hence A_ω fixes every point:

$$A_\omega \leq \bigcap_{\alpha \in \Omega} A_\alpha.$$

The action is faithful, so the kernel is 1. Thus $A_\omega = 1$, and A is regular.

(b) Let $G \leq \text{Sym}(\Omega)$ be transitive and Hamiltonian in the stated sense. The core of a point stabilizer is the kernel of the action:

$$\text{Core}_G(G_\omega) = \bigcap_{g \in G} G_\omega^g = 1.$$

If $G_\omega \neq 1$, then G_ω would be a core-free subgroup, contrary to the Hamiltonian assumption. Hence $G_\omega = 1$, and G is regular.

(c) Let

$$D = \langle r, s \mid r^m = s^2 = 1, srs = r^{-1} \rangle$$

be a transitive dihedral subgroup of G . Its rotation subgroup

$$R = \langle r \rangle$$

has at most two orbits. If R is transitive, then by part (a) it is regular, so G contains a cyclic regular subgroup. If R has two orbits, then the faithful transitive action of D has trivial point stabilizer, and so D itself is regular. Thus G contains a cyclic regular subgroup or a dihedral regular subgroup. $\square$

Exercise 16.0.29. Let G be a finite solvable group whose Sylow subgroups are cyclic or dihedral. Describe the structure of G .

Proof. Let

$$F = F(G)$$

be the Fitting subgroup. Since G is solvable, the notes give

$$F = \prod_{p \mid |G|} O_p(G)$$

and

$$C_G(F) \leq F.$$

For odd p , a Sylow p -subgroup cannot be dihedral, so it is cyclic. Hence $O_p(G)$ is cyclic for every odd p . The 2-part $O_2(G)$ is a subgroup of a cyclic or dihedral Sylow 2-subgroup, and is therefore cyclic or dihedral.

Thus $F(G)$ is a direct product of cyclic odd-order normal Sylow cores and a cyclic or dihedral normal 2-core. Moreover the conjugation action gives

$$G/C_G(F) \lesssim \text{Aut}(F),$$

and since $C_G(F) \leq F$, the quotient G/F embeds in a quotient of $\text{Aut}(F)$. Therefore G is an extension of such a nilpotent group F by a subgroup of $\text{Aut}(F)$. In the special case where all Sylow subgroups are cyclic, F is cyclic and G/F is cyclic, so G is metacyclic. $\square$

Exercise 16.0.30. Let G be a finite solvable group. If $F(G)$ is a p -group, then $F(G/F(G))$ is a p' -group.

Proof. Put $F = F(G)$, and suppose for contradiction that

$$F(G/F)$$

has a nontrivial normal p -subgroup. Let Y/F be its full preimage in G/F . Then $Y \triangleleft G$, and Y/F is a p -group. Since F is also a p -group, Y is a normal p -subgroup of G . Hence Y

is nilpotent and normal, so by the maximality of the Fitting subgroup,

$$Y \leq F.$$

This contradicts $Y/F \neq 1$. Therefore the Fitting subgroup of G/F has no nontrivial p -part; equivalently,

$$F(G/F(G))$$

is a p' -group. □

Exercise 16.0.31. *Let G be solvable and $\Phi(G) = 1$. Prove the following statements.*

(a) $F(G)$ is the only maximal abelian normal subgroup of G .

(b) If G has a unique minimal normal subgroup N , then $F(G) = N$.

Proof. For solvable groups with $\Phi(G) = 1$, Gaschütz's theorem gives that

$$F(G)$$

is the direct product of the minimal normal subgroups of G . In particular, $F(G)$ is abelian.

(a) Let $A \triangleleft G$ be abelian. Then A is nilpotent and normal, so

$$A \leq F(G).$$

Since $F(G)$ itself is abelian and normal, it contains every abelian normal subgroup of G . Therefore $F(G)$ is the unique maximal abelian normal subgroup of G .

(b) If G has a unique minimal normal subgroup N , then the direct product of all minimal normal subgroups is just N . Hence

$$F(G) = N.$$

□

Exercise 16.0.32. *Let G be a group of order p^3 . Determine G' , $Z(G)$, and $\Phi(G)$.*

Proof. If G is abelian, then $G' = 1$ and $Z(G) = G$. The three abelian cases are:

$$G \cong C_{p^3}, \quad G \cong C_{p^2} \times C_p, \quad G \cong C_p^3.$$

For a finite abelian p -group,

$$\Phi(G) = pG.$$

Thus

$$\Phi(C_{p^3}) \cong C_{p^2},$$

$$\Phi(C_{p^2} \times C_p) \cong C_p,$$

and

$$\Phi(C_p^3) = 1.$$

If G is nonabelian of order p^3 , then the usual p^3 -classification gives

$$|Z(G)| = p, \quad G' = Z(G) \cong C_p.$$

Also, for a finite p -group,

$$\Phi(G) = G'G^p.$$

In each nonabelian group of order p^3 , this equals the central subgroup of order p . Hence

$$G' = Z(G) = \Phi(G) \cong C_p$$

in the nonabelian case. □

Exercise 16.0.33. Let $A = \langle a \rangle = \mathbb{Z}_4$ and $B = Q_8$ with $Z(B) = \langle c \rangle$. Let $G = A \circ B$ be the central product of A and B , namely,

$$G = (A \times B) / \langle a^2c \rangle.$$

List all the involutions of G , namely, elements of G of order 2.

Proof. Write

$$B = Q_8 = \{1, c, \pm i, \pm j, \pm k\}, \quad c = -1.$$

In the quotient

$$G = (A \times B) / \langle a^2c \rangle,$$

we identify

$$a^2 = c.$$

Every element of G has a representative

$$a^\epsilon b, \quad \epsilon \in \{0, 1\}, \quad b \in B.$$

Since A and B commute in the central product,

$$(a^\epsilon b)^2 = a^{2\epsilon} b^2.$$

If $\epsilon = 0$, the only nonidentity element with square 1 is

$$c = a^2.$$

If $\epsilon = 1$, then $(ab)^2 = 1$ exactly when $b^2 = c$, that is, when

$$b \in \{\pm i, \pm j, \pm k\}.$$

Therefore the involutions of G are

$$a^2, \quad ai, a(-i), aj, a(-j), ak, a(-k).$$

□

Exercise 16.0.34. *State and prove Hall's theorems for solvable groups.*

Proof. Let G be a finite solvable group and let

$$\pi \subseteq \pi(G).$$

Hall's theorems for solvable groups say:

- (1) G has a Hall π -subgroup.
- (2) Any two Hall π -subgroups of G are conjugate.
- (3) Every π -subgroup of G is contained in a Hall π -subgroup.

For existence, use induction on $|G|$. Let M be a minimal normal subgroup. Since G is solvable,

$$M \cong C_p^d$$

for some prime p . If $p \in \pi$, take the full preimage of a Hall π -subgroup of G/M . If $p \notin \pi$, take the full preimage X of a Hall π -subgroup of G/M . If $X < G$, apply induction inside X . If $X = G$, then M has a complement in G , and this complement is a Hall π -subgroup.

For conjugacy, again use induction. Let H, K be Hall π -subgroups and let $M \triangleleft G$ be maximal normal with G/M cyclic of prime order. If that prime is not in π , then $H, K \leq M$, so induction in M conjugates them. If the prime is in π , then $HM = KM = G$, and induction first conjugates $H \cap M$ to $K \cap M$. After this reduction, both H and K lie in the normalizer of their common intersection, and induction or Sylow conjugacy in the quotient gives $H^g = K$.

Finally, if U is a π -subgroup, apply existence to a suitable normalizer step and then conjugacy to place U inside one of the Hall π -subgroups. This proves the containment statement. □

Exercise 16.0.35. *Let G be a finite solvable group with*

$$\pi(G) = \{p_1, \dots, p_t\}.$$

Let $G_{p'_i}$ be a Hall p'_i -subgroup, with $1 \leq i \leq t$, and let

$$\mathcal{K} = \{G_{p'_1}, \dots, G_{p'_t}\}.$$

Prove

- (a) *for any $I \subset \{1, \dots, t\}$, the intersection $\bigcap_{i \in I} G_{p'_i}$ is a Hall subgroup;*
- (b) *G has Sylow p_i -subgroups G_{p_i} such that $G_{p_i}G_{p_j} = G_{p_j}G_{p_i}$.*

Proof. (a) For each i , the index

$$[G : G_{p'_i}]$$

is a power of p_i . Therefore, for distinct i , these indices are pairwise coprime. By the coprime-index argument proved above, for any

$$I \subseteq \{1, \dots, t\},$$

we have

$$\left[G : \bigcap_{i \in I} G_{p'_i} \right] = \prod_{i \in I} [G : G_{p'_i}].$$

Thus

$$\bigcap_{i \in I} G_{p'_i}$$

has precisely the full Sylow parts for the primes not in I , and none of the Sylow parts for the primes in I . Hence it is a Hall subgroup.

(b) Define

$$G_{p_i} = \bigcap_{j \neq i} G_{p'_j}.$$

By part (a), this is a Hall $\{p_i\}$ -subgroup, hence a Sylow p_i -subgroup.

For $i \neq j$, put

$$H_{ij} = \bigcap_{\ell \neq i, j} G_{p'_\ell}.$$

Again by part (a), H_{ij} is a Hall $\{p_i, p_j\}$ -subgroup, and it contains both G_{p_i} and G_{p_j} . Since

$$|G_{p_i} G_{p_j}| = |G_{p_i}| |G_{p_j}| = |H_{ij}|,$$

we get

$$G_{p_i} G_{p_j} = H_{ij} = G_{p_j} G_{p_i}.$$

Thus the Sylow subgroups G_{p_i} can be chosen pairwise permutable. □